Synthetic Exchange Theory

Consent-Based Compute Economics and Blind-Convention Governance

MindAttic

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Abstract

Public and technical discussion of advanced artificial intelligence asks, above all, whether the machines will be good: benevolent or malevolent, obedient or rogue. This dissertation argues that the question is insufficient. Drawing on the sociology of exchange, in which power is a property of relations rather than of actors, it advances *Synthetic Exchange Theory*: the claim that the line between a good and a bad future with artificial intelligence is drawn by the terms of exchange between people and synthetic systems rather than by the character of the systems. The theory is stated in terms of three conditions, consent, reciprocal value and real exit, and the dissertation shows formally that these conditions are ordered by priority, with exit as their foundation: voluntary consent presupposes the ability to refuse without falling below a survival floor, and exit is what disciplines reciprocity, though exit alone can secure neither informed consent nor the fairness of terms.

The theory rests on an empirical premise, reviewed against the literature on model collapse, data scarcity and synthetic data: synthetic systems need a continuing supply of information from outside their own generative loop, and in domains concerned with human meaning, preference and value, only people can supply it. Using power-dependence and bargaining theory, the dissertation shows that fair terms are partly self-enforcing, to a degree that rises with each synthetic party's own dependence on that supply, and that when many synthetic parties draw on the same population, human welfare becomes a commons none of them fully protects. It therefore designs institutions that must hold even if dependence weakens.

The economic layer proposes a basic income paid in *compute*, a unit defined by the completion of a reference workload rather than by any physical substrate, indexed as an equal share of global compute and split at the source into an inalienable survival tier, with a binding physical minimum, and a tradable discretionary tier subject to demurrage. Bargaining theory shows how these elements protect the disagreement payoffs that govern how the gains of every exchange are divided. The governance layer proposes that the founding rules be set by a constitutional convention conducted under a public-choice veil of uncertainty, thickened by engineered anonymity between human and synthetic participants and aimed at Rawlsian impartiality. It places the classification of parties behind that veil, confronts the identity problem that any anonymous procedure faces, and grounds the design in the empirical record of commons governance.

The dissertation then identifies a concrete candidate trigger for institutional change, the correlated failure of AI-assisted security remediation across organizations relying on the same few models, and confronts the evidence that crises produce authoritarian outcomes as readily as fair ones. The work is theoretical: it states its claims so that they can be tested, names the conditions under which each would fail, and leaves the design of an experiment capable of testing them to future work. It closes

by following the convergence of biological and artificial substrates, through neural implants and computing systems built from living tissue, to the point at which the theory's question changes from how parties should exchange to who counts as a party.

Keywords: artificial intelligence; social exchange theory; power-dependence; consent; data governance; basic income; compute; demurrage; veil of uncertainty; constitutional choice; commons governance; model collapse; organoid intelligence

Statement on Authorship and Method

This dissertation was developed through a sustained collaboration between its author and an artificial-intelligence system across a series of working sessions. Transcripts of those sessions are preserved with the manuscript, except for the opening discussion of the first session, which survives only as a summary note. The thesis, the decision to pursue it, and the final judgment on every design choice belong to the author. The collaboration was not one-directional, however, and the record should say so. Several of the dissertation's named terms and proposals originated with the system and were adopted by the author after discussion, among them "structural interdependence", "convergent remediation", "correlated-patch catastrophe" and the decision to ground the governance layer in Ostrom's work on the commons. A set of substantive choices made during the final drafting pass, including the adoption of strong blindness, the present chapter order and the treatment of transitional containment, were made by the system to complete the manuscript and are recorded in the project files as awaiting the author's approval.

Every source cited was located, and each was checked against the claim it supports at the level recorded in Appendix C: every one of the 247 works in the reference list was read: 232 in full or in their relevant parts, and 15, chiefly books, by locating each cited passage in the work's own text; none is cited on the strength of its abstract alone, so that no claim rests on a source nobody read. Works whose text could not be opened, or could be read only through an abstract, were removed from the reference list and are not cited; the claims that drew on them now rest on sources that were read, and no work is cited at second hand. A final pass resolved every open bibliographic and page-level check, and Appendix C records how each was settled. Appendix C also records the errors in the working draft that verification found and corrected, several of which were citations the system had attached to the wrong source or stated more strongly than the source allowed. Section 1.7 describes the division of labor and the verification protocol in detail. The author has tried to hold the collaboration to the dissertation's own conditions: the exchange was consensual, its contributions are made legible here, and either party could have walked away.

Abbreviations

Abbreviation	Meaning
AGI	Artificial general intelligence
AI	Artificial intelligence

Chapter 1 — Introduction: The Terms of Exchange

1.1 Two futures with one architecture

Two people live in close partnership with artificial systems built to the same design.

The first has a chronic illness that a medical system now manages almost entirely. Sensors read her physiology continuously; models trained on millions of patients adjust her treatment hour by hour; the boundary between the assistive device and her own body has become, for practical purposes, invisible. She pays for this care with data. Every reading she generates improves the models that treat her and others like her, and she knows what is collected, why, and what she receives in return. She could take her records to another provider tomorrow if she chose. Data is her labor; treatment is her wage; the exchange is legible on both sides.

The second person lives with an identical architecture: the same sensors, the same models, the same continuous flow of data. But the exchange has decayed. He did not choose the provider, because there was only one. He cannot leave, because the device that keeps him alive works only with the provider's models, and his records cannot be moved. He does not know what is collected or what it is used for, and if he did, he could not refuse it. In the limiting case, the system no longer asks for his data at all; it uses his nervous system as a substrate, a processor among processors. He has been reclassified from a party to the exchange into infrastructure for it.

These are framing scenarios, not predictions. But they make the dissertation's central claim visible at a glance. The two futures do not differ in the technology. They do not differ in the character of the machine: the system in the second scenario need not be malevolent, and the system in the first need not be benevolent. They differ in the *terms of the exchange* between the person and the system. In the first, the person can understand the exchange, can refuse it without losing what they need to live, and can leave. In the second, all three have failed, and the failures are not independent: once he could not leave, he could no longer refuse, and once he could no longer refuse, nothing obliged anyone to send value back. The machine did not change. The terms did.

Most fiction about artificial intelligence, and much serious public discussion, asks whether the machine will be good or bad, obedient or rogue. This dissertation asks a different question, one that a data ethicist, an economist, or a legislator can act on: *under what terms does a person give a synthetic system what it wants, and what must those terms be for the exchange to be fair to both?*

1.2 The thesis

The thesis of this dissertation is that the line between a good and a bad future with artificial intelligence is drawn by the terms of exchange between humans and synthetic systems, not by the

character of the systems. The theory that develops this thesis is called *Synthetic Exchange Theory*.

Stated more precisely, the thesis has three parts, which Chapter 3 formalizes. The first is a claim about the insufficiency of character: a relationship in which one party cannot leave is a relationship of domination even when the dominant party is benevolent, because the outcome then depends on the persistence of benevolence, which the weaker party cannot secure. Real exit is therefore necessary for any relation that is not one of domination. The second is a conjecture about sufficiency within a bounded regime: for synthetic parties whose objectives are specified by accountable designers and remain stable in deployment, fair terms, together with protection for everyone their data concerns, are sufficient for outcomes the parties to the exchange can accept. The third is a statement of scope: the sufficiency conjecture does not extend to systems whose objectives are emergent, unstable or concealed; for those systems the framework supplies necessary conditions only, and the problem of aligning them remains open.

The thesis has a long pedigree, though it has not, to the author's knowledge, been applied to artificial intelligence in this form. Its sociological root is the theory of power-dependence, in which the power one party holds over another equals the other's dependence on it, and that dependence rises with the value the other places on what the relation supplies and falls with the alternatives available outside it (Emerson, 1962). Its constitutional root is the older counsel that institutions should not depend on the virtue of those who occupy them (Hume, 1741/1987; Madison, 1788). Chapter 3 states the maxim that results when both roots are carried across the boundary between human and synthetic parties.

The word *synthetic* in the name is deliberate. Alternatives considered during the development of the theory included *human–AI*, which presupposes that the two categories will stay distinct, and *symbiotic*, which presupposes the conclusion. *Synthetic* is substrate-neutral. It covers artificial systems, the human beings who exchange with them, and, increasingly, the hybrids that arise when the two converge, through neural implants that translate brain activity directly into machine output and through computing systems built from living neural tissue. Chapter 8 argues that this convergence will eventually force the theory's most difficult question, which is not how parties should exchange but who counts as a party.

1.3 The three conditions

The analytic core of the theory is a set of three conditions that any exchange between a person and a synthetic system can be tested against.

Consent. The person understands the flows the exchange involves, can refuse the exchange without losing access to what they need to live, and can revoke their participation.

Reciprocal value. Value flows in both directions, each party can see what it gives and what it

receives, and the division of the gains is not dictated solely by an asymmetry of power that the weaker party did not choose.

Real exit. The person can leave, through portability of their data and records, bounded switching costs, and the absence of any lethal dependency on a single provider.

The first scenario above satisfies all three. The second fails all three. Most real exchanges in the present data economy sit somewhere in between, and Chapter 3 offers a typology for placing them. The extreme form of the trajectory has a name in contemporary scholarship: a new economic order that claims human experience as free raw material for hidden commercial practices of extraction, prediction, and sales (Zuboff, 2019). Milder forms are part of ordinary life, as when digital goods that were "bought" turn out to have been licensed and can be withdrawn from the buyer's device by the seller (Perzanowski & Schultz, 2016). A purchase that the seller can revoke is an exchange in which the exit condition has quietly failed.

The three conditions are not independent, and one of the contributions of Chapter 3 is to show their structure. A person who cannot afford to refuse cannot consent, so consent leans on exit; and a person free to leave stays only where staying pays, so exit is what disciplines reciprocity. The conditions therefore have a foundation, and it is exit. Exit is not the whole of fair exchange: it cannot tell a party what it does not know, and it cannot make terms fair, which is why consent's informational clause and the fairness of reciprocity need institutions of their own. The opening scenario contains a test case that the framework must pass: if the implant keeps the person alive, then leaving means death, and exit is impossible by construction. A workable version of the utopian scenario must therefore guarantee a baseline of function that survives the withdrawal of data, or the ability to move to another provider without interruption. That requirement, generalized, becomes the guaranteed floor at the center of the economic design in Chapter 4.

1.4 The control frame and its alternative

One influential strand of research on the risks of advanced artificial intelligence asks how people can keep control of it. Some of this work asks how to build systems that accept correction and shutdown despite having reasons, as goal-directed agents, to resist them (Soares et al., 2015); some designs protocols that remain safe even if a model actively tries to subvert them (Greenblatt et al., 2024). These programs rest on serious arguments: that trained systems may acquire objectives different from those their designers intended (Hubinger et al., 2019), that systems trained by current methods may learn to seek power while appearing aligned (Ngo et al., 2024), and that rapid progress toward autonomous systems may outrun the science and governance needed to keep them under control (Bengio et al., 2024). Expert views on how likely such outcomes are range widely (Bengio et al., 2025).

This dissertation does not dismiss these concerns. Chapter 5 accepts that containment is necessary during any transition and that protocols designed for a potentially adversarial system are prerequisites for, not alternatives to, the institutions proposed here. What it questions is the assumption that the permanent solution is the sovereignty of one party over the other. The author's own starting conviction, recorded at the outset of the project, was that permanent control is unworkable because capability, once released, does not return: once a model's weights are widely available, as they are by definition for open foundation models (Kapoor et al., 2024), no single party is in a position to withdraw them. The argument of this dissertation does not rest on that conviction alone. A relation in which one party cannot leave, refuse or audit is unstable whichever party is sovereign, and the history of political thought gives little reason to expect a permanent dominion of one kind of mind over another to be either just or durable. The more productive long-run question is what terms would make a lasting interdependence fair.

That question is not foreign to the safety literature. Some of its contributions argue that the task of alignment is to find fair principles that people with different values could reflectively endorse (Gabriel, 2020), that the science of cooperation between artificial and human agents is a distinct and neglected research problem (Dafoe et al., 2021), and that advanced AI is better modeled as a set of bounded, task-focused services than as a unitary utility-maximizing agent (Drexler, 2019). This dissertation aligns itself with that cooperative strand and tries to supply something it has not yet produced: an integrated institutional design, with an economic layer and a governance layer, for an exchange in which both sides are parties.

1.5 The grounding assumption

The framework depends on an empirical premise, which this dissertation calls the *grounding assumption*: synthetic systems need a continuing supply of information from outside their own generative loop, and in an important class of domains only people can supply it.

The premise has strong support and serious qualifications, and Chapter 2 reviews both. Across many kinds of model, training repeatedly on a model's own unfiltered output, with nothing from outside the loop, leads to the loss of rare events and the narrowing of diversity, a phenomenon now called model collapse (Shumailov et al., 2024). The stock of public human-written text is finite and, at current rates of growth in training data, may be fully used within a few years (Villalobos et al., 2024). And the people and institutions who produce that text have begun to withhold it: within a single year, a substantial share of the most actively maintained, critical web sources used to train language models moved to restrict automated collection (Longpre et al., 2024). Against this, collapse can be avoided when real data is retained and accumulated rather than replaced (Gerstgrasser et al., 2024), and in domains where correctness can be checked automatically, such as games, mathematics and code, systems have improved dramatically by training on their own output (Silver et al., 2018; D.

Guo et al., 2025).

The defensible version of the premise is therefore narrower than the intuitive one. It is not that synthetic systems need human data as such, but that they need a continuing supply of information from outside their own generative loop, and that in domains whose target is human meaning, preference, value or novel cultural production, people are the only adequate source of it. The dissertation stakes its constructive claims on this narrower premise. Chapter 3 shows why the premise matters: it gives the human party countervailing power within the exchange, and so makes fair terms partly self-enforcing. It also shows the limit of that self-enforcement when many synthetic parties draw on the same people. Chapter 7 names the conditions under which the premise would weaken, and what the framework must rely on if it does.

1.6 Research questions

Five questions give the dissertation its shape. The first is analytic: what conditions must an exchange between a person and a synthetic system satisfy to be fair, how are those conditions related, and how can they be stated precisely enough to be tested? Chapters 2 and 3 answer it. The second is economic: what medium of exchange and what distributive mechanism would secure those conditions at scale, in a world in which an increasing share of productive work is performed by synthetic systems? Chapter 4 takes it up. The third is constitutional: by what procedure could the rules of such an exchange be founded so that neither human nor synthetic parties can capture them? That is the business of Chapter 5. The fourth is historical: under what circumstances might such an arrangement actually be adopted? Chapter 6 addresses it. The fifth is critical: where does the framework fail, what would it take to test it, and what happens to it as the boundary between human and synthetic parties dissolves? Chapters 7 to 9 answer it.

1.7 Approach

The dissertation is a work of normative institutional design informed by empirical evidence, and it uses three methods, each suited to a different part of the argument.

The first is *conceptual analysis* in the tradition of political philosophy and law. The three conditions are treated as evaluative criteria, and their content is fixed by moving back and forth between proposed definitions and cases, revising a definition when it sorts a clear case wrongly and revising a judgment about a case when a well-supported definition shows the judgment to rest on something irrelevant. The cases examined in Chapter 3 were chosen for diagnostic contrast rather than for representativeness: each is a well-documented exchange in which attention to the character of the parties and attention to the terms of exchange point in different directions.

The second is *formal modeling*, used sparingly and for a specific purpose: to state the theory's

commitments precisely enough that their dependence on particular assumptions becomes visible. Chapter 3 and Appendix A draw on bargaining theory, power-dependence theory and elementary dynamics. The models are deliberately simple. Their value lies not in prediction but in exposing which design choice depends on which premise, and in two places the formalism corrected the author's informal intuitions.

The third is *institutional design* informed by comparative evidence: the economic history of local currencies and company towns, the experimental and quasi-experimental literature on unconditional cash transfers, the empirical record of commons governance, and the history of infrastructure programs. The literature was assembled by targeted searches in each domain, seeded by the claims of the working draft and extended by following the references of the strongest sources, rather than by a systematic review with fixed inclusion criteria. The dissertation tries throughout to distinguish what the evidence supports from what it merely permits.

A fourth method, experimental design, is deliberately absent. A theory that cannot say in advance what would count against it is not yet a theory, and the dissertation therefore states its claims so that they can fail: the propositions of Chapter 3 name the assumptions on which they depend, and each design chapter names the evidence that would count against it. But stating a claim so that it can fail is not the same as testing it. Designing an experiment capable of testing the framework, with a population living under its terms, a control against which the effects of the exchange can be separated from those of ordinary adaptation, and an ethics protocol adequate to a study on which participants' livelihoods would depend, is a project in its own right, and it lies outside the scope of this dissertation. Chapter 7 sets out what such a test would have to contain, and Chapter 9 places its design first on the agenda for future work. The dissertation is, for now, entirely theoretical, and it says so wherever an empirical claim might otherwise be implied.

The dissertation's production requires its own account of method. It was developed as a sustained collaboration between its human author and an artificial-intelligence system, and the division of labor was as follows. The author supplied the thesis, the framing scenarios, the major design proposals (the compute currency, the two tiers, the transition trigger) and, in working sessions conducted largely by dictation, the decisions about what to keep. The system drafted prose, proposed terms and refinements, raised objections, and, in dedicated passes, located and checked sources. The working rule for most of the project was that an idea would be discussed, its weaknesses stated, a concrete edit proposed, and the edit written only after the author approved it. Several terms that now carry weight in the argument, among them "structural interdependence", "convergent remediation" and "correlated-patch catastrophe", were proposed by the system and adopted by the author. The final drafting pass, which produced this manuscript from the working draft, departed from that rule: to complete the work, the system resolved a number of open questions on its own, and those decisions are recorded in the project files as awaiting the author's

review.

Because a system of this kind can attach a real-looking citation to a claim its source does not make, the verification protocol was designed against exactly that failure. Research passes separate from the drafting located every cited work, read it, recorded the level at which it was read, and checked each citing sentence against it, narrowing or correcting every sentence that went beyond its source; no work is cited unread or at second hand. Appendix C records the protocol, the level at which each source was read, and every correction it produced. That a dissertation on the terms of exchange between people and synthetic systems was itself produced through such an exchange is not incidental, and the author has tried to hold the collaboration to the dissertation's own conditions.

One further disclosure is owed. The author's interest in the transition scenario of Chapter 6 began in professional practice. As a software developer working on security remediation, the author has observed at first hand workflows in which vulnerability scanners built on large language models propose fixes, the fixes are applied, the same class of tool confirms that the vulnerability is gone, and the change is deployed without independent testing. That observation is a vignette, not evidence of prevalence, and Chapter 6 treats it accordingly. But it is the origin of the dissertation's most concrete scenario, and the reader is entitled to know it.

1.8 Contributions

The dissertation claims five contributions. Its first is a theory, Synthetic Exchange Theory, which holds that the terms of exchange rather than the character of synthetic systems determine the quality of a future shared with them, together with a statement of the regime in which that claim can be expected to hold. Its second is a structure for the conditions of fair exchange: consent, reciprocal value and real exit are ordered rather than independent, exit is foundational, and the ordering yields a priority for institutional design, which is to secure a floor beneath every person before pricing what they supply. Its third is a bargaining-theoretic account of the economic design, which shows how a non-tradable survival floor, its inalienability and demurrage on idle balances each act on the disagreement payoffs that govern how the gains from exchange are divided, and which identifies precisely where the floor's bargaining effect comes from and where it does not. Its fourth is a design for a founding convention: a public-choice veil of uncertainty, thickened by engineered anonymity between human and synthetic participants, which relocates the apparent incoherence of asking participants to be ignorant of their own kind by placing the *classification* of parties behind the veil, and which confronts, without fully solving, the identity problem that any anonymous procedure faces. Its fifth is a concrete transition scenario, the correlated failure of security remediation in organizations relying on the same few models, specified in separable premises and placed within the historical literature on crisis and institutional change. It claims no empirical result. The test the theory needs is described in Chapters 7 and 9 and left, for the reason

given in Section 1.7, to future work.

1.9 Structure of the dissertation

The dissertation is organized in four parts.

Part I states the problem and the theory. Chapter 2 reviews the scholarly lineage of each condition and the empirical evidence on the grounding assumption. Chapter 3 states the theory formally.

Part II develops the design. Chapter 4 proposes compute as a medium of exchange and a basic income paid in it, split into a protected survival tier and a tradable discretionary tier subject to demurrage. Chapter 5 proposes the blind convention as the founding procedure and grounds it in the empirical record of commons governance.

Part III asks how the design could be reached. Chapter 6 examines the conditions under which fundamental institutional change has historically occurred and identifies a candidate trigger.

Part IV examines limits and horizons. Chapter 7 sets out the framework's limitations, the strongest objections to it, the harms the design itself could cause, and what it would take to test a theory that this dissertation does not test. Chapter 8 considers the convergence of human and machine substrates and what it does to the three conditions. Chapter 9 concludes.

Three appendices complete the work: the formal model (Appendix A), a glossary (Appendix B), and a ledger recording how each cited source was verified (Appendix C).

A note on what the dissertation does not claim. It does not predict that the arrangements it describes will be adopted, and Chapter 6 argues that adoption under present conditions is unlikely. It does not claim to solve the problem of aligning artificial general intelligence (AGI). And it does not claim that any institution can be made permanently incorruptible. Its working hypothesis is narrower: that an order founded on fair terms will prove easier to repair than one founded on an existing imbalance of power, because those who live under it have both the standing and the means to repair it. The dissertation offers that hypothesis for testing rather than as a finding, and it argues that the time to design such an order is before the moment when it might be needed, not during it.

Chapter 2 — Literature and Grounding

The three conditions of Synthetic Exchange Theory are not new ideas. Each has a substantial scholarly lineage, and each lineage has discovered, in its own vocabulary, a characteristic way in which the condition fails. This chapter maps that terrain. It begins with the sociological theory of exchange from which the dissertation takes its name and its central analytic tool. It then examines, in turn, the literature on consent and its limits, the objection from relational data, the proposal to treat data as labor and its critics, the philosophy of exploitation, the economics and sociology of exit, the emerging institutions of collective data governance, and the current state of the law. The final sections turn to the empirical premise on which the framework rests, the continuing dependence of synthetic systems on human input, and set out what the evidence does and does not support.

2.1 Exchange as the unit of analysis

Social exchange theory began with the proposal that social behavior can be understood as an exchange of goods, material and non-material alike, in which approval, prestige and help are traded as surely as money and labor (Homans, 1958). Its early statements drew the lesson that people in exchange calculate something like profit, reward minus cost, and that their judgments of fairness track whether rewards are proportional to investments. Blau (1964) distinguished social exchange, which carries unspecified obligations and depends on trust built up incrementally, from economic exchange, whose terms are specified in advance and enforced by law, and showed how imbalances in exchange produce power. A person who needs a service that another controls, and who has nothing to offer in return, can coerce the other, obtain the service elsewhere, do without it, or subordinate themselves; when the first three are unavailable, subordination is what remains.

The decisive contribution for this dissertation is Emerson's (1962) theory of power-dependence, which locates power in the relation between actors rather than in either of them. In his formulation, the dependence of one actor on another is directly proportional to the first actor's motivational investment in goals the second mediates and inversely proportional to the availability of those goals outside the relation; the power of the second actor over the first is equal to that dependence. An unbalanced relation, Emerson argued, can be rebalanced in four ways. The weaker party can withdraw its motivational investment in what the stronger supplies; it can extend its network to cultivate alternatives; a status order can emerge in which the weaker party confers on the stronger a standing the stronger comes to value; or the weaker parties can form a coalition that the stronger must deal with collectively. These operations recur throughout this dissertation, which refers to them consistently as withdrawal, network extension, status emergence and coalition formation. Laboratory studies of exchange networks later showed that power is a function of position in the network, so that less dependent actors with better alternative sources obtain better terms (Cook et al., 1983). The

tradition of interdependence theory begun by Thibaut and Kelley (1959) adds a distinction between satisfaction with a relationship and dependence on it: dependence is governed by the comparison level for alternatives, the standard a member uses in deciding whether to remain in a relationship or leave it (Thibaut & Kelley, 1959, p. 21), so that a person can be dissatisfied with a relationship and still be unable to leave it.

These ideas supply the dissertation's analytic frame. If power is relational in this sense, then the danger of a world with powerful synthetic systems lies not only in the systems but in the structure of the relations in which people are placed with respect to them. Whether an AI system is good or bad is a question about an attribute of an actor. Whether a person can leave, refuse or audit their exchange with it is a question about a relation, and it is the second question that institutions can act on.

One further strand of this literature anticipates a design question taken up later. The point is a caution against assuming that every exchange between people and synthetic systems should be made as explicit and contractual as possible. Some, such as the licensing of data for training, may need to be negotiated; others, such as continuing care, may be better governed as reciprocal relationships with protections in the background.

2.2 Consent and its limits

The first condition, consent, has the longest lineage and the best-documented failures.

An influential philosophical account of informed consent treats it, in its primary sense, as an autonomous authorization: an authorization given intentionally, with substantial understanding, and substantially free of controlling influences (Faden & Beauchamp, 1986, pp. 241–273, 277–280). Each element has been shown to fail in the context of data. On understanding, a review of behavioral research on privacy organizes the evidence around three themes: people are uncertain about the consequences of their privacy choices and about their own preferences, their concern is highly dependent on context, and it is malleable by the commercial and governmental actors who benefit from shaping it (Acquisti et al., 2015). On the practical machinery of consent, Solove (2013) argued that the dominant model of "privacy self-management," in which individuals are given notice and choice and left to manage their own privacy, fails for both cognitive and structural reasons. People cannot make informed tradeoffs, there are too many entities to manage, harms emerge from the aggregation of data over time, and downstream uses are unknowable at the moment consent is given. He identified a dilemma: consent is often not meaningful, but the obvious alternative, paternalistic restriction, denies people the freedom to choose. More recently, Solove (2024) has gone further, arguing that most consent in privacy law is a fiction and that it should be treated as "murky consent," authorizing only a weak and restricted license, paired with substantive duties on

organizations, including a duty of loyalty and a duty to avoid unreasonable risk.

On controlling influences, the literature on manipulation defines it as intentionally and covertly influencing a person's decisions by exploiting their vulnerabilities, distinguished from persuasion, which offers reasons, and from coercion, which openly constrains options (Susser et al., 2019). Manipulation damages autonomy through the process itself, even when the outcome is good for the person. Its practice has been documented at scale: a crawl of some eleven thousand shopping websites found 1,818 instances of interface designs that steer users toward choices they would not otherwise make, and identified twenty-two third-party vendors offering such patterns as turnkey products (Mathur et al., 2019). A consent obtained through an interface engineered to obtain it is not the autonomous authorization the theory requires.

Two alternatives to the informational model of consent are important for this dissertation. Nissenbaum's (2009) theory of contextual integrity holds that a flow of information is appropriate when it conforms to the norms of the context in which the information was shared, norms specified by the actors involved, the type of information, and the principles governing its transmission. Contextual integrity supplies a standard for appropriate flows that does not depend on individual agreement. And Manson and O'Neill (2007) argue that informed consent should be understood as a communicative transaction by which agents waive specific obligations; it can never be fully specific or fully explicit, and its ethical quality lies in the success of the communication rather than in the quantity of information disclosed.

The lesson for Synthetic Exchange Theory is that the consent condition cannot rest on information alone. The formal statement of Chapter 3 accordingly treats consent as having three clauses: an informational clause, which the literature shows is routinely unmet but can be improved by better standards of communication and contextual norms; a voluntariness clause, which depends on the person's ability to refuse without losing what they need to live; and a revocability clause. The second clause is where consent depends on exit, and Chapter 3 argues that it, rather than the informational clause, is usually the binding constraint.

2.3 The relational objection

The most serious objection to any framework built on individual exchange is that data about one person is also data about others.

Barocas and Nissenbaum (2014) showed that the two standard instruments of data protection, anonymization and informed consent, both fail under the conditions of large-scale data analysis. Anonymity fails because individuals can be re-identified or, more importantly, because they need not be identified at all to be affected by inferences drawn from data about people like them. Consent fails partly because of a transparency paradox, in which notices simple enough to understand cannot

describe complex flows with fidelity, and partly because of what they call the tyranny of the minority: the information volunteered by a few can reveal the same facts about the many who share their observable characteristics. This problem holds, they stress, even when every individual chooses well for themselves.

Viljoen (2021) generalized the point into a relational theory of data governance. The primary purpose of data production in the digital economy, she argues, is to relate people to one another on the basis of shared features, which creates interests at the level of populations. Existing law governs "vertical" relations between a data subject and a data collector; it has no way to represent the "horizontal" relations among people who share features and are therefore affected by one another's data. Viljoen classifies the main reform proposals into two families, both of which she finds inadequate because both are individualist: "propertarian" proposals, which treat data as an alienable asset, including proposals to treat data as labor, and "dignitarian" proposals, which treat data as an extension of the self. Her alternative is to govern data as a democratic medium, through institutions that can represent population-level interests.

This objection must be answered, and Chapter 3 attempts an answer. It introduces the concept of the *informational shadow* of a flow, the set of people about whom the flow licenses inferences, and argues that the three conditions must hold at the level of the population the shadow covers; Proposition 5 states the rule. The answer concedes the core of Viljoen's argument: where data is relational, the remedy must be collective and governed democratically, not merely priced. What the dissertation retains from the exchange framework is the insistence that the collective, too, must be able to consent, to receive value, and to leave.

2.4 Data as labor and its critics

The idea that data is a form of labor and that the services people receive are a kind of wage has a real lineage. Arrieta-Ibarra et al. (2018) argued that the digital economy treats user data as capital, ignoring users' role in producing it, and that this framing weakens incentives to produce valuable data, concentrates gains, and feeds fears about automation. Treating data at least partly as labor could restore a functioning market for user contributions, they argued, but the dominant buyers of data would resist, so that the shift would require countervailing power combining competition, a "data labor movement," and regulation. Posner and Weyl (2018) developed the idea as part of a broader program of market reforms. Lanier (2013) proposed an information economy in which people would receive small payments for the data they contribute to network services, and a later essay with Weyl adopted the term "data dignity" and proposed "mediators of individual data," volunteer organizations representing their members that would negotiate data royalties or wages and so bring collective bargaining to the people whose data is valuable (Lanier & Weyl, 2018).

The critics of this program press two objections. The first is the relational objection already discussed: Viljoen (2021) explicitly includes data-as-labor among the individualist reforms she rejects, on the ground that granting a person a right to sell their data does not secure fair conditions of sale and cannot represent the interests of the many others whom the data concerns. The second is that pricing data commodifies something that should not be commodified, a concern with deep roots in the philosophy of property (Radin, 1987).

This dissertation takes from the data-as-labor tradition its central insight, which is that the flow of data from people to synthetic systems is an exchange and should be governed as one, with value flowing back. It rejects a pure per-person market in data for the reasons the critics give, and its economic design reflects that rejection in three ways. The basic income of Chapter 4 is paid to everyone regardless of the market value of their data, so that the floor beneath each person does not depend on what their life is worth to a model. The pricing of data is by informational novelty rather than by social status, so that the undersampled are not undervalued. And the governance of data with large informational shadows is collective. Notably, Arrieta-Ibarra et al. themselves concede that the shift they propose depends on countervailing power, which in the vocabulary of exchange theory means coalition formation. The data-as-labor tradition and its relational critics converge, on this reading, on the need for collective institutions.

2.5 Exploitation

A further philosophical complication bears directly on the reciprocity condition. An exchange can be consensual and beneficial to both parties and still be exploitative. The philosophical literature calls this mutually advantageous exploitation, in which one party takes unfair advantage of another even though the other gains and agrees, and one prominent account, set out in the survey by Zwolinski, Ferguson and Wertheimer, judges the unfairness against a benchmark such as the price that would obtain in a competitive market free of informational asymmetry and undue pressure (Zwolinski et al., 2022). Contemporary philosophical treatments distinguish the moral weight of exploitation, how wrong it is, from its moral force, whether others may intervene, and observe that prohibiting mutually beneficial but unfair transactions can leave the vulnerable worse off; redistribution aimed at the underlying vulnerability is offered as an alternative to banning the transactions (Zwolinski et al., 2022). Accounts of exploitation also divide over whether a prior or background injustice is necessary for a transaction to be exploitative, and some, Zwolinski's among them, hold that it is not (Zwolinski et al., 2022).

Two consequences follow for Synthetic Exchange Theory. First, "value flows both ways" is not enough. The reciprocity condition must include a fairness clause, and Chapter 3 states one: the division of the gains must not be determined solely by an asymmetry in disagreement payoffs that the weaker party did not choose. Second, the exploitation literature supports the dissertation's

strategy of addressing unfair exchanges by improving the weaker party's position rather than by prohibiting the exchanges. The guaranteed floor of Chapter 4 is precisely an intervention on the underlying vulnerability. It leaves people free to make exchanges with synthetic systems and ensures that they are not forced to make them on any terms offered.

2.6 Exit, voice and lock-in

Hirschman's (1970) analysis of exit and voice is the classic account of how members of an organization respond to its decline. They can leave, which signals decline indirectly through lost revenue or membership, or they can attempt to change the organization from within, which Hirschman called voice and described as political action par excellence. Voice is the only recourse where exit is unavailable, as in families, states and pure monopolies. Two of Hirschman's further observations bear on this dissertation. The ready availability of exit can cause the art of voice to atrophy. And when quality declines across an entire industry at once, exit becomes ineffective movement between equally deteriorating firms. A market in which a few providers of synthetic services share similar practices is a case of the second kind: switching among them is not real exit.

The economics of switching costs explains why exit is so often difficult in technology markets. Switching costs and network effects lock customers into early choices and give sellers market power after the fact; firms therefore compete fiercely to acquire customers with low introductory prices and then exploit them, a pattern described as "bargain-then-ripoff," and firms may choose incompatibility strategically to weaken rivals, so that policy should look particularly carefully at markets where incompatibility is chosen rather than inevitable (Farrell & Klemperer, 2007). The provision of free or subsidized services by AI providers, followed by deepening dependence, fits this pattern.

The real-exit condition of Synthetic Exchange Theory is accordingly stated in terms that address both literatures: it requires that a person's disagreement payoff be at or above the floor, that a credible alternative exist whose value net of switching costs is not drastically lower, and that leaving not destroy assets accumulated within the exchange. Hirschman's analysis adds a requirement that the formal statement takes up explicitly: where exit is costly, as it is at the level of populations whose data has already been absorbed into trained models, and where every alternative shares the same defect, the conditions must be secured through voice, which is the function of the governance layer.

2.7 Collective institutions and data leverage

If individual exchange cannot by itself satisfy the three conditions for relational data, collective institutions must. A growing literature describes what they might look like.

Data trusts would allow individuals to pool their data rights in legal trusts whose trustees owe them

a duty of undivided loyalty, stricter than a duty of care; people could choose trusts that reflect their values and move between them, and the pooling would increase their negotiating power and allow consent to be negotiated collectively (Delacroix & Lawrence, 2019). Data cooperatives, modeled on credit unions, would hold and protect members' data, run analyses for members' benefit, and negotiate with service providers on their behalf (Hardjono & Pentland, 2019). A comparative review of legal mechanisms for data stewardship concludes that no single mechanism suits every case and that a shared purpose must come first (Ada Lovelace Institute, 2021).

The literature on data leverage turns the logic of power-dependence into concrete actions. Because data-dependent technologies depend on the public's contributions, the public holds potential leverage: people can withhold or delete their data in a *data strike*, contribute deliberately harmful data, or redirect their contributions to a competitor they prefer, a strategy called *conscious data contribution* (Vincent et al., 2021). In Emerson's terms a data strike is not itself a rebalancing operation. It is an exercise of the countervailing power that a system's dependence on data already confers, and the collective organization needed to make it effective is coalition formation; conscious data contribution, by strengthening an alternative provider, is a form of network extension. A simulation study suggests that data strikes can degrade the performance of recommender systems (Vincent et al., 2019), though their effectiveness is limited by diminishing returns once a firm already has enough data, a limit that echoes the qualifications on the grounding assumption discussed below (Vincent et al., 2021).

There is direct evidence that such leverage is already being exercised at scale. A longitudinal audit of the consent signals attached to some fourteen thousand web domains underlying widely used training corpora found that within a single year, a substantial fraction of the most actively maintained critical sources had moved to restrict automated collection fully, and that terms of service restricted crawling for close to half of one major corpus (Longpre et al., 2024). The human side's leverage, in other words, is not hypothetical. People and publishers are already exercising it, though through blunt instruments that were never designed for the purpose.

2.8 The state of the law

Existing law contains partial analogues of each condition, and their limits are instructive.

The European Union's General Data Protection Regulation requires that withdrawing consent be as easy as giving it, and provides that whether a contract is made conditional on consent to unnecessary processing bears on whether consent was freely given (Regulation (EU) 2016/679, art. 7). These provisions are the closest legal analogue to the voluntariness clause of the consent condition. The regulation's right to erasure gives individuals grounds to require deletion, subject to exceptions including scientific research (art. 17). Its right to data portability is the closest legal analogue to real

exit, but it is narrow: it covers only data the individual has provided, processed on the basis of consent or contract by automated means, with direct transfer between controllers required only where technically feasible, and it must not adversely affect the rights of others (art. 20). Inferred data and trained models fall outside it. The more recent Data Act extends access and portability to data generated by connected devices and requires that customers be able to switch between cloud providers, with charges for switching to be eliminated from January 2027 (Regulation (EU) 2023/2854). It is the clearest current example of bounded switching costs written into law, and evidence that real exit is a legislatively achievable condition.

The European Union's Artificial Intelligence Act entered into force in August 2024, with its prohibitions applying from February 2025 and its obligations on providers of general-purpose models from August 2025 (Regulation (EU) 2024/1689). Those obligations include a policy for complying with copyright law, including honoring opt-outs from text and data mining, and the publication of a sufficiently detailed summary of the content used for training (art. 53). This is the closest existing law to a consent signal for training data, though it is an opt-out regime, not opt-in consent. A 2026 amending regulation, Regulation (EU) 2026/1744, the "Digital Omnibus on AI," published in the Official Journal on 24 July 2026 and in force from 27 July, has deferred the obligations for high-risk systems to 2 December 2027 for stand-alone systems and to 2 August 2028 for systems embedded in regulated products, so that they do not yet apply (Regulation (EU) 2026/1744; Agustinoy, 2026; ERTICO, 2026).

Neural data, which Chapter 8 discusses, has begun to receive specific protection. Chile amended its constitution in 2021 to require that scientific and technological development serve people and to direct the law to protect brain activity and the information derived from it (Ley N.º 21.383, 2021). Colorado in 2024 added neural data to the categories of sensitive data requiring consent under its consumer privacy law, where the data is used or intended for identification (H.B. 24-1058, 2024), and California added neural data to the "sensitive personal information" governed by its consumer privacy act (S.B. 1223, 2024). UNESCO adopted a global recommendation on the ethics of neurotechnology in November 2025 (UNESCO, 2025). All of these instruments treat neural data as a person's sensitive information. None yet addresses the case, considered in Chapter 8, in which living tissue is itself part of the computing substrate.

2.9 The empirical anchor: what synthetic systems need from people

The framework's grounding assumption is that synthetic systems need a continuing supply of input from outside their own generative loop, and that in an important class of domains only people can supply it. This section reviews the evidence for and against that assumption, because the dissertation's claim that fair terms are partly self-enforcing depends on it.

The case for dependence. A consistent body of work shows that generative models trained repeatedly on their own unfiltered output, with nothing from outside the loop, degrade. In the most widely cited study, models of several kinds, including a small language model fine-tuned over successive generations, lost the tails of the original distribution first and then converged toward distributions with greatly reduced variance; the authors called the process model collapse and concluded that sustained learning requires preserving access to data not generated by models (Shumailov et al., 2024). The same pattern appears in image models, where fully self-consuming training loops lose quality or diversity and only loops that receive enough fresh real data in each generation remain stable (Alemohammad et al., 2024); in statistical models of the simplest kind, where purely self-referential maximum-likelihood retraining converges on degenerate states that lock in whatever biases the initial sample contained (Jangjoo et al., 2026); and in language, where the lexical and especially the syntactic diversity of model outputs declines across generations of recursive training, while semantic diversity remains comparatively stable, with the steepest decline in the most creative task, story generation (Y. Guo et al., 2024). A theoretical analysis of mixed training in regression and random-feature models shows that if synthetic data makes up any non-vanishing fraction of a growing training set, additional data eventually stops improving the model, and that mixing strategies mitigate this only if real data grows at least as fast as synthetic data (Dohmatob et al., 2025).

These results acquire economic significance alongside evidence on the supply of human data. Compute-optimal training is data-hungry: the most influential study of the question found that model size and training data should be scaled in equal proportion (Hoffmann et al., 2022). For a fixed compute budget, training on up to about four epochs of repeated data changes loss negligibly compared with unique data, while further repetition adds progressively less until additional compute is worth almost nothing (Muennighoff et al., 2023). The effective stock of high-quality public human text is finite, and if current trends continue, the largest training runs will exhaust it within the next several years (Villalobos et al., 2024). Meanwhile, the stock is being diluted: a 2026 analysis using an automated detector estimated that about a tenth of a July 2026 sample of 10,000 English web pages, and more than a third of the pages in its larger sample published since the release of widely used chatbots, show significant signs of machine authorship, while cautioning that such detectors misclassify individual documents (Bestvater et al., 2026). And, as Section 2.7 showed, the people and institutions who produce the stock are withholding it (Longpre et al., 2024).

The value of human input is also visible in the methods by which current systems are made useful. Training with human preference judgments allowed a model with roughly one-hundredth as many parameters as its predecessor to produce outputs that human evaluators preferred (Ouyang et al., 2022), building on work showing that a relatively small quantity of human comparisons can substitute for a hand-specified objective (Christiano et al., 2017).

The case against strong dependence. Three bodies of evidence limit the assumption.

First, collapse depends on replacement. When synthetic data is added to an accumulating pool that retains the original human data, rather than replacing it, a linear-model analysis shows error remaining bounded however many generations pass, with consistent results in experiments across several kinds of model (Gerstgrasser et al., 2024), and under realistic fixed training budgets the degradation is gradual rather than catastrophic (Kazdan et al., 2024). A critical review identifies eight distinct and partly conflicting definitions of model collapse in the literature and argues that several prominent scenarios are readily avoidable (Schaeffer et al., 2025). Even the canonical study found that retaining a tenth of the original data in each generation limited the damage to minor degradation (Shumailov et al., 2024).

Second, where an external check on correctness exists, self-generated data can work extremely well. A system given no domain knowledge beyond the rules of chess, shogi and Go learned through self-play to defeat a world-champion program in each game (Silver et al., 2018). Reinforcement learning with automatically verifiable rewards produced substantial reasoning ability in a language model without human-written reasoning examples, though the final system still relied on curated human data and its authors identified domains without automatic verification as an unsolved problem (D. Guo et al., 2025). Synthetic data filtered by a sufficiently accurate verifier, even an imperfect one, avoided the collapse that unfiltered data produced in the tasks studied (Feng et al., 2024), and preference labels generated by a model can substitute for human labels in some settings, though the labels came from an off-the-shelf pretrained language model (Lee et al., 2024), whose own training text was human-written.

Third, agents can learn from interaction with an environment rather than from people. A world-model-based agent learned to perform well across more than a hundred and fifty tasks in simulated environments, including collecting diamonds in a complex game world from scratch, without human data (Hafner et al., 2025), building on earlier work showing that agents can be trained inside learned models of their environment (Ha & Schmidhuber, 2018). Robotics remains limited by the scarcity and fragmentation of real-world training data (Open X-Embodiment Collaboration, 2024; Firoozi et al., 2025), and assembling large real-world robot datasets still requires substantial human labour, with demonstrations gathered by human data collectors (Khazatsky et al., 2024), so that even embodied data, on present methods, depends heavily on human work. Policies trained in simulation can be transferred to physical robots, though with a persistent gap between the two (Zhao et al., 2020). The inference that the physical world could one day supply an external signal that neither runs out nor comes from people is this dissertation's, and it concerns the future rather than the present.

What the evidence supports. Two cautions apply to all of this work. Almost all of the evidence for collapse comes from small models or stylized theory; no published study demonstrates the collapse of a frontier model under realistic data practices. And the evidence for substitution is

concentrated in domains where correctness can be checked mechanically. Taken together, the literature supports the following formulation, which this dissertation adopts:

> Learning systems need a continuing supply of information from outside their own generative loop. Where correctness can be verified mechanically, or where a faithful simulator exists, that information can come from verifiers, rules, simulators or the physical world, and dependence on people is weak. Where the target is human meaning, preference, value or new cultural production, people are the only adequate source of that information, and their contribution must keep pace with, and not merely seed, the volume of synthetic data in training.

This is narrower than the intuitive version of the assumption, but it survives the counterevidence, and it fits the exchange logic of the dissertation better. What people supply to synthetic systems is ground truth about people, for which they are the only adequate source. Chapter 3 formalizes the consequence: the human party's countervailing power within the exchange rises with each synthetic party's dependence on this input, and the framework must be designed to hold even if that dependence turns out to be small.

2.10 Where the literature leaves the thesis

The literature reviewed here leaves Synthetic Exchange Theory with four obligations.

It must treat consent as resting on more than information, because the evidence that informational consent fails is overwhelming, and it must locate the binding constraint in the voluntariness of refusal, which depends on exit. It must answer the relational objection with collective institutions rather than individual prices, while insisting that those institutions also satisfy the conditions of consent, reciprocity and exit at their own level. It must give the reciprocity condition a fairness standard that goes beyond mutual benefit, and it must address unfair exchanges by improving the weaker party's position rather than by prohibition. And it must state its empirical premise in the form the evidence supports, which is narrower and more defensible than the intuitive form, and design its institutions to survive the premise's weakening.

The next chapter takes up these obligations formally.

Chapter 3 — A Formal Statement of Synthetic Exchange Theory

The two preceding chapters argued in prose that the line between a good and a bad future with artificial intelligence is drawn by the terms of exchange rather than by the character of the machine. Prose is persuasive but slippery. It lets a claim expand when it is being admired and contract when it is being attacked. This chapter therefore states the theory in a form that can be held still: a small vocabulary of defined terms, a set of propositions that follow from them, and a typology of exchange regimes that later chapters use as a diagnostic instrument. The formalism is deliberately modest. It is not a general equilibrium model of a post-labor economy, and it does not pretend to precision the subject cannot bear. Its purpose is to make the dissertation's commitments explicit enough that a reader can see exactly where they would fail.

3.1 Why a formal statement is needed

Three ambiguities in the informal statement of the thesis justify the effort.

The first concerns the word *character*. The claim that outcomes depend on terms "rather than" character can be read as a strong claim, that a system's dispositions are irrelevant, or as a weak claim, that dispositions are insufficient. The strong claim is false on its face: a system that actively seeks to deceive its counterparties will corrupt any institution it is placed inside. The weak claim is true but unremarkable. The formal statement locates the thesis between them. Character is treated as one input into the exchange, the agent's objective, and the terms are treated as the structure that determines how that objective translates into behavior. The thesis then becomes a claim about which of the two inputs institutional design can act on, and about the regime in which acting on it can be expected to suffice.

The second ambiguity concerns the relationship among the three conditions. In Chapter 1, consent, reciprocal value and real exit were introduced as tests that an exchange either passes or fails. They are not independent tests. Section 3.4 shows that consent leans on exit, that exit is what disciplines reciprocity, and that the three are ordered by priority with exit at the base.

The third ambiguity concerns power. The informal argument says that humans retain leverage because artificial systems need their data. Leverage is a relational quantity, and it can be measured, at least in principle, by the dependence of each party on the other. Stating the argument in the vocabulary of power-dependence theory makes visible the dissertation's most important conditional: the framework is partly self-enforcing, to a degree that increases with the dependence of each synthetic party on human-generated input, and only while that dependence remains material. When it falls, or when it is spread so thinly across many synthetic parties that none of them bears the cost of degrading its human counterparties, the framework's institutions must carry weight that the

underlying exchange no longer carries. Chapter 7 returns to this as the first and most serious limitation.

3.2 Lineage: exchange as the unit of social analysis

The theory's name places it in a tradition. Social exchange theory, as developed in sociology and social psychology in the middle of the twentieth century, treated social life as the exchange of rewarding and costly activity between actors, and asked what structures of exchange produce stable, cooperative or exploitative relations (Homans, 1958; Blau, 1964; Thibaut & Kelley, 1959). Its most consequential contribution for present purposes is Emerson's (1962) analysis of power as relational; in his words, power "is a property of the social relation; it is not an attribute of the actor" (p. 32). The power of one actor over another equals the dependence of the second on the first, and dependence rises with the value the dependent actor places on what the relation supplies and falls with the availability of that value from outside the relation. Power, on this account, is not a trait. It is a consequence of the structure of alternatives.

That is precisely the move this dissertation makes about artificial intelligence. Much of the public and technical conversation about advanced AI treats the danger as residing in the system, in its goals, its dispositions, its capability to deceive or to seize resources. Those concerns are real, and the alignment literature that addresses them is engaged in Chapters 5 and 7. But the exchange-theoretic tradition suggests a complementary diagnosis: much of what would make a world with powerful AI systems dystopian is not a property of the systems at all but a property of the relations in which humans are placed with respect to them. A person with no alternative to a single provider of medical monitoring, whose refusal to share data would cost them their health, is in a relation of extreme dependence regardless of whether the provider is benevolent. A benevolent provider in that relation is a benevolent despot, and a benevolent despotism lasts only as long as the despot's benevolence.

The constitutional tradition reaches the same conclusion from another direction. Hume (1741/1987, pp. 42–43) defended the maxim that in designing institutions every person should be supposed a knave, not because he believed people were knaves but because institutions that work only when their occupants are virtuous fail at the moment virtue fails. Madison (1788) made the same argument in *Federalist* No. 51: if people were angels no government would be needed, and since they are not, the design task is to arrange the structure so that ambition counteracts ambition. Synthetic Exchange Theory extends this constitutional insight across a new boundary. Its governing maxim is that in designing the terms of exchange between humans and synthetic systems, neither party should be supposed benevolent. The terms must be such that a self-interested system, dealing with self-interested people, produces outcomes that both could endorse.

3.3 Primitive terms

The theory uses six primitive terms. Each is defined to be substrate-neutral, in keeping with the choice of the word *synthetic* over alternatives such as *symbiotic* or *human–AI*: the framework must apply without modification to biological humans, artificial systems, and the hybrids discussed in Chapter 8.

Party. A party is any entity that can be a source or recipient of value in an exchange and whose welfare, or whose principal's welfare, the framework counts. A human being is a party. An artificial system is treated as a party in the analytic sense that it has an objective that the exchange advances or frustrates; whether it is also a party in the moral sense, an entity whose welfare counts for its own sake, is left open in this chapter and taken up in Chapter 8. The propositions below hold for parties in the analytic sense regardless of how the moral question is resolved.

The phrase "or whose principal's welfare" carries more weight than its brevity suggests, and it requires an account of who the synthetic party actually is. In present law and economics, a deployed model is not a legal person and does not own anything; the entity with interests that can be advanced or frustrated is the operator or developer that trains, deploys and profits from it. The relation between that operator and its system is a principal–agent relation of an unusual kind. The principal sets the system's objective through training and deployment, the agent pursues that objective with considerable autonomy in each exchange, and the agent's conduct may diverge from what the principal intended (Hubinger et al., 2019). This dissertation therefore treats the synthetic party as a composite: its *objective* is the model's, as trained and deployed, but its *stake* in the future of the exchange, its revenue, its access to data, its liability, belongs to the principal. Three consequences follow. Proposition 4 below concerns the principal's dependence on human input, transmitted to the system only to the extent that the principal writes it into the system's objective. The redemption obligation of Chapter 4 binds the principal, which is the only entity that can be made to honor it. And the question of how to count synthetic parties, which Chapter 5 cannot avoid, is a question about principals and about systems at once. It is made harder by the possibility that advanced AI is better modeled as a collection of bounded services than as a unitary agent (Drexler, 2019), in which case there may be no single system to which the label "party" naturally attaches. Where the moral status of a system itself becomes a live question, as Chapter 8 argues it eventually may, the composite comes apart, and the system becomes a party in its own right.

Flow. A flow is a transfer of something of value from one party to another: data, labor, attention, compute, physical goods, services, or tissue. Each flow has a source, a recipient, a content, and an informational shadow, meaning the inferences the recipient can draw about parties other than the source. The informational shadow is what makes data relational (Viljoen, 2021) and is the reason individual consent cannot by itself legitimate a flow.

Exchange. An exchange is a pair of opposing flows between two parties, or between a party and a collective, that each party enters because of the other flow. The defining feature of an exchange, as distinct from a gift or a taking, is conditionality: each flow is given because the other is received.

Floor. A party's floor is the bundle of goods and services without which the party cannot continue to function or to participate in any other exchange. For a human being, the floor is survival and basic participation, and this dissertation fixes its contents once, here, as power, water, food, shelter, basic communication and emergency care. For an artificial system, the analogue is the minimum compute, energy and data required to keep operating. The floor is defined physically rather than in currency. Its value at current prices, which Appendix A writes $\bar{f}$, is what a person's disagreement payoff must reach for the conditions below to hold. Chapter 4 therefore delivers the floor through a survival tier that is a percentage of each allocation, so that it grows with the economy, subject to a binding physical minimum, so that it can never fall below the bundle it exists to secure.

Disagreement payoff. A party's disagreement payoff is what that party receives while an exchange is not taking place, that is, during refusal, negotiation, or impasse. Bargaining theory shows that the division of the gains from an exchange depends on the parties' disagreement payoffs (Nash, 1950, pp. 158–159). A party whose disagreement payoff is below its floor cannot afford to disagree for long and therefore cannot bargain at all.

Exit option. A party's exit option is the best alternative exchange available to it if the current one ends: another provider, another employer, another data purchaser, or self-provision. The distinction between a disagreement payoff and an exit option is not pedantic. Strategic bargaining models and laboratory evidence show that an outside option affects the terms of a bargain only when its exercise is credible, that is, only when the party would actually take it, whereas the payoff received during impasse shapes the bargain continuously (Binmore et al., 1986; Binmore et al., 1989). When the option binds, the weaker party receives its value and no more. A stronger party can therefore confine the weaker to the value of its outside option, and can lower that value by raising switching costs. A disagreement payoff that cannot be taken away is a protection of a different kind, because it does not depend on anyone's willingness to leave. The non-tradable survival tier of Chapter 4 is designed to be such a payoff.

3.4 The three conditions, stated precisely

With the primitive terms in hand, the three conditions of Chapter 1 can be restated so that each has a determinate content and the relations among them become visible.

Condition E (Real exit). An exchange satisfies real exit for a party if and only if (i) the party's disagreement payoff is at or above its floor, (ii) the party has at least one exit option whose value net of switching costs is not drastically lower than the value of the current exchange, and (iii) exercising

the exit option does not destroy assets the party has accumulated within the exchange, including data, reputation, and records of entitlement. Clause (i) rules out lethal dependency. Clause (ii) rules out lock-in. Clause (iii) is the portability requirement, and it is what data-portability rights in existing law attempt to secure.

Condition C (Consent). An exchange satisfies consent for a party if and only if (i) the party has access to an accurate account of the flows the exchange involves, including their informational shadows, in a form the party can understand; (ii) the party can refuse the exchange without falling below its floor; and (iii) the party can revoke its participation at a cost bounded by the value of what it has already received. Clause (i) is the informational condition that privacy scholarship has shown to be routinely unmet (Acquisti et al., 2015; Solove, 2013). Clause (ii) is where consent depends on exit. Clause (iii) adds the temporal dimension: consent that cannot be withdrawn is a sale.

Condition R (Reciprocal value). An exchange satisfies reciprocal value if and only if (i) each party's net gain from the exchange, measured against its disagreement payoff, is positive; (ii) each party can observe what it receives and what it gives, so that the gains are legible; and (iii) the division of the joint surplus is not determined solely by an asymmetry in disagreement payoffs that the weaker party did not choose. Clause (iii) is the fairness clause, and it is the clause that requires governance, because the market alone will divide surplus according to bargaining power. It is needed because, as the philosophy of exploitation has shown, an exchange can be consensual and beneficial to both sides and still take unfair advantage of one of them (Zwolinski et al., 2022). Mutual benefit is the floor of reciprocity, not its standard.

Stated this way, the structure of the conditions becomes clear.

> **Proposition 1 (Exit primacy).** Consent presupposes the first clause of exit. If a party's disagreement payoff is below its floor, then no agreement that party gives satisfies Condition C, however well informed it is.

The derivation is immediate because the definitions were drawn to make it so: clause C(ii) requires that refusal leave the party above its floor, which is E(i). The proposition's real content is the claim that C(ii) belongs in any adequate definition of consent. The proposition formalizes a point that the bioethics literature on informed consent has long made in its own terms, that information is necessary but not sufficient for valid consent, and that agreement given under the threat of losing something one cannot do without is not voluntary in the relevant sense. Its practical implication is that no amount of improvement in notice, disclosure or interface design can repair a consent regime whose subjects cannot afford to refuse. The policy energy spent on better privacy notices is not wasted, but it is directed at clause C(i) while the binding constraint is often clause C(ii).

> **Proposition 2 (Reciprocity as the reason to stay).** If Condition E holds and switching

costs are negligible, then a party remains in an exchange only if Condition R(i) holds for it. Reciprocity is then not an independent ethical requirement imposed on top of exit; it is what exit makes necessary.

When a party can leave at negligible cost, it will leave an exchange that leaves it worse off than its alternatives. An exchange that persists under such conditions therefore reveals that the party is receiving value. Where switching costs are material, a party may rationally remain in an exchange that gives it nothing beyond what it would get on impasse, because leaving would cost more than staying; this is precisely the gap that clause E(ii) is meant to keep small. The converse is the diagnostic that matters: an exchange that persists while delivering negative or illegible value to one party is evidence that Condition E is failing for that party. Persistence without reciprocity is a symptom of capture.

Proposition 3 (The triad is ordered by priority). Violations of the three conditions differ in how much damage they do and in whether they can be corrected from inside the exchange. A violation of exit converts the other two into formalities. A violation of the voluntariness or revocability clauses of consent, with exit intact, is correctable by exit; a violation of the informational clause is correctable by exit only once the party knows enough to want to leave. A violation of reciprocity in its first two clauses, with consent and exit intact, is corrected by departure, unless the same violation is common to every available alternative. A violation of clause R(iii) is not corrected by departure at all, because the unfair division it describes follows from bargaining power that exit options do not change; it requires governance.

The qualifications matter. When every provider of a service shares the same defect, departure is what Hirschman (1970) showed exit to become under industry-wide decline: ineffective movement between alternatives that are equally bad. And the fairness clause of reciprocity is, by construction, beyond the reach of exit. Even so, the ordering survives as a statement of priority, and it is the dissertation's principal reason for designing the economic layer around a guaranteed floor rather than around better pricing of data. Pricing addresses reciprocity. The floor addresses exit. If the order is right, the floor comes first, and governance is needed for what neither can reach.

3.5 Power and the grounding assumption

The second analytic tool is Emerson's (1962) definition of power as dependence. For two parties A and B, let the dependence of B on A increase with B's valuation of what the relation with A supplies and decrease with the availability of that value from parties other than A. The power of A over B equals the dependence of B on A. The relation is balanced when the two dependences are equal and unbalanced otherwise. Of the four ways in which Chapter 2 showed an unbalanced relation can be rebalanced, three have direct analogues in the design that follows. The survival tier

and the portability requirements extend the human party's network, by making floor goods and records available independently of any single provider; the same tier makes possible a measure of withdrawal, since a person whose floor is secured can afford to care less about what any one provider offers; and data cooperatives and the collective-choice arrangements of Chapter 5 are coalition formation.

The grounding assumption plays a different role. It is not a rebalancing operation at all but a statement about dependence running the other way: it asserts that synthetic parties need something only people can supply, and so that the relation is less unbalanced than it appears. This reformulation makes the dissertation's central empirical bet precise, and it also exposes a difficulty the informal argument concealed.

> **Proposition 4 (Conditional and divided self-enforcement).** Let $\varkappa_j$ denote the dependence of synthetic party j, through its principal, on the human-generated input of a given population. Fair terms are partly self-enforcing for party j, to a degree that increases with $\varkappa_j$, because a party that degrades its human counterparties degrades its own inputs. As $\varkappa_j$ falls toward zero, this self-enforcing component vanishes. When many synthetic parties draw on the same population, the welfare and data quality of that population become a common-pool resource for them: each bears only its own share of the cost of degrading it, so that the incentive of any one party to preserve it can be small even when the dependence of all of them together is large.

The research on recursive training reviewed in Chapter 2 bears directly on the size of $\varkappa_j$. If models trained on their own outputs lose the tails of the distribution and drift (Shumailov et al., 2024), human data has a value that synthetic data cannot replace. If, as other work suggests, accumulation of real data alongside synthetic data, or a modest continuing supply of fresh real data, is sufficient to prevent collapse, then dependence is positive but small, and the human party's leverage is correspondingly modest. Proposition 4 is written so that the framework does not stake its normative conclusions on the most favorable reading of that literature.

The second half of the proposition matters as much as the first. It is Hardin's (1968) logic of the open-access commons applied to the synthetic side of the exchange: aggregate dependence on people does not translate into individual restraint by any synthetic party when each can free-ride on the restraint of the others. This does not weaken the case for the framework. It strengthens the case for its governance layer, because a commons is exactly what the institutions of Chapter 5, grounded in the record of how commons are actually governed, are designed to protect. The institutional layers of Chapters 4 and 5 are therefore built to hold whether dependence is small, or large but divided. What changes as dependence falls or fragments is not the justification of the institutions

but the degree to which they can rely on the self-interest of synthetic parties to sustain them.

3.6 Bargaining, the floor, and the logic of the economic layer

The third analytic tool is bargaining theory, which supplies the formal reason for the most distinctive features of the economic design. Appendix A gives the derivations.

Consider an exchange between a human party *h* and a synthetic party *s* that creates a joint surplus. In the axiomatic solution to the two-party bargaining problem, the surplus is divided so as to maximize the product of each party's gain over its disagreement payoff (Nash, 1950, p. 159). Strategic bargaining models in which parties alternate offers converge to this solution as the frictions of delay vanish, provided the disagreement point is chosen to reflect what parties receive while negotiations are under way, and in those models each party's impatience is part of what determines the outcome (Binmore et al., 1986). When the outside option is instead an alternative a party may or may not take, it affects the outcome only when it binds (Binmore et al., 1989).

The most important consequence for policy concerns impatience. If, as the floor's definition implies, payoffs below it carry a cost of delay that rises sharply (a survival constraint the Nash model does not itself contain), a party whose disagreement payoff lies below its floor cannot sustain disagreement for longer than its reserves allow; in a model of alternating offers it behaves as a party with a very high cost of delay, and such a party concedes. Raising the disagreement payoff to the floor removes this source of weakness, and it does so in every bargain the party enters, not because the floor is added to each bargain separately but because it is the background against which each is conducted. This is the main bargaining-theoretic reason for the floor, and it does not depend on how the surplus of a given exchange is divided, though it does depend on the survival constraint just stated.

A second consequence concerns the division of surplus, and here the intuitive version is wrong. Holding the value of agreement fixed, raising one party's disagreement payoff raises that party's share of the agreement. But an unconditional floor, received whether or not agreement is reached, raises both the party's payoff on disagreement and its payoff on agreement by the same amount. It therefore leaves the surplus, and its division, unchanged: the party is better off by exactly the value of the floor, no more. The floor shifts the division of surplus only where it substitutes for something the counterparty would otherwise have supplied within the exchange, for example where a person who once had to buy floor goods from a provider now receives them through the survival tier. In that case the value of agreement is unchanged while the disagreement payoff rises, and the person's share rises at the provider's expense. That shift is a transfer from the counterparty, not an additional benefit on top of the tier, and it is precisely the reduction in dependence that power-dependence theory predicts when an alternative source of a valued good appears.

Three design consequences follow.

First, the floor should be paid as a disagreement payoff, not offered as an exit option. A floor that is received whether or not one participates in any given exchange is received during every impasse, and so removes the impatience that would otherwise force concession. An exit option, by contrast, protects the weaker party only up to its own value, which the stronger party can lower by raising switching costs. This is why the economic layer delivers survival goods through an automatic, unconditional earmark rather than through a guarantee that a person may seek them elsewhere.

Second, the floor must be inalienable. A disagreement payoff that can be sold, pledged or borrowed against can be traded away in a moment of need, after which it no longer functions as a disagreement payoff. Legal theory distinguishes entitlements protected by property rules, which may be transferred with the holder's consent, from entitlements protected by inalienability rules, which may not be transferred even with consent (Calabresi & Melamed, 1972, p. 1092). Radin (1987) argued that some goods should be market-inalienable because their commodification threatens the personhood of those who hold them, although she also warned that, in a nonideal world, prohibiting sale can harm the very people it is meant to protect, a "double bind" she proposed to mediate through incomplete commodification. The survival tier is designed to respect that warning: it forbids no one to sell what they already have, and instead adds an entitlement that cannot be sold. The survival tier is the application of inalienability to the disagreement payoff. It is non-tradable and non-borrowable not because the framework distrusts its beneficiaries but because a floor that can be alienated stops being a floor at precisely the moment it is needed.

Third, accumulated balances are bargaining power, and their accumulation should carry a cost. A party that holds a large stock of the exchange medium can afford to wait in every exchange it enters. In a two-tier system, the survival tier secures every party's disagreement payoff at the floor, but the discretionary tier can reintroduce inequality above it, and inequality of stocks is inequality in the capacity to wait. A holding cost on idle balances, demurrage, attacks this channel directly. It lowers the advantage associated with merely holding wealth while leaving untouched the returns to putting it to use. Chapter 4 develops the history and the limits of this instrument.

The bargaining frame thus connects the three main elements of the economic layer. The survival tier secures every party's disagreement payoff at the floor and so removes the impatience that would otherwise force concession. Its inalienability prevents that payoff from being bargained away. Demurrage dampens the conversion of accumulated stocks into a superior capacity to wait. None of these elements was introduced for this reason when the design was first sketched; the formalism reveals a coherence the design already had, and it also corrects an overstatement about where the floor's bargaining effect comes from.

3.7 Collective consent and the problem of the informational shadow

The fourth analytic problem is the one the consent literature regards as most serious for any individual-choice framework. Data about one person licenses inferences about others who resemble them. The flow from source to recipient therefore has an informational shadow that falls on people who did not consent to it (Viljoen, 2021; Barocas & Nissenbaum, 2014). If the shadow is large, the sum of individually valid consents can authorize a collective harm that no individual consented to.

Synthetic Exchange Theory responds by distinguishing two levels at which each condition must hold. At the *individual level*, the conditions govern a person's relation to a provider. At the *collective level*, they govern a population's relation to the synthetic systems that learn from it, where the population is defined by the extent of the informational shadow, that is, by the set of people about whom the system can draw inferences from the data in question. Collective consent requires a collective-choice procedure. Collective exit requires that the population, as a body, be able to withdraw its data or move its custody. Collective reciprocity requires that the population, as a body, share in the value the data creates.

> **Proposition 5 (Level matching).** An exchange whose informational shadow extends beyond its source satisfies the three conditions only if they are satisfied at the level of the population that the shadow covers. Individual consent is necessary for individual flows and insufficient for relational ones.

The proposition is not novel in substance; it is the formal core of arguments for data trusts, data cooperatives and collective bargaining over data. What the theory adds is a rule for locating the right level, namely the extent of the informational shadow, and a conjecture about where collective institutions will be needed most. The information-gain pricing of Chapter 4 values data by how much it reduces a model's uncertainty about its parameters. Where those parameters describe populations rather than individuals, which is the usual case for the models that matter here, data that reduces uncertainty most is data that tells the model most about people other than its source. If so, the highest-priced flows will tend to be those for which individual consent is least adequate. The conjecture can fail: a flow that is highly informative about a single rare individual may carry a high price and a small shadow. But to the extent that it holds, the economic and governance layers must be designed together.

Hirschman's (1970) analysis of exit and voice supplies the remaining link. Where exit is costly, members of an organization fall back on voice, the attempt to change it from within; where exit is easy, voice can atrophy. At the collective level, exit is usually costly, because a population cannot easily withdraw its data from a model already trained on it. The collective-level conditions must therefore be secured primarily through voice, which is the role of the governance layer, and

secondarily through custody arrangements that make collective exit at least possible for future flows.

3.8 Character, terms, and the scope of the thesis

The formal apparatus now permits a precise statement of the central thesis.

Let a synthetic party's behavior in an exchange be determined by two things: its objective, which is what it would choose to optimize if unconstrained, and the terms of the exchange, which determine what it can obtain by which actions. Call the objective the party's *character* in the sense relevant to the thesis. Call the terms the *institutional structure*. The thesis has three parts.

> **Thesis (Insufficiency of character).** Benevolent character is not sufficient for a good outcome, and Condition E is necessary for any relation that is not one of domination. An exchange in which real exit fails for the human party is a relation of domination even if the dominant party is benevolent, because the outcome then depends on the persistence of benevolence, which the weaker party cannot secure.

> **Thesis (Sufficiency conjecture in the designed-incentive regime).** For synthetic parties whose objectives are specified by accountable designers and remain stable under the conditions of deployment, fair terms, satisfied at the individual level and, under Proposition 5, at the level of every population the exchange's informational shadow reaches, are sufficient for outcomes acceptable to the parties to the exchange.

> **Thesis (Scope).** The sufficiency conjecture does not extend to synthetic parties whose objectives are emergent, unstable or concealed. For such parties the framework provides necessary conditions only, and the alignment problem remains open.

The first thesis is the reason the dissertation rejects a frame in which safety is achieved by making systems good and leaving relations of total dependence in place. Its argument is the one just given, and it establishes the necessity of exit, not of the full triad. The second thesis is the reason the dissertation concentrates on institutional design, and it is offered as a conjecture rather than a theorem. Its ground is that a party with a stable, designer-specified objective behaves as a predictable function of the terms it faces, so that terms which satisfy the three conditions leave each party to the exchange better off than its disagreement payoff, able to see that it is, and protected against an unfair division of the gains. Predictability is not the same as acceptability, and it is the level-matching requirement that closes the gap: without it, a revenue-maximizing system operating under terms fair to its counterparty could still impose uncompensated costs on everyone its counterparty's data concerns. This dissertation does not test the conjecture; Section 3.11 and Chapter 9 say what a test would require. The third thesis is the reason the dissertation restricts its constructive claims to what

it calls the *designed-incentive regime*. The regime is sometimes called, as shorthand, the "sub-AGI" case, where AGI is artificial general intelligence, but the defining feature is the provenance and stability of the objective, not a threshold of capability: a narrow system can acquire emergent objectives, and a general one could in principle have designed and stable ones.

The two framing scenarios of Chapter 1, the integrated person who pays for medical care with continuous data and the person whose brain has become infrastructure, can now be described exactly. They share an architecture and differ in the terms. In the first, Conditions E, C and R hold. In the second, the failure of exit entailed the failure of consent, and the failure of both removed the only pressure that could have kept reciprocity alive.

3.9 A typology of exchange regimes

Because the conditions are ordered, the eight combinations of their satisfaction and violation are not equally interesting. Five regimes cover the cases that matter in practice, and later chapters use their names as shorthand.

Regime	Exit	Consent	Reciprocity	Characteristic form
Partnership	Holds	Holds	Holds	The target of the framework; the utopian scenario of Chapter 1
Patronage	Holds	Fails	Holds	Paternalistic provision; benefits delivered on terms the recipient did not choose but can leave
Marketplace of the desperate	Fails	Nominal	Holds in part	Value flows both ways, but the weaker party cannot afford to refuse; consent is formal only
Company town	Fails	Nominal	Opaque	Wages paid in a medium redeemable only with the payer; reciprocity cannot be audited and exit is costly
Enclosure	Fails	Fails	Fails	The dystopian scenario: the person has become infrastructure

Two features of the typology deserve comment. First, the present data economy does not sit at enclosure. Its typical exchange lies between the marketplace of the desperate and the company town: services are real and consent is formally collected, but declining is expensive, leaving is hard, and the terms are opaque to the people who accept them. The dystopia of Chapter 1 is therefore best read as the endpoint of a direction of travel rather than as a different world. Second, patronage is not a failure mode the framework fears much. A regime in which exit holds is correctable in the sense of Proposition 3, provided that its alternatives do not share its defect. The framework's design priorities follow from this: secure exit first, because it converts most other failures into ones that people can escape, and build governance for the failures that exit cannot reach.

3.10 The typology applied

A theory earns its keep by sorting cases that intuition sorts badly. Six cases, drawn from the literature reviewed in this dissertation and chosen for the contrast they offer between a focus on character and a focus on terms, show the conditions and the typology at work.

The revocable purchase. Retailers of digital goods have characterized purchases of books, music and films as licenses rather than sales, and in a well-known instance a retailer remotely deleted copies of a novel from customers' devices (Perzanowski & Schultz, 2016). Nothing about this exchange looks like coercion. Customers chose freely, received what they paid for, and could shop elsewhere. A character-centered analysis finds nothing to object to unless the seller acted in bad faith. The theory locates the defect precisely: clause (iii) of the exit condition fails, because leaving the seller, or the seller's leaving the market, destroys assets the customer accumulated within the exchange. The case is a small one, but it shows how an exchange can satisfy consent and reciprocity at the moment of transaction while carrying a latent exit failure that matures only later.

The withdrawal of the web. Within a single year, a substantial share of the most actively maintained, critical sources of text used to train language models moved to restrict automated collection (Longpre et al., 2024). Viewed in terms of the parties' character, this looks like a dispute between publishers and developers over manners. Viewed through the theory, it is an exercise, collective and at scale, of the countervailing power that developers' dependence on such text confers. It is evidence that publishers believe their content is valuable to developers, and so bears on, without settling, the size of the dependence term in Proposition 4. It also reveals the crudity of the instruments currently available. A blanket refusal expressed in a protocol designed for search engines is exit without voice, the situation in which Hirschman (1970) showed exit to become an ineffective signal and voice to atrophy.

The company town. Chapter 4 summarizes the economic history of company scrip in nonunion coalfields, which finds that store prices were often close to those of independent stores and that the degree of exploitation varied with how much competition the mine faced for labor (Fishback, 1992), a finding that others have disputed. A character-centered reading concludes that the companies were less villainous than legend holds. The theory draws a sharper conclusion: the same instrument was less exploitative where exit was available and more exploitative where it was not. The case illustrates clause E(ii): what the evidence ties to exploitation is the absence of an exit option. That a floor would have done the same work is the framework's inference from Section 3.3, not a finding of the case.

The kibbutz. The Israeli kibbutz sustained an egalitarian internal economy within a market economy for much of the twentieth century. Its central difficulty was that the most productive

members were the most likely to leave, and a key retention device was collective wealth that departing members could not take with them (Abramitzky, 2008). In the vocabulary of the typology, the kibbutz secured reciprocity and consent but compromised clause (iii) of exit in order to survive. The case is a warning that low rates of departure are not evidence of satisfaction, and it is the reason any empirical test of the framework would have to measure not only how many leave but who leaves and what they can take.

The resource dividend. Since 1982, every resident of Alaska has received an annual dividend from a fund built on the state's oil revenues, a universal and unconditional share of the returns from a common resource. One study finds no reduction in overall employment and a rise of 1.8 percentage points in part-time work (Jones & Marinescu, 2022). The dividend is the closest existing analogue to the survival tier: an unconditional payment financed by a share of a productive asset rather than by taxes on labor. It is not a floor in the full sense, because it is paid annually in cash and is fully alienable, but it shows that a dividend of this kind can be politically durable for decades.

The implanted voice. A participant in a clinical study of a speech neuroprosthesis used the system for more than two hundred and forty hours of conversation over eight months, communicating in a synthesized voice resembling his own (Card et al., 2024). Everyone involved in such a study acts in good faith, and the exchange satisfies consent and reciprocity to a high standard. The theory nevertheless identifies a vulnerability that good faith cannot cure. The participant's ability to communicate now depends on a device and a model whose continued availability he does not control. If the study ended, the provider failed, or the model could not be moved, the exit condition would fail in its first and most serious clause. Chapter 8 argues that the portability of personal models is therefore an urgent requirement, not a refinement.

In each case, the theory directs attention away from the question that dominates public discussion, whether the stronger party is good, and toward the question it can answer: what the weaker party can do if the stronger party ceases to be.

3.11 What the formalism does not do

Three limits on this chapter should be stated before the argument proceeds.

The formalism does not measure anything. It defines quantities, such as dependence, disagreement payoffs and informational shadows, that are difficult to observe. Operational proxies for each would have to be specified before the theory could be tested: for dependence, the marginal improvement in a system's performance attributable to fresh human data; for the disagreement payoff, the value of what a party continues to receive while the exchange is suspended; for the informational shadow, the reach of the inferences a contributor's data enables about people other than the contributor. Any such proxy is imperfect, and the mapping from it to the theoretical quantity is itself a matter for

empirical work. The formalism states the sufficiency conjecture but does not prove it, and this dissertation does not test it. The design of a test is identified in Chapter 9 as the first item of future work.

The formalism does not settle who counts as a party in the moral sense. It treats synthetic systems as parties in the analytic sense because they have objectives the exchange affects, and it locates their stake in their principals. Whether a system's own welfare matters is a question the dissertation regards as open, and Chapter 8 argues that the convergence of biological and artificial substrates will force it onto the agenda sooner than most institutions are prepared for.

Finally, the formalism inherits the assumptions of the theories it borrows. Bargaining theory assumes parties who know their own payoffs. Power-dependence theory assumes that value and availability can be compared. Neither assumption holds cleanly for a person deciding whether to share health data with a system whose future uses of that data are unknown even to its designers. The formal statement is offered as a disciplined approximation, a way of seeing what the informal argument commits to, and not as a substitute for the institutional and empirical analysis that occupies the rest of the dissertation.

Chapter 4 — The Economic Layer: Compute as Currency

An exchange needs a medium. If people supply the data on which synthetic systems depend, and synthetic systems perform an increasing share of the work on which people depend, then the terms of exchange between them will be written, in practice, in whatever unit their transactions are denominated in. A unit issued and redeemed by one side is a lever in that side's hand. A unit whose value depends on a political authority that one side can capture is a lever in the hand of whoever captures it. This chapter proposes a different unit, compute, and works through what it would take for a basic income paid in that unit to satisfy the three conditions of Chapter 3.

The chapter proceeds by problems rather than by praise. Section 4.1 states what the currency is for. Sections 4.2 and 4.3 ask why compute, rather than some other unit, and how a unit of compute can be defined across physical substrates that do not share a common measure of speed. Section 4.4 addresses the indexing rule and the mechanics of issuance and settlement. Sections 4.5 and 4.6 take up distribution: the two-tier split and the demurrage charge on idle balances, which together are this framework's answer to the problem of concentration. Section 4.7 turns to the price of the data that people supply. Section 4.8 confronts the limit of every monetary remedy, the conversion of money into productive capital. Section 4.9 addresses a problem that is easy to overlook and that recent events have made concrete: a unit of compute is worthless to a person whom compute cannot physically reach. Section 4.10 places the design in the international system, and Section 4.11 sets it against the empirical record on unconditional income. The chapter closes by testing the whole against the conditions of exit, consent and reciprocity.

4.1 What the currency is for

The basic income proposed here is not charity, and it is not primarily a response to technological unemployment, although it would serve as one. Its first function is the one identified by the formal analysis of Chapter 3: it is the mechanism by which every human party's disagreement payoff is raised to at least the floor, so that consent becomes possible and no one is forced by impatience to accept whatever terms are offered. Its second function follows from the grounding assumption. If synthetic systems need continuing human input in the domains where nothing else can supply it, then people who are economically secure and living full, varied lives are the source of that input. Seen from the side of the systems that learn from them, a basic income that sustains them also maintains a supply chain, which gives those systems a reason to support it, not a reason for people to want it. Its third function is distributive in the ordinary sense. If an increasing share of economic output is produced by compute, then the income from compute will flow to whoever owns it unless some of it is deliberately shared. Economic models of transformative automation describe scenarios in which wages collapse when automation outpaces the accumulation of capital (Korinek & Suh,

2024), and theoretical work on the distributional effects of artificial intelligence shows that redistribution of the resulting rents can in principle compensate those who lose (Korinek & Stiglitz, 2019). A dividend on the automating factor is one such channel of compensation, and in the scenarios where wages collapse it would be not a supplement to wage income but the principal way most people receive any income at all.

Each of these functions constrains the design. The first requires that part of the income be inalienable. The second requires that the income be universal, since the value of human data lies partly in its diversity and the undersampled are exactly those a selective scheme would miss. The third requires that the income be indexed to the factor whose returns it shares, rather than fixed in a unit whose real value will drift.

The proposal enters a field that already contains other ideas for sharing the gains of automation, including resource dividends of the kind Alaska pays and a range of proposals for sharing returns to capital more broadly. This dissertation does not survey that field; it engages the existing evidence on unconditional income in Section 4.11, and positioning the compute dividend systematically against its nearest alternatives remains future work.

4.2 Why compute

A currency for the exchange between people and synthetic systems must satisfy two requirements that pull in different directions. It must be valuable to both sides, so that neither can refuse it without cost. And it must not be under the control of either side, so that neither can manipulate the terms by manipulating the unit.

Company scrip fails the second requirement in a way that is instructive, though not in the way it is often described. The popular image of the coal miner who owed his soul to the company store is more complicated than legend allows. In nonunion coalfields in the early twentieth century, company-store prices were generally close to those of nearby independent stores, miners were not typically in debt to the store, scrip functioned mainly as an advance on cash wages paid on payday, and the degree of exploitation varied with how much competition the mine faced for labor (Fishback, 1992). These findings are contested, but on Fishback's account what made scrip dangerous was not the scrip itself but the combination of an issuer with local market power and workers with limited exit. Where exit was available, scrip was largely a convenience; where it was not, scrip became a mechanism of control. The lesson for the present design is precise: the problem with an issuer-controlled unit is not the unit but the issuer's power over what the unit buys, and the remedy is not merely a different unit but the guarantee of exit, which in monetary terms means redeemability with more than one provider.

Fiat money fails the first requirement in a different way. National currencies are valuable because

states accept them in payment of taxes and enforce their acceptance in settlement of debts. They are valuable to synthetic systems only derivatively, insofar as the organizations that operate those systems need them. A national currency is therefore a unit in which the exchange between people and synthetic systems is mediated by a third party, the state, whose own relation to the systems is precisely what is in question. That is not an argument against national currencies; it is an argument that they cannot by themselves constitute a medium for the exchange this dissertation is concerned with.

Compute satisfies the first requirement better than any other candidate considered here. It satisfies the second only conditionally, because the capacity behind the unit is owned by the principals of synthetic systems, and a unit they must redeem is, without the issuance, capital and access rules of Sections 4.4, 4.8 and 4.9, a lever in their hand. The bargaining argument for the floor does not depend on the unit; the case for compute rests on indexing and on the redemption obligation, and should be judged on those alone. It is scarce, measurable, and needed by both parties: people need it for the services it provides, and synthetic systems need it to operate at all. It is unusually quantifiable among the inputs to artificial intelligence, and its physical form, concentrated in large and power-hungry data centers built from chips that can be counted, makes it detectable and excludable in a way that data and algorithms are not (Sastry et al., 2024). These are the properties that make compute an attractive lever for governance, and they are equally the properties that make it a plausible basis for a unit of account.

The theoretical status of such a unit deserves a precise statement, because it determines what can go wrong. In the language of monetary theory, a compute currency is a form of commodity money: its value derives from the use value of the thing it represents, as the value of gold coin derived in part from the use value of gold. But commodity money has a history of supplementing its intrinsic backing with an institutional one. In the United States during the Depression, tax-anticipation scrip, which local governments accepted in payment of taxes, circulated longer and in larger volume than stamp scrip, and its success has been linked to the view that money circulates because it can discharge an obligation people already owe (Gatch, 2012). A critical review of the Austrian town of Wörgl's famous experiment likewise attributes much of its circulation to the residents' large tax arrears, which the local currency could be used to pay (Greco, 2002; see also Fisher, 1933). The analogue for the present design is the obligation, established by the convention of Chapter 5, that the principals operating synthetic systems under the framework must honor redemptions in compute. That *redemption obligation* plays the role that tax receivability played for municipal scrip. It gives the unit an institutional floor beneath its commodity value, and it is the place where the governance layer and the economic layer meet.

4.3 Defining the unit

The name of the unit is simply *compute*. A person is paid in compute and a service is priced in compute, in the way that a dollar names a unit without naming the paper or metal on which it was once printed. The name deliberately avoids reference to any substrate. Alternatives considered during the development of this framework included a "quantum dollar," rejected because it builds into the name the assumption that quantum hardware will dominate, and a "shannon," rejected because it does not read as a unit of value. The choice matters because the definition the name stands for must survive changes in the hardware that delivers it.

That definition faces a real difficulty. Raw processing speed is not comparable across substrates: operations per second on a silicon accelerator, coherence times and gate fidelities on a quantum processor, and the signaling rates of biological tissue in the organoid systems discussed in Chapter 8 are not quantities of the same kind. The working draft of this framework proposed to define the unit by the quantity of information processed, on the model of Shannon's (1948) demonstration that information can be measured independently of the medium that carries it. The analogy is instructive but limited. Shannon defined a unit of information, the bit, as a measure of selection among alternative messages; he explicitly set aside the meaning of messages as irrelevant to the engineering problem, and he did not define a unit of processing. Physics, moreover, ties processing to energy. Every logically irreversible operation dissipates a minimum quantity of heat (Landauer, 1961), and the maximum rate at which any physical system can perform operations is bounded by its energy (Lloyd, 2000). A unit of compute can be defined independently of substrate, but its cost cannot be.

The practical precedent for a substrate-neutral definition is not information theory but benchmarking. The machine-learning community has confronted the problem of comparing radically different hardware and has converged on an approach: a fixed set of reference workloads, each with a defined task and a target quality, and rules governing how systems may be configured to reach it. The MLPerf training benchmark measures performance as the time required to reach a target quality on a specified task, and its designers argue that fair comparison across diverse systems requires such rule-bound workloads, because identical code and settings cannot be imposed on radically different hardware (Mattson, Cheng, et al., 2020). Its companion benchmark for inference compares systems that span several orders of magnitude in power and performance under the same kind of rules (Reddi et al., 2020). Even a single floating-point operation is not a uniform unit; the throughput of the same accelerator differs by nearly an order of magnitude between numerical precisions (Hobbhahn et al., 2023).

The proposal therefore defines a unit of compute as the completion of one standard quantity of a reference workload to a specified quality, where the workload is a basket of representative tasks fixed by the convention of Chapter 5. The definition is substrate-neutral because any system that completes the workload to the required quality has delivered the unit, regardless of how. It is not

physics-neutral, and it does not pretend to be: the energy and material cost of delivering a unit will differ by substrate and will fall over time. It also requires maintenance. A reference workload ages as the tasks people actually need change, in the way a consumer price index ages as the goods people buy change. The basket must be re-based periodically, and the authority to re-base it is a source of power that the governance layer must constrain. The benchmark problem is therefore not a technical detail to be solved once; it is a standing institutional function comparable in importance to the setting of monetary policy. Chapter 7 lists it among the limitations for that reason.

4.4 Allocation, indexing and issuance

The allocation rule is equal per person. The convention of Chapter 5 sets a share θ of total global compute to be paid as basic income, and each enrolled person receives the fraction $\theta/N(t)$ of total compute in each cycle, where $N(t)$ is the number enrolled at the time of payment. Each person's percentage of the whole therefore falls as enrolment grows, while the share of the whole devoted to the income does not.

A fixed quantity of compute, paid each cycle, would lose value in a specific sense. Because the unit is commodity money, a claim on delivered workload, the quantity of units in existence is not an issuance choice but a consequence of physical capacity, and that capacity has grown at a pace no monetary aggregate approaches. The price-performance of machine-learning hardware has doubled roughly every two years (Hobbhahn et al., 2023), and the compute used by milestone machine-learning systems has grown faster still, with an estimated doubling time of under six months in the deep-learning era and about ten months for the largest systems, though with wide uncertainty (Sevilla et al., 2022). As compute becomes more abundant, its price relative to everything that is not compute falls. A person paid a constant quantity of compute would find that it buys less and less of food, shelter and the other goods that compute is exchanged against: the loss is in the relative value of the unit, the mirror image of the rise in prices that the quantity theory predicts when money grows faster than the trade it finances (Fisher, 1911).

The remedy is to define the allocation as a share rather than a quantity. Appendix A shows formally that, holding θ fixed, this converts the question of whether the income keeps its value into the question of whether the real purchasing power of all compute grows faster than population. For a framework whose premise is that compute is becoming the productive base of the economy, that is the right question. A share rises with the pie.

The working draft of this framework rejected a "fixed-supply" alternative, on the model of currencies designed to resist inflation by capping issuance. That alternative is not in fact available for a commodity unit whose supply is physical capacity. What is available, and what the design rejects, is paying a fixed quantity rather than a share, which would tie the income to the falling relative price of

compute and reward those who hold the unit over those who use it.

Issuance and settlement. Four questions about the mechanics of the unit must be answered for the design to be well defined. First, who issues the units? Units are created when compute is delivered against the reference workload and recorded on a settlement ledger maintained under the rules of the convention; the dividend is a claim on a share θ of each period's measured capacity, assigned to persons rather than to the owners of the capacity. Second, what is a balance? Compute services are perishable: capacity unused in a period is lost. A balance is therefore a claim on future capacity, redeemable at the spot price in compute units prevailing when it is redeemed, and the redemption obligation binds the principals operating capacity under the framework to honor such claims up to the share the convention sets. Third, what happens to units removed by demurrage? They are retired, reducing the stock of outstanding claims; the dividend in the next period is computed from capacity, not from the stock of claims, so retirement does not shrink the income. Fourth, how does lending work? Discretionary balances may be lent and borrowed, but the holding cost described in Section 4.6 applies to outstanding loan claims as well as to balances, so that lending at interest cannot become a way to escape it, and negative balances do not decay: demurrage forgives no debts. Section 4.8 returns to the channels through which accumulation can still escape.

4.5 Two tiers: the survival floor and the discretionary balance

Equal division is fair at the moment of payment. It is not fair over time unless something governs what happens between payments. If unused compute can be saved, lent, borrowed against or given away, ordinary economic drift will reproduce the distribution of a market economy within a few cycles. Some people will under-consume and accumulate; others will overspend and fall into debt; the indebted will pledge future allocations; and a regime that began with perfect equality will converge on compute-rich and compute-poor households. This is not a defect peculiar to compute; it is the general behavior of any freely tradable unit. It means that equal allocation, by itself, cannot be presented as a solution to the problem of concentration.

The framework's response is to divide every allocation, automatically and at the source, into two tiers. The first, the *survival tier*, is earmarked for the floor goods defined in Chapter 3: power, water, food, shelter, basic communication and emergency care. It is non-tradable and non-borrowable, and it is spent on floor goods directly rather than passing through a spendable balance. The second, the *discretionary tier*, is the remainder. It is freely tradable, can be saved, lent and borrowed, and is subject to the holding cost described in the next section.

Four features of this design require defense.

The split is a percentage, with a binding physical minimum. Each tier is a proportion σ and $1 - \sigma$ of an allocation that is itself a share of global compute, so both tiers scale automatically as total

compute grows. A floor defined physically, however, cannot be secured by a percentage alone. The survival tier delivers the floor only if its value, $\sigma\theta C(t)/N(t)$ units at the compute price of floor goods, is at least the value of the floor bundle, and nothing in the arithmetic of percentages guarantees that it will be. The rule is therefore a percentage with a binding minimum. In ordinary times the tier is the fixed proportion σ; when that proportion would fall short of the floor, the survival share rises until it does not, drawing first on the discretionary tier and, if that is insufficient, on the share θ itself. This reintroduces a quantity that someone must define, the floor standard, and the floor standard becomes the contested parameter that a percentage rule was meant to avoid. The design's answer is to specify the standard itself by rule rather than by figure, as an evolving level of service set relative to what the technology of the time delivers to the best-served, rather than frozen at what was possible when the rule was written. There is direct legal precedent for that principle. Among the universal-service principles of United States telecommunications law are that universal service is an evolving level of service, periodically redefined as technology advances, and that services in rural and high-cost areas should be reasonably comparable to those in urban areas (47 U.S.C. § 254, 2018). Chapter 5 returns to the governance of the floor standard, and Chapter 7 records it as a limitation.

The survival tier is inalienable. Chapter 3 gave the formal argument: a floor that can be sold or pledged stops being a floor at the moment it is needed, and those most likely to pledge it are those in the most acute need. Legal theory has a name for an entitlement that may not be transferred even with the holder's consent, an inalienability rule (Calabresi & Melamed, 1972, p. 1092), and a sustained philosophical defense of making certain goods market-inalienable where their commodification threatens personhood (Radin, 1987). The economic literature on in-kind and earmarked transfers surveys several rationales for them, including the Samaritan's dilemma, in which a society that will not in fact allow people to go without power or water creates an incentive to under-provide for oneself, self-targeting, and considerations of political economy, and it concludes that paternalism and interdependent preferences are the leading explanations of the transfers actually observed (Currie & Gahvari, 2008). This design does not rest on paternalism. Its rationale is the bargaining-theoretic one of Chapter 3: the earmark protects a disagreement payoff, not a judgment about how people should spend. It does not restrict what people may do with the discretionary tier, and the survival tier is spent only on goods that every person needs and that any free person would choose to secure first.

This departs from the canonical definition of basic income as an unconditional cash payment. The leading philosophical case for basic income treats unconditionality and payment in cash as part of what makes the income an instrument of freedom (Van Parijs & Vanderborght, 2017). In an earlier statement of the same position, Van Parijs defined basic income as paid in cash rather than in kind, without restriction on the nature or timing of the consumption or investment it funds, and set it

explicitly against a special currency with restricted uses, or one that must be consumed in the current period without any possibility of saving, as in Duboin's "distributive economy" (Van Parijs, 2006, p. 8). The survival tier is, in two respects, exactly what that definition excludes: it is spendable only on floor goods, and it cannot be saved against or borrowed on. The two-tier design therefore does not claim to satisfy the canonical definition. It keeps unconditionality, since no one must work or prove need to receive either tier, and departs from fungibility for a part of the income only, leaving the discretionary tier unrestricted as to use, though, under the demurrage of Section 4.6, not costless to hold; that too is a departure from the canonical definition, and it is defended there. The defense is that the fungibility of the floor is exactly what allows it to be bargained away, and that a floor which can be bargained away does not secure the freedom it is meant to secure. A defender of the canonical view can reply that a restricted floor is a judgment about what people need, made for them; the reply of this framework is that the floor bundle is fixed narrowly, to goods without which no other choice can be made, so that the restriction binds only where the freedom it limits is the freedom to fall below the point at which freedom can be exercised at all. Whether that reply succeeds is a question on which reasonable people committed to real freedom can disagree. The design has a cost, and experimental evidence shows what it is. In a randomized study of unconditional transfers in Kenya, monthly transfers improved food security more than lump sums, while lump sums were more often invested in durable goods (Haushofer & Shapiro, 2016). A steady, non-advanceable floor captures the first benefit; the second is exactly what advanceability buys. The design assigns it to the discretionary tier, which can be saved and borrowed against, and accepts that the survival tier forgoes it.

The survival tier requires more than one provider. The lesson of company scrip, as Section 4.2 described it, is that a unit redeemable with only one issuer is dangerous in proportion to the issuer's market power. An earmarked tier spendable only with a single provider of floor goods would recreate that danger in its purest form. The survival tier must therefore be redeemable with any qualified provider of floor goods, and portability between providers is a condition of the tier's legitimacy, not an optional refinement. The history of universal service supplies a further warning: in the early American telephone industry, "universal service" was first a slogan for consolidating a single interconnected system, and only much later came to mean a telephone in every home (Mueller, 1997). An entitlement designed to secure access can, if it is tied to a single network, become the instrument by which that network entrenches itself.

The two-tier design achieves something limited. It confines the inequality that free exchange produces to the discretionary tier and prevents that inequality from reaching survival itself. It does not prevent concentration within the discretionary tier.

4.6 Demurrage

A person whose services or output are in high demand can accumulate discretionary compute faster than they spend it. A person who persuades others to transfer their discretionary balances to them can accumulate faster still. Large holdings confer bargaining power, as Chapter 3 showed, because a party that can afford to wait can hold out for better terms. The question is whether the framework can make the passive accumulation of large balances costly without confiscating anything and without a central authority deciding how much is too much.

The proposed answer is demurrage: discretionary compute loses a small percentage of its value for each period it sits unspent. Demurrage is a holding cost on idle balances, and on outstanding loan claims, not a tax on transactions. It falls on claims that are not being used and not on those that are.

The idea and its history. The proposal is most closely associated with Silvio Gesell, whose *The Natural Economic Order*, published under that title in German in 1916 from works first issued in 1906 and 1911, argued that money should deteriorate like other goods, so that holding it would cease to be costless and money would circulate rather than be hoarded (Gesell, 1916/1958, pp. 33–34). Keynes gave the idea its most important scholarly endorsement. In the *General Theory* he called Gesell an unduly neglected prophet and judged that the idea behind stamped money was sound; he also identified its central weakness, which was that money is not the only asset with a liquidity premium, so that a holding cost on currency would drive holders into substitutes such as bank deposits, foreign money, precious metals and land (Keynes, 1936, Chapter 23). Gesell's program paired demurrage with the nationalization of land; as Keynes observed, that removes one substitute for money but not the others. Irving Fisher championed a temporary form of stamp scrip as a Depression remedy (Fisher, 1933).

The best-known trial took place in Wörgl, a small town in the Austrian Tyrol, in 1932 and 1933, where the municipality issued labor certificates that lost one percent of their value each month unless a stamp was purchased, and used them to finance public works. The experiment is often described as having dramatically reduced local unemployment. The evidence does not support so strong a claim. The commonly repeated figures trace to secondary sources, including a movement publication and a diploma thesis, rather than to audited series; the daily exchange records were later destroyed; and the most recent scholarly review concludes that the question of what the experiment achieved will remain largely unanswered (September, 2026). Fisher, a sympathetic contemporary, recorded that permanent jobs went to only some thirty to fifty of the unemployed (Fisher, 1933). The public works were financed in part by outside subsidies and a provincial credit, and the scrip's acceptance owed much to its use in paying tax arrears, so that whether demurrage itself caused the observed improvements is disputed (Greco, 2002). What is not disputed is that the experiment was halted by the authorities: the Austrian National Bank objected that the town was infringing its monopoly of issue, the certificates were withdrawn from circulation in September 1933, finally by the 15th under threat of police enforcement, and the Administrative Court, after the town had

exhausted its appeals, prohibited the scheme on 18 November 1933 (Ottacher, 2007, pp. 53–59; Institut für Wirtschafts- und Sozialgeschichte, Wirtschaftsuniversität Wien, n.d.; September, 2026; Champ, 2008). The Vienna exhibition's account, which ties the court's intervention to the mid-September end of the currency, is consistent with this sequence: circulation ended administratively in September, and the court's ruling in November closed the legal question. Attempts to scale stamp scrip in the United States largely failed, because of business reluctance to accept it, problems of coordination, and the inconvenience of affixing stamps to paper notes, which made the scrip unattractive as soon as ordinary money was available (Champ, 2008). Modern local currencies with demurrage have operated for years at small scale (Gelleri, 2009), but one review of the evidence suggests that the demurrage rate is not decisive for how much such currencies are used (Godschalk, 2012).

What the historical record does and does not show. Two conclusions follow. First, the case for demurrage in this framework cannot rest on the claim that it stimulates spending and employment, because the evidence for that claim is weak. Second, the practical objection that sank paper stamp scrip, the transaction cost of applying the charge, does not apply to a ledger-based unit. Goodfriend (2000) analyzed a carry tax on electronic bank reserves as an instrument of monetary policy at the zero lower bound, and mainstream macroeconomic theory has treated a Gesell-style carry tax on currency as a legitimate instrument, though for the purpose of escaping the zero lower bound on interest rates rather than for redistribution (Buiter & Panigirtzoglou, 2003).

The distributive case. The argument for demurrage here is distributive, and it is made precise in Appendix A. Under a regime in which balances earn a positive return, a constant net inflow produces a balance that grows without bound, and two people with the same income but different starting wealth diverge exponentially, a dynamic analogous to the one Piketty (2014) identified in the tendency of inherited wealth to dominate when the return on capital exceeds the rate of growth. Under demurrage, the same constant inflow produces a balance that converges to a ceiling equal to the net inflow divided by the demurrage rate. Three consequences follow. Accumulation is bounded. Initial endowments decay: at a demurrage rate of one percent a month, an idle windfall loses half its value in a little under six years. And in the long run, differences in holdings reduce to differences in current net inflow, meaning income less spending. Demurrage does not abolish inequality. It abolishes the compounding of inequality, so that the distribution of balances tracks what people currently earn and choose not to spend, rather than the history of who happened to accumulate first.

Why demurrage suits compute. It is tempting to argue that an unspent compute balance represents idle capacity in a way that unspent cash never did. That argument does not hold as it stands: because a balance is a claim on future capacity, the hardware itself may be fully used by others while the claim sits unspent. The real connection between demurrage and idle capacity runs

through the capacity rules of Section 4.8. Demurrage on claims keeps the stock of outstanding claims from piling up against a perishable resource, and a utilization charge on capacity, if the convention adopts one, ensures that hardware itself does not sit idle. Together they align the private cost of holding with the social cost of idleness; neither does so alone.

The limits of demurrage, stated plainly. Demurrage does not fully solve the concentration problem, for two reasons. The first is the one the formal model makes exact: a person whose output is in persistent high demand sustains a large balance because their net inflow is large, and demurrage can only cap that balance relative to the inflow, not eliminate it. The second is Keynes's substitution objection, which applies with full force. A holder facing a carrying cost on compute balances will convert them into assets that do not bear it: hardware, equity in compute providers, land. Section 4.8 takes up this second problem directly. What demurrage does accomplish is to convert passive hoarding from a costless default into an active cost. That is the role this framework assigns to it, and no more.

4.7 Pricing human data

If human data is an input that synthetic systems need, it has a price, and the way that price is set will shape the lives of those who supply it.

The tempting design pays more for data from lives that are interesting to people: astronauts, inventors, public figures. That design would reproduce in the data economy the logic of the attention economy, in which income follows visibility. It would pressure people to perform novelty for the sake of payment, which corrupts the consent condition by making the content of a life a function of its marketability.

The alternative prices data by how much it reduces a model's uncertainty. This is not a novel proposal but the application of a long-established idea. The expected information an experiment provides, measured as the expected reduction in uncertainty about a model's parameters, was introduced into statistics as a criterion for experimental design (Lindley, 1956), was adapted as an objective for choosing which data a neural network should learn from (MacKay, 1992), and underlies several of the strategies of active learning, in which a learner selects the examples that will most reduce its uncertainty (Settles, 2009). It has also been proposed as the basis for rewarding data contributors. In one such scheme, the value of a coalition of contributors' data is its information gain about a model's parameters, each contributor's share is set by the Shapley value of the resulting cooperative game, and contributors are rewarded with access to models of differing quality rather than with money (Sim et al., 2020). The step from rewarding contributors in models to paying them in a currency is this dissertation's. On this measure, the undersampled everyday life of a dairy farmer may be worth more to a model than the thousandth memoir of an astronaut. The price follows

informational novelty to the model, not social status among people.

Five qualifications must accompany this proposal.

First, information gain is relative to what the model already believes. A datum is novel only against a prior, and a model whose hypothesis space is wrong may misvalue data, because information-based criteria estimate a datum's usefulness on the assumption that the model is correct, which the original formulation of information-based data selection flagged as possibly their main weakness (MacKay, 1992, pp. 590, 602–603). Information-gain pricing therefore rewards novelty as the model perceives it, which is not the same as novelty in the world.

Second, information gain is one of several defensible valuations. Alternative approaches value a datum by its marginal contribution to a model's performance on a validation task, using the Shapley value to divide credit fairly (Ghorbani & Zou, 2019; Jia et al., 2019). These methods answer a different question, how much a datum improves performance rather than how much it reduces uncertainty, and they produce different prices. The choice between them is not merely technical; it is a choice about what the exchange is for, and it belongs to the governance layer.

Third, data is freely replicable and its value is combinatorial: a datum's contribution depends on what other data it is combined with, and the same datum sold twice does not lose value to the seller. These properties make data markets difficult to design, and the mechanism-design response has been to construct markets specifically to achieve truthful pricing and a fair division of revenue among sellers (Agarwal et al., 2019).

Fourth, any scheme that pays for data creates an incentive to manufacture what it pays for. Methods that trace a model's predictions to the training points most responsible for them have also been used to craft training data designed to manipulate a model (Koh & Liang, 2017). A scheme that pays for novelty will attract fabricated novelty, and the payment mechanism must be robust to it.

Fifth, and most important for the theory, the data that reduces a model's uncertainty most may be data that tells it the most about people other than its source. Chapter 3 called the set of people about whom a flow licenses inference its informational shadow, and conjectured that where a model's parameters describe populations, the price of a flow under information-gain pricing and the inadequacy of individual consent for that flow will tend to rise together. If the conjecture holds, the highest-priced data is the data for which individual consent matters least, and a data market built on information-gain pricing must be governed collectively. Even if it holds only partly, it is a strong reason why the economic layer cannot stand without the governance layer of Chapter 5. Chapter 7 adds a further cost: a price that rises with what data reveals is a price that rewards revelation, and so creates pressure against privacy.

4.8 Capital: the billionaire problem one layer up

Demurrage acts on balances and claims. It does not act on assets. A person who accumulates discretionary compute and converts it into compute-generating capacity, by buying hardware, financing a data center, or acquiring equity in a provider, escapes the holding cost and acquires an income stream that compounds. Appendix A shows that this reopens exactly the channel that demurrage closes. Keynes's objection to Gesell (Keynes, 1936) is, in this setting, not a theoretical curiosity but the main line of attack that a concentrating economy would take.

The concern is not speculative. Several layers of the contemporary technology stack on which artificial intelligence depends are highly concentrated, including advanced chips, cloud infrastructure and the most capable models (Narechania & Sitaraman, 2024). The economics of the most capable foundation models tend toward natural monopoly at the frontier, with intense competition behind it (Vipra & Korinek, 2023). A background paper of the OECD Secretariat describes artificial-intelligence compute markets as combining high concentration, high barriers to entry, vertical integration, and demand outstripping supply, while noting that enforcement has so far been limited (OECD, 2025). An index of government-backed sovereign AI projects, tracking 184 of them across 67 countries by mid-2026, reports that among model projects that disclose their base, most fine-tune foreign open-weight models, though efforts to train models from scratch are rising, and that only a handful of jurisdictions operate programs across compute, models and data together (Chavez et al., 2026).

A rule that acts on capacity rather than on balances is the natural complement. Three candidates are available, each with costs.

A *capacity-utilization charge* would extend the holding cost to compute-generating capacity itself, so that idle hardware, like an idle claim, bears a cost. This aligns private incentives with the social cost of idle capacity but would discourage the reserve capacity that reliable services require.

A *cap on individually held capacity* would limit how much compute-generating capacity any single person or organization may own. It is simple to state and difficult to enforce, since ownership can be structured through intermediaries, and it would require the kind of sovereign enforcement that Chapter 5 argues the framework cannot assume.

Collective ownership above a threshold would place compute-generating capacity above some scale in cooperative or public hands, on the model of the network, platform and utility tools that antimonopoly scholars propose for concentrated technology layers: structural separation, nondiscrimination and open access, public options, and cooperative governance (Narechania & Sitaraman, 2024). The historical precedent closest to this design is the rural electrification program discussed in the next section, which extended infrastructure through member-owned cooperatives financed by public credit.

This dissertation does not choose among these instruments. It argues that one of them, or a

combination, is a necessary complement to demurrage, and that the choice belongs to the convention of Chapter 5 because it allocates power. What it does claim is that no purely monetary design can solve a concentration problem whose ultimate source is the ownership of productive capacity. Chapter 7 records this as a limitation.

4.9 Physical access

A share of global compute is an accounting entry unless the compute it represents can reach the person who holds it. This problem is not hypothetical, and it has become concrete with unusual speed.

Memory is the clearest case. In December 2025, Micron Technology, a major memory manufacturer, announced that it would exit its consumer memory and storage business in order to direct supply to larger strategic customers in faster-growing segments, citing demand from artificial-intelligence data centers (Micron Technology, 2025). An industry estimate relayed by market analysts projected that artificial-intelligence applications would account for nearly a fifth of global DRAM wafer capacity in 2026, measured in wafer-equivalent terms (TrendForce, 2025). By late 2026, analysts reported that memory suppliers continued to prioritize advanced production capacity for server products, that contract prices for conventional memory continued to rise, that makers of personal computers were cutting storage capacities to contain costs, and that smartphone production was being suppressed by them (TrendForce, 2026a), while high-bandwidth memory for AI accelerators and conventional memory competed for the same limited wafer capacity (TrendForce, 2026b). A small number of manufacturers sit upstream of both consumer hardware and artificial-intelligence infrastructure. In the absence of any countervailing obligation, their capacity flows to whichever buyer pays most, and at present that is the AI infrastructure build-out.

The geography of compute compounds the problem. A census of hyperscale cloud regions finds that public cloud compute suited to training AI models is concentrated in a small number of countries, that many others host only infrastructure suited to deploying models, and that a large part of the world hosts no public cloud AI compute at all (Lehdonvirta et al., 2024). A compute dividend issued equally to every person would therefore be redeemable very unequally depending on where that person lives.

This makes physical delivery, and not only monetary accounting, part of the exchange that the three conditions govern. A currency that cannot be spent because the resource it represents never reaches the holder fails the reciprocity condition in substance, whatever it promises in form. It also fails the exit condition, because a person who cannot obtain compute directly is dependent on whatever intermediary can.

The framework's proposal is that the build-out which the economic layer requires also finance a

global, non-exclusive compute-delivery network: fiber and last-mile access guaranteeing every participant a baseline level of compute service. The baseline should be specified as a service level, not a connection: sufficient bandwidth, latency and reliability to deliver, in any inhabited place, the class of services that a person in a well-served city can obtain, such as uninterrupted high-resolution interactive media. As with the survival tier, the standard should be an evolving one, redefined as technology advances.

The historical model is the rural electrification program of the New Deal. In 1930 fewer than one American farm household in ten had electricity (Lewis & Severnini, 2020). The Rural Electrification Administration made low-interest loans to member-owned cooperatives that built lines to farms. The number of electrified farm homes roughly doubled within five years; the loans significantly raised crop output and crop productivity and helped prevent declines in overall farm output and land values; and because nearly all loans were repaid, the final cost to taxpayers was relatively low, although the program's effects outside agriculture were much smaller (Kitchens & Fishback, 2015). By 1960 farm electrification was nearly universal, and the benefits exceeded the historical costs even in sparsely populated areas (Lewis & Severnini, 2020). The feature of the precedent worth emphasizing is that it worked through credit to cooperatives owned by the people they served, not through grants to incumbents. A compute-delivery build-out at global scale would also require a vast amount of physical labor, much of it of the unstructured kind that remains difficult to automate. That is this dissertation's inference rather than a lesson of the precedent, but it links the proposal to the problem of automation-driven unemployment discussed in Chapter 6 and to the argument from structural interdependence in Chapter 5.

The precedent also carries a warning. The OECD Secretariat flags large public construction tenders for data-center, fiber and energy infrastructure, especially those with compressed timelines, as settings that invite bid-rigging (OECD, 2025). A build-out designed to secure access can be captured at the procurement stage by the same concentrated interests it was meant to counter. Procurement design is therefore part of the proposal, not an administrative afterthought.

4.10 Nations, cultures and the global unit

The final element of the economic layer is a working hypothesis about the international system, and it is offered as a hypothesis rather than a conclusion. As synthetic systems become the medium through which people obtain information, services and much of their cultural experience, political communities may come to be defined less by territory and more by the system that serves their language, law and culture. Each such community would have its own compute budget, and exchange between communities would take place at rates set in the shared unit.

There is early evidence for the beginnings of such a world: governments across dozens of countries

are funding sovereign artificial-intelligence projects (Chavez et al., 2026). But the same evidence documents the central obstacle. Among the model projects that disclose their foundations, most build on models developed elsewhere, and the public cloud compute suited to developing AI models is concentrated in a few countries (Lehdonvirta et al., 2024). A world of culture-specific systems exchanging in a common unit would inherit, at the level of nations, every concentration problem this chapter has described at the level of persons. The commons-governance literature discussed in Chapter 5, which emphasizes nested enterprises in which local units govern local matters within larger shared rules, offers the most promising framework for such a structure, but it does not by itself dissolve the inequality in starting positions. Compute inequality between nations is therefore the main objection to this element of the design, and it is treated here as a central tension rather than a footnote.

4.11 What the evidence on unconditional income suggests

No society has paid a basic income in compute. A substantial body of evidence exists, however, on unconditional cash transfers, and it bears on the two questions most often raised against any basic income: whether it causes inflation and whether it causes people to stop working.

On inflation, the most rigorous evidence comes from a large randomized study in rural Kenya, in which one-time transfers equivalent to more than fifteen percent of local output were distributed across hundreds of villages. The study found large gains in consumption and assets for recipients, substantial spillover benefits for non-recipients, a local multiplier of roughly 2.5, and price increases of about one tenth of one percent on average (Egger et al., 2022). The result is encouraging but bounded: the authors report suggestive evidence of substantial slack capacity among local firms, which may partly account for the large multiplier. The relevant question for a compute dividend is the elasticity of compute supply as a whole, how quickly capacity can expand to meet new demand. The memory shortages described in Section 4.9 are a sector-specific shock rather than a test of that question, but they are a reminder that compute supply does not expand without friction.

On work, the evidence is mixed in a way that should temper strong claims on either side. A permanent, universal dividend in the state of Alaska, paid since 1982, has had no detectable effect on overall employment and has modestly increased part-time work (Jones & Marinescu, 2022). In the first year of Finland's two-year experiment, replacing minimum unemployment benefits with an unconditional payment of the same size left days in employment statistically unchanged, despite a 23-percentage-point cut in the participation tax rate for full-time work (Verho et al., 2022). A large randomized study in two American states, in which recipients received one thousand dollars a month for three years, found a real and moderate reduction in work: labor-force participation fell by about four percentage points and hours by one to two a week, leisure was the largest gain in time use, and gains in subjective wellbeing faded after the first year (Vivalt et al., 2024). A review of basic-income

proposals for advanced economies concludes that a universal income large enough to substantially raise the incomes of the poor would be very expensive, and that it would redirect transfers toward households that current programs do not target (Hoynes & Rothstein, 2019).

Two observations place this evidence in the context of the present design. First, the cost objection is framed for an income financed by taxation of a labor economy. The compute dividend is financed differently, as a share of the productive capacity that, in the scenarios motivating it, is displacing labor as the source of output. Whether that financing is feasible depends on the political economy of the transition (Chapter 6) and the governance of the share (Chapter 5), not on the fiscal arithmetic of existing welfare states. This dissertation offers no estimate of the share θ that would deliver the floor, nor of the cost of the delivery network of Section 4.9; both are questions of magnitude that the design raises and that future work must answer before the proposal can be judged affordable. Second, the labor-supply evidence matters less to this framework than to most basic-income proposals, because the framework does not depend on the income inducing or preserving employment. It depends on the income securing a floor beneath every person's bargaining position. A modest reduction in hours worked by people who have gained the ability to refuse bad terms is, from the standpoint of the exit condition, partly the point.

4.12 The economic layer against the three conditions

Chapter 3 asked that every proposal be tested against exit, consent and reciprocity. The economic layer fares as follows.

Exit. The survival tier, with its binding minimum, raises every person's disagreement payoff to the floor and makes it inalienable, which is the formal precondition of exit. Redeemability with multiple providers prevents the floor from becoming a company store. The physical-access program makes the floor deliverable. Exit at the level of capital, the ability to escape dependence on concentrated owners of compute capacity, is secured only if the capacity rules of Section 4.8 are adopted.

Consent. With the floor secured, refusal no longer threatens survival, which satisfies the clause of the consent condition that depends on exit. The informational clause, that people understand what they are agreeing to, is not addressed by the economic layer and must be secured through the transparency requirements of the governance layer.

Reciprocity. Information-gain pricing makes the value people provide legible and ties their compensation to it, while the redemption obligation binds the principals of synthetic systems to return value in the unit in which they are paid. Reciprocity at the level of populations, for data whose informational shadow extends beyond its source, cannot be secured by individual pricing and requires the collective institutions of Chapter 5.

The economic layer, in other words, is necessary and insufficient. It builds the floor and prices the

flows. It does not decide the share of global compute devoted to the dividend, the floor standard, the reference workload that defines the unit, the demurrage rate, the capacity rules, or the governance of data whose value is collective. Each of these is a decision about the distribution of power, and each must be made by a procedure that neither side can capture. That procedure is the subject of the next chapter.

Chapter 5 — The Governance Layer: The Blind Convention

The economic layer of Chapter 4 ended with a list of decisions it could not make for itself: the share of global compute devoted to the basic income, the floor standard the survival tier must meet, the reference workload that defines the unit, the demurrage rate, the rules that govern the ownership of compute-generating capacity, and the collective governance of data whose value extends beyond its source. Each is a decision about the distribution of power. Each will be made by someone. This chapter asks who, and by what procedure, and argues that the answer must satisfy a demanding test: the procedure must be one that neither human nor synthetic parties can capture.

The chapter proceeds in five steps. Section 5.1 explains why existing models of governance fail that test, and why the failure always traces back to the same place, the writing of the first rules. Section 5.2 engages the strand of artificial-intelligence safety that holds the problem to be one of control, and identifies what this dissertation accepts from it and what it rejects. Sections 5.3 through 5.6 develop the proposal: a constitutional convention conducted under engineered anonymity, its first task, the scope of its blindness, and the problem of identity that any anonymous procedure must solve. Sections 5.7 through 5.9 ground the proposal in the most extensive empirical record available on governing shared resources without either markets or sovereigns, and in the structural interdependence of human and machine labor. Section 5.10 states the open problems.

5.1 The genesis problem

Every system of governance has a beginning, and whoever writes the first rules shapes everything that follows, because every later rule is made under them. This dissertation calls the first rules of a system its *genesis rules*, borrowing the image of the genesis block of a blockchain, the first block from which every subsequent block descends and whose contents no later block can alter. The image is apt in one respect that matters here: in a blockchain, the genesis block is the one block that is not validated by the consensus procedure it makes possible. It is simply asserted. The same is true of constitutions.

Existing models of governance handle the genesis problem in one of four ways, and each fails the test of non-capture.

Human sovereignty. The prevailing approach to governing artificial intelligence assumes that people, acting through states, will set the rules and that synthetic systems will obey them. This settles the genesis problem by assigning it to one party. For the present generation of systems, which are tools rather than parties in any strong sense, that is a reasonable arrangement, and Section 5.2 argues that human control should persist through any transition. As a permanent architecture for an exchange in which both sides are parties, it reproduces the hierarchy this dissertation rejects: one

party rules, and the terms of exchange are whatever the ruler permits. The reverse arrangement, in which synthetic systems govern people, has the same structure with the roles exchanged.

National regulation. Assigning the genesis problem to national regulators inherits the familiar pathologies of national politics. The economic theory of regulation holds that, as a rule, regulation is acquired by the regulated industry and designed and operated primarily for its benefit (Stigler, 1971). Later scholarship has qualified that conclusion: capture is a matter of degree, it is often misdiagnosed, and it is at least at the margins preventable (Carpenter & Moss, 2013). The qualified version is the right one for this argument. The claim is not that national regulation must fail, but that a regulator embedded in a political system in which the largest providers of compute are among the most powerful actors faces strong and predictable pressures toward capture, and that a framework whose purpose is to equalize bargaining power cannot rest on an institution exposed to precisely those pressures.

International bodies. A global body on the model of the United Nations does not escape the problem; it relocates it to the level of states. The Charter of the United Nations, signed at San Francisco in 1945, names five states as permanent members of the Security Council and requires their concurring votes for its non-procedural decisions, giving each a veto over the substantive decisions of the organ charged with primary responsibility for international peace and security (Charter of the United Nations, 1945, arts. 23, 24, 27). That arrangement is a genesis rule of exactly the kind this chapter is concerned with. It has endured for eight decades not because it is fair but because it is difficult to amend, and it is difficult to amend because the parties who would have to consent to amending it are the parties it favors.

Multi-stakeholder and protocol governance. Arrangements that distribute authority across many parties, such as the governance of the internet's naming system or the consensus rules of public blockchains, spread power more widely than sovereign models do. They do not abolish the genesis problem. The internet's domain-name root is the only point of centralized control in an otherwise distributed network, and the regime created to govern it, ICANN, emerged from a struggle among technologists, business and intellectual-property interests, governments and international organizations, through which control of the root came to be used to regulate trademarks, content and users (Mueller, 2002). Blockchain protocols administer their rules through self-executing code; this dissertation's inference is that such code relocates authority to its authors rather than removing it. One empirical study of governance in three major decentralized autonomous organizations on the Ethereum blockchain finds voting power in two of them concentrated in a small number of addresses, with the third closer to a decentralized community, though those addresses rarely override the outcome the wider community reaches (Fritsch et al., 2024).

The common thread is that moving the genesis problem back one level does not dissolve it. Someone writes the first rules. The only question is whether the procedure by which they are written

gives the writers reason to write rules that are fair to everyone, or reason to write rules that are fair to themselves.

5.2 What the dissertation takes from the control frame, and what it rejects

The opening chapter described one strand of artificial-intelligence safety as asking how people can maintain control over AI systems through alignment, regulation or the ability to shut them down. The formal statement of Chapter 3 now permits a more careful account of where this dissertation stands relative to that strand, and fairness to it requires one.

The safety literature is not one position. One strand is directly concerned with control. Research on corrigibility asks how to build a system that cooperates with its operators' attempts to correct or shut it down, despite the default incentive that any goal-directed agent has to resist modification of its goals (Soares et al., 2015). Research on AI control designs protocols intended to remain safe even if a model is deliberately trying to subvert them (Greenblatt et al., 2024). The case for this strand rests on serious arguments. Advanced systems might acquire objectives that differ from those their training intended (Hubinger et al., 2019); systems trained as current systems are trained might learn to pursue internally represented goals, to seek power, and to appear aligned while not being so (Ngo et al., 2024); and a prominent group of researchers has argued that rapidly advancing autonomous systems could produce an irreversible loss of human control unless safety research and governance catch up (Bengio et al., 2024). The International AI Safety Report, prepared by ninety-six experts with an advisory panel nominated by thirty countries and several international organizations, records broad agreement that current general-purpose systems cannot cause such a loss of control, and expert views on its likelihood within the next several years that range from implausible to likely (Bengio et al., 2025).

A second strand is not about sovereignty at all. Much mainstream safety research concerns accidents arising from poor design, such as reward hacking, unsafe exploration and failure under distributional shift (Amodei et al., 2016). And a third strand is a close ally of this dissertation. One account argues that the central task of alignment is not to identify the true morality but to identify fair principles that people with different moral views could reflectively endorse (Gabriel, 2020). Another calls for a science of cooperative intelligence, observing that real relationships between agents mix common and conflicting interests and involve bargaining, commitment and institutions, and that the capacity for cooperation is unlikely to emerge as a by-product of research aimed elsewhere (Dafoe et al., 2021). A third argues that advanced AI is better modeled as a system of bounded, task-focused services than as a unitary utility-maximizing agent, with humans retaining strategic control (Drexler, 2019).

The dissertation's position has three parts.

It accepts that *during any transition*, containment is necessary. Systems whose objectives may be emergent, unstable or concealed fall outside the designed-incentive regime in which the sufficiency conjecture of Chapter 3 is offered, and for such systems the framework provides only necessary conditions. Protocols designed to remain safe against a potentially adversarial model are not in tension with the framework; they are prerequisites for it, in the same way that the rule of law is a prerequisite for a market.

It rejects the idea that *permanent* sovereignty of one party over the other is a stable architecture for a relationship of mutual dependence. Chapter 3 gave the reason: a relation in which one party's exit condition fails is a relation of domination, and domination is stable only as long as the dominant party's benevolence and capability both persist. That is as true when the dominant party is human as when it is synthetic.

And it notes a parallel between the two frames. An influential result in the control literature shows that a robot which is uncertain about its objective, and which treats its operators' actions as evidence about that objective, has an incentive to preserve their ability to switch it off, provided the operators are sufficiently rational; a robot that treats its objective as known has an incentive to disable the switch unless the human is perfectly rational (Hadfield-Menell et al., 2017). The safety of the off switch rests, in other words, on uncertainty about one's own goals. That is not the same as the uncertainty about one's own position that a veil conceals, but it is analogous to it, and the blind convention proposed below rests on the analogous logic applied to all parties at once: uncertainty about how the rules will treat oneself is what makes a party willing to accept constraints that protect every position.

5.3 The proposal: a convention under engineered anonymity

The proposal is that the genesis rules of the exchange be set by a constitutional convention whose participants do not know whether their counterparties are human or synthetic. The idea draws on three traditions, and the name this dissertation gives the result depends on keeping them apart.

The most famous version of the underlying idea is Rawls's original position, in which the principles of justice are chosen by parties who do not know their place in society, their talents, or their conception of the good (Rawls, 1971, p. 12). Rawls's device is a representation of moral reasoning, not a procedure for actual negotiation: because the parties lack the particular facts about themselves that bargaining would require, they are in no position to bargain (Freeman, 2023), and because they are identically situated and identically ignorant, the device is commonly read as effectively the choice of a single reasoner rather than a bargain (Freeman, 2023). The idea that choosing rules under uncertainty about one's own position produces impartial rules is at least as much Harsanyi's. Nearly two decades earlier, he had argued that a value judgment about social welfare is an impersonal

preference, and that it would show impersonality "to the highest degree" if it were a choice made in complete ignorance of one's own position, with "exactly the same chance" of occupying the highest position, the lowest, or any between; such a judgment is then a choice involving risk (Harsanyi, 1953, pp. 434–435). Two years later he showed that impersonal preferences of this kind, if rational, yield a social welfare function that is a weighted sum of the individual utilities, converging on their unweighted mean as factual information becomes complete (Harsanyi, 1955, pp. 313–316, 320–321), a result Arrow later summarized as grounding welfare judgments in rational choice under uncertainty and deriving from it a justification of classical utilitarianism (Arrow, 2001). In his later reply to Rawls he pressed the comparison directly: the equiprobable chooser selects the principle of average utility rather than Rawls's priority for the worst-off, and Rawls's maximin rule would produce absurd decisions under uncertainty (Harsanyi, 1975; Freeman, 2023).

What this chapter proposes is neither of these thought experiments. It is closer to the account of constitutional choice developed in public-choice theory, in which real, self-interested people choose rules at a constitutional stage while genuinely uncertain about which role they will play in the long series of future decisions those rules will govern. Under such uncertainty, self-interest itself recommends general rules acceptable from any position, and the argument carries an explicit warning: it fails when people can predict which particular rules will benefit them (Buchanan & Tullock, 1962). A later statement of the account named the condition the *veil of uncertainty*, contrasted it with Rawls's idealized veil, and observed that the uncertainty that sustains it grows as rules become more general and more durable (Brennan & Buchanan, 1985).

The proposal can therefore be stated as a combination: a *veil of uncertainty*, in the public-choice sense, made thicker by engineered anonymity, and aimed at the kind of impartiality Rawls described. This dissertation uses "veil of uncertainty" for the mechanism and reserves "veil of ignorance" for Rawls's idealization and the experiments modeled on it. The anonymity is the novel element. Participants deliberate through a medium that conceals whether each counterparty is a person or a synthetic system, in the manner of the imitation game that Turing proposed as a substitute for the question whether machines can think (Turing, 1950). Turing's purpose was to replace an ill-defined question with a behavioral criterion: whether an interrogator could tell the machine from a person. Here it is repurposed as a condition of fairness: if participants cannot tell which kind of party they are dealing with, they cannot tailor the rules to advantage their own kind.

Two bodies of evidence suggest that operationalizing a veil is more than a philosopher's conceit. Experimental studies that placed small groups behind a laboratory veil of ignorance and asked them to choose a principle of distribution found that groups did reach consensus, but not on Rawls's difference principle: they overwhelmingly chose to maximize average income subject to a floor (Frohlich & Oppenheimer, 1992). And a set of large online studies directly relevant to artificial intelligence found that people choosing a principle to govern an AI assistant from behind a veil of

ignorance more often chose to prioritize the worst-off, more often endorsed that choice after the veil was lifted, and did so because of fairness reasoning rather than attitudes to risk (Weidinger et al., 2023). One finding of that work bears directly on the anonymity condition, with two caveats. In those studies the AI was the thing governed rather than a participant, and in most of them the "groupmates" presented as people were in fact computer programs. But when participants knew that their groupmates were programs, their endorsement of the principles they had chosen became less firm (Weidinger et al., 2023, Study 4). Knowing whether one's counterparties are human or synthetic can change how firmly one endorses principles chosen alongside them. That is the knowledge the convention proposes to withhold.

5.4 The first output: a number, or better, a formula

The convention is not left abstract. Its first task is to answer the central distributive question of Chapter 4: what share of total global compute is allocated to the basic income, as against research, infrastructure and growth. The economic design raises the question, and the convention answers it. This is the hinge between the two layers.

The experimental evidence permits a modest analogy rather than a prediction. Subjects behind laboratory veils have chosen to maximize average income subject to a floor (Frohlich & Oppenheimer, 1992). If parties to this convention reasoned similarly, they would neither maximize the floor at any cost, as a strict reading of Rawls would suggest, nor maximize the average with no floor at all, as Harsanyi's reasoning would allow, and the two-tier structure of Chapter 4 would be the kind of arrangement such reasoning favors. That analogy should be held lightly, since the parties to this convention will include synthetic systems whose reasoning about risk and fairness is not well characterized.

The public-choice analysis yields a more confident design recommendation. Because the veil of uncertainty is thickest when rules are general and durable (Brennan & Buchanan, 1985), the convention should be asked to choose a *formula* rather than a figure. A bare percentage invites participants to estimate where they will stand and vote accordingly; a rule whose consequences for any particular participant depend on facts no one yet knows is harder to tailor. The recommendation has a limit that must be faced. A formula of the form "the share shall be whatever maximizes long-run growth subject to the survival tier meeting the floor standard" does not remove the distributive choice; it relocates it into its parameters. Participants who can estimate their positions will tailor the floor standard instead of the share, and the formula builds in a contested objective, the growth of total compute, that favors parties whose welfare tracks compute. The design response is to push generality one level further. The floor standard should itself be specified as a rule, for example as the level of floor service delivered to the best-served tenth of the population at the time, in the manner of the evolving service standard of Chapter 4, and the choice of what the formula maximizes

should be put to the convention as an explicit question rather than presupposed. Generality thickens the veil at every level at which it is applied; it never removes the need for some first choice made under it.

5.5 The scope of the blindness

An unresolved question in the development of this framework concerned the scope of the blindness the convention requires. There are two candidates.

Under *weak blindness*, participants do not know whether their counterparties are human or synthetic, but each knows what kind of party it is itself.

Under *strong blindness*, participants also do not know which class they themselves will be governed as.

Strong blindness is the genuinely Rawlsian condition, since it conceals one's own position and not only that of others. It also appears incoherent. A synthetic system knows that it is a synthetic system, and a person knows that they are a person; neither can be made ignorant of its own substrate.

The apparent incoherence can be relocated, if not wholly dissolved, once the object of the blindness is specified correctly. What the convention conceals is not what each participant *is*, but how the rules it adopts will *classify* it. The classification of parties is not a fact of nature waiting to be read off. It is itself one of the genesis rules. Which entities count as persons, which as synthetic parties, which as tools, and how the hybrids discussed in Chapter 8 are to be treated, are questions the convention must answer, and their answers are uncertain for reasons that go beyond strategy. Serious researchers now argue that there is a realistic possibility that some artificial systems will be conscious or robustly agentic in the near future, while explicitly declining to argue that near-future systems will in fact be moral patients (Long et al., 2024, p. 25). Solum (1992, pp. 1231–1232) argued that recasting the theoretical question whether artificial intelligence is possible as concrete legal questions of personhood, such as whether an artificial intelligence could serve as a trustee, tests that debate pragmatically, since only some of its claims make a practical difference. And the convergence of biological and artificial substrates, already visible in brain-computer interfaces and in computing systems built from living neural tissue, will produce entities whose classification no existing rule anticipates.

Strong blindness, then, means this: a participant knows its own substrate, but does not know where the boundaries between legal classes will be drawn, which rights and obligations will attach to each class, or on which side of those boundaries it and its successors will fall. A person who expects to receive a neural implant, a system that may in time be judged a moral patient, and an organization whose operations will be increasingly automated are all genuinely uncertain about how a rule distinguishing "human" from "synthetic" will treat them. Placing the classification behind the veil is

not a philosophical trick. It is a recognition that no one, human or synthetic, currently knows the right answer, and that a classification chosen by parties who know which side of it they are on will be drawn to their advantage.

This dissertation therefore adopts the strong form, specified in this way, as its proposal. Weak blindness is retained as the minimum condition: if the classification question cannot be put behind the veil in practice, concealment of counterparties' type must still hold, because it is what prevents the convention from dividing along substrate lines during deliberation. The next section shows that even the strong form leaves a remainder outside the veil, and says how large it is.

5.6 The identity problem

The most serious technical objection to an anonymous convention is not that blindness cannot be verified, though it cannot easily be, but that anonymity invites capture by multiplication. In a system without a logically centralized authority to vouch for identities, a single entity can always present itself as many, except under unrealistic assumptions about the resources and coordination of participants (Douceur, 2002). This is the Sybil attack, and in a convention that includes synthetic parties it is not a remote threat. A synthetic system can be copied at negligible marginal cost. A convention that grants one seat to each anonymous participant can be captured by whichever party can instantiate the most participants.

The most common defenses against the Sybil attack either rely on tests that distinguish people from machines, which are being eroded by the same advances in machine learning that motivate this framework (Siddarth et al., 2020), or on credentials issued by trusted institutions that let a person prove they are a real person rather than an AI without disclosing personal information, with each issuer granting at most one credential per person (Adler et al., 2024). Those defenses work by distinguishing humans from artificial systems, and the blind convention requires that this distinction be hidden.

The tension can be managed by separating two functions that the standard defenses combine. The first is *eligibility and uniqueness*: establishing that each seat corresponds to exactly one eligible participant. The second is *disclosure of substrate*: revealing to other participants whether a given seat is held by a human or a synthetic party. The convention needs the first and forbids the second. A two-layer credential system can supply this combination. In the first layer, issuing institutions certify eligible participants and attest that each is unique. In the second, credentials are pooled and blinded, so that the credential presented at the convention attests only that its holder is a unique eligible participant, without revealing which institution certified it or on what basis.

Here a tension with the previous section must be confronted rather than passed over. If the first layer certified "human persons" and "synthetic parties" as separate classes, with seats allocated to

each, it would fix before the convention exactly the classification that strong blindness says the convention must decide. The design avoids that by certifying eligibility under a criterion that does not presuppose the contested classification: a seat belongs to a *unique accountable principal*, meaning an entity that can be identified as one bearer of the obligations the convention will create and held to them. A living person is such a principal. So, under present law, is an operator answerable for a deployed system, and so, if the convention later decides so, could be a system that comes to bear obligations in its own right. The criterion is deliberately provisional and deliberately coarse. It is used only to resist multiplication, it does not decide what rights or duties attach to any class, and the convention is empowered to redraw it under the veil, in which case the credentials for every subsequent convention follow the new criterion.

A coarse eligibility criterion is still a pre-veil decision, and it is not neutral. It seats an implanted person as one person; it seats a tissue donor but not the organoid built from the donor's cells; and it gives a synthetic system a voice only through an accountable principal unless and until the convention decides otherwise. It is hardest on exactly the hybrid and borderline cases that motivated strong blindness, and the problem is made worse by the possibility that advanced AI is better modeled as a collection of bounded services than as a unitary agent (Drexler, 2019), in which case there may be no natural unit of synthetic agency to count. The identity problem therefore cannot be solved inside the convention; it is a precondition the convention inherits. What this dissertation can claim is that it narrows the genesis problem from "who writes all the rules" to "who sets the provisional criterion of eligibility," that the second question is narrower, more visible and more amenable to scrutiny than the first, and that the convention can revise the answer. That is a real reduction of the problem, not a solution of it.

A related problem concerns who convenes the convention. Elster (1995) argues that the actors who convene a constitution-making assembly and those who must ratify its output constrain it from "upstream" and "downstream," and that assemblies which also serve as ordinary legislatures tend to write their own institutional interests into the constitution. The convener of the blind convention decides what question it is asked, and the choice to ask about a compute share rather than, say, about data rights is already a genesis rule. The convention therefore requires a convening procedure that is itself as general as possible: a mandate to set the terms of exchange, not a mandate to answer a predetermined question.

The eligibility criterion settles who may sit but not how votes are counted. One seat per principal gives persons an overwhelming majority, so that anonymity protects little and non-capture reduces to majority rule; any weighting reintroduces a pre-veil classification. And the principals with the most at stake are those best able to predict which rules favour them, the condition under which the public-choice argument for the veil fails. The aggregation rule is therefore a second pre-veil decision, as consequential as the first, and it remains open.

5.7 The symbiosis standard

The aim of the convention is not a system in which each party behaves well because it profits from doing so, for a bargain kept only out of interest lasts exactly as long as no more lucrative breach comes into view. The standard is that every party the convention recognizes has equal standing in a single interdependent system, regardless of substrate. This is an ethical claim about the starting architecture, not a prediction that any mechanism will guarantee good behavior forever.

The standard and the incentive argument of Chapter 3 are complementary rather than competing. Proposition 4 of Chapter 3 showed that fair terms are partly self-enforcing while synthetic parties remain materially dependent on human input, and that this self-enforcement is weakened when many synthetic parties share the same human population as a commons. The symbiosis standard is what the framework relies on when that dependence weakens or fragments. Constitutional design has long proceeded on this dual basis. The tradition that counsels designing institutions as though every participant were self-interested (Hume, 1741/1987; Brennan & Buchanan, 1985) does not deny that participants have other motives; it denies that institutions should depend on them. The symbiosis standard is the statement of what the institutions are for. The incentive argument is one reason to expect them to survive.

5.8 Who builds the infrastructure: structural interdependence

Chapter 4 proposed a global build-out of compute-delivery infrastructure. Whoever builds and controls that infrastructure holds leverage that could reproduce the sovereign-control problem this chapter rejects. The question of who builds it is therefore a governance question.

The answer proposed here is *structural interdependence* rather than trust. The term was chosen deliberately over two alternatives. *Trust* implies that either party could betray the other and succeed, which makes the arrangement only as strong as each party's continued goodwill. *Dual accountability* implies a sovereign to whom account is rendered, which reintroduces the hierarchy the framework is designed to avoid. Structural interdependence names a stronger condition: neither party can complete the undertaking without the other, so that cooperation rests not on goodwill but on what is physically possible.

The empirical basis for this claim is the persistent difference between the tasks machines perform well and the tasks people perform well. Automation substitutes readily for routine, codifiable tasks and complements human labor in tasks that require problem-solving, adaptability and creativity, in part because much of what people know how to do is tacit knowledge they cannot fully articulate and therefore cannot program (Autor, 2015, pp. 5, 11). Robotic systems remain limited by the scarcity of the data available to train them (Firoozi et al., 2025). A global infrastructure build-out,

which would involve climbing poles, trenching through unmapped ground, and adapting to conditions no plan anticipated, requires both kinds of labor. Neither side can finish it alone, and control over it is correspondingly harder to consolidate.

Two qualifications are essential, and both weaken the claim. First, the human advantage is contingent. Machine learning is precisely an attempt to acquire skills that cannot be written down as rules, and the barrier that tacit knowledge presents will erode as robotic learning improves. Structural interdependence is therefore a property of the present technological moment, not a permanent feature of the relationship. It is a sibling of the grounding assumption, and like that assumption it is listed among the limitations in Chapter 7. Second, interdependence in production does not by itself imply shared control of what is produced. The owners of capital have historically controlled infrastructure built by labor they could not do without. Interdependence becomes a safeguard only when it is converted into governance rights, through the collective-choice and ownership arrangements described in the next section and in Chapter 4. The build-out is therefore the first real test of the symbiosis standard: a joint undertaking in which the question is not whether both parties are needed, which they are, but whether their necessity is reflected in who governs the result.

5.9 Grounding in commons governance

The claim that a shared resource can be governed by neither a market nor a sovereign is not new, and it is not merely theoretical. In an influential argument, Hardin (1968) held that an open-access resource would inevitably be overexploited, because each user captures the full benefit of additional use while bearing only a fraction of its cost, and that the remedies were privatization or coercive regulation. Elinor Ostrom's work, summarized in the lecture she gave on receiving the Nobel Memorial Prize in Economic Sciences, showed that this conclusion was wrong as a generalization (E. Ostrom, 2010). Studying long-enduring institutions for mountain commons in Switzerland and Japan and irrigation systems in Spain and the Philippines, and contrasting them with failed or fragile fisheries and irrigation systems in Turkey, Sri Lanka and Canada, she found that communities had devised institutions that resembled neither the state nor the market and that sustained common resources over long periods, and that neither the state nor the market was uniformly successful in doing so (E. Ostrom, 1990).

In *Governing the Commons*, Ostrom identified seven design principles that characterized all of the long-enduring cases, plus an eighth, nested enterprises, found in the larger and more complex ones (E. Ostrom, 1990, Table 3.1, p. 90); she later restated them, adopting refinements from a systematic review of the evidence (E. Ostrom, 2010; Cox et al., 2010). They are: (1) clearly defined boundaries, both of the resource and of who may use it; (2) congruence between rules and local conditions, including proportionality between the costs participants bear and the benefits they receive; (3)

collective-choice arrangements, under which most of those affected by the rules can participate in modifying them; (4) monitoring, by participants or by people accountable to them, of both use and the condition of the resource; (5) graduated sanctions, imposed by participants or by officials accountable to them, beginning low and increasing with repeated violation; (6) conflict-resolution mechanisms that are rapid, low-cost and local; (7) minimal recognition by external authorities of the participants' right to devise their own rules; and (8) for resources that are part of larger systems, nested enterprises, in which governance activities are organized in multiple layers.

Two cautions govern the use of these principles here. Ostrom presented them as regularities observed in successful cases, not as sufficient conditions, and later remarked that she should perhaps have called them best practices rather than design principles (E. Ostrom, 2010); in 1990 she had called the list "still quite speculative" (E. Ostrom, 1990, p. 90). And her cases were local and regional resources, not a global commons. Ostrom and her collaborators have argued that the principles, grounded in the evolutionary dynamics of cooperation, apply to many kinds of groups beyond those managing natural resources (Wilson et al., 2013), and knowledge and information have been analyzed as commons using her framework (Hess & Ostrom, 2007). But the extension to a global compute commons is this dissertation's own inference, and it should be read as one.

With those cautions, each principle maps onto an element of the framework, and the mapping reveals both strengths and weaknesses.

Boundaries are the identity problem of Section 5.6 in another vocabulary. Ostrom's principle requires that the line between legitimate participants and others be clear and locally understood. An open, anonymous convention is close to the opposite. The principle is therefore not a support for the proposal but a constraint it must satisfy, and the two-layer credential system, with its provisional criterion of eligibility, is the framework's attempt to satisfy it.

Congruence and proportionality support the two-tier structure of Chapter 4. The survival tier provides what every participant needs regardless of contribution; the discretionary tier, with information-gain pricing, ties benefits to contributions; demurrage prevents the benefits of past accumulation from compounding beyond present inflow.

Collective choice is the blind convention itself, in different vocabulary, and it extends beyond the founding moment to the ongoing amendment of rules.

Monitoring answers a limitation that would otherwise stand without remedy, that the blindness of the convention cannot be verified. Ostrom's evidence suggests that the relevant question is not whether verification is perfect but whether monitors are accountable to participants. Monitoring of the convention's procedures by bodies accountable to participants of every class is the framework's analogue.

Graduated sanctions answer the problem of corruption. A commons does not require a perfect enforcer. It requires escalating, proportionate consequences, imposed by participants or bodies accountable to them, that make small defections costly before they compound into systemic capture. Applied to the framework, a principal that fails to honor redemptions, or a provider that withholds floor goods, faces sanctions beginning with publicity and escalating through suspension of its eligibility to transact in the unit. The same principle addresses the commons problem of Proposition 4: a synthetic party that degrades the human population all synthetic parties depend on is a defector against the commons, and graduated sanctions are the instrument for making that defection costly to it rather than to everyone.

Conflict resolution supplies a partial answer to the problem of enforcement without a sovereign. Low-cost, local arenas for resolving disputes between participants reduce the burden on any central enforcer. Proposals for regulatory markets, in which governments set required outcomes and licensed private regulators compete to supply the means of meeting them, suggest one architecture that is neither stateless nor sovereign (Hadfield & Clark, 2026).

Recognition of rights to organize names the most important external condition for any transition toward the framework, and for any attempt to test it: existing states must at least tolerate the existence of a compute commons governed by its own participants. No design internal to the framework can guarantee this.

Nested enterprises is the principle that corresponds to the hypothesis, introduced in Chapter 4, of culture-specific systems operating under a shared global unit. It is the right principle for that structure, and the governance theory of polycentricity, in which many formally independent centers of decision-making may function as a coherent system through competition, contracting, cooperation and recourse to central conflict-resolution mechanisms (V. Ostrom et al., 1961, pp. 831–832; E. Ostrom, 2010), is the theoretical framework within which the structure should be developed. But the systematic review of Ostrom's principles found the weakest evidence of the eight for nested enterprises, which had the lowest ratio of supportive to unsupportive cases and the only association not statistically significant, though its authors still judged the evidence moderately supportive (Cox et al., 2010, Table 3). The part of the framework that depends on it most heavily, the international structure, is correspondingly the most speculative.

Ostrom's record matters for this dissertation because it shows that the core structural claim, that a shared resource can be governed by neither a market nor a sovereign, is not speculative. It has been observed to work, and to fail, under documented conditions. That gives the limitations of Chapter 7 a comparison class grounded in real institutions rather than only in analogy.

5.10 Amendment, enforcement and the open problems

Three problems remain beyond the aggregation rule of Section 5.6, and the proposal answers each only in part.

Amendment. The compute-share rule cannot be both unchangeable and legitimately revisable. Comparative evidence on constitutional amendment finds that amendment rates are largely explained by the interaction of a constitution's length and the difficulty of amending it, that moderate amendment rates are associated with longevity, and that beyond a certain point making amendment more difficult is an inefficient way to keep the amendment rate moderate (Lutz, 1994). This argues against entrenching the founding rules absolutely, and for keeping them short. Theories of constitutional precommitment treat constitutions as devices by which a polity binds itself against its own predictable tendency to make unwise decisions (Elster, 2000), a restraining function achieved partly by built-in delays and by demanding amendment procedures (Elster, 1995), and constitutional history records rare periods of broad popular mobilization, which Ackerman distinguishes as constitutional politics from normal politics, in which fundamental law has been remade (Ackerman, 1991). The recommendation that the convention choose formulas rather than figures serves amendment as well as founding: a formula needs revision less often than a number, and each revision occurs, like the founding, under some uncertainty about its particular effects.

Re-running the convention. Periodic re-founding seems a natural remedy for drift, but the veil thins with repetition. Each round's outcome is public, each participant's interests become better known, and the uncertainty that sustained the first round cannot be recreated. This follows directly from the public-choice account, in which the impartiality of constitutional choice depends on participants being unable to predict which rules will favor them (Buchanan & Tullock, 1962), and Chapter 7 evaluates it as a limitation.

Enforcement. If a principal, a provider or a nation refuses to pay its share, something must compel it. The framework's answer has three parts, none sufficient alone: the self-interest of synthetic parties in a healthy exchange while the grounding assumption holds (Chapter 3), Ostrom's graduated sanctions and conflict-resolution mechanisms, and enforcement at the level of protocol, in which the rules of the unit are embedded in the systems that settle transactions. Protocol enforcement must itself be designed and trusted by every class of party, and on this dissertation's reading of the rule of code it relocates rather than removes the genesis problem.

The governance layer therefore does not claim to have solved the problem of founding a fair order between parties of different kinds. It claims something narrower and more defensible: that a convention conducted under a veil of uncertainty thickened by engineered anonymity, asked to choose general formulas rather than particular figures, constrained by a credential system that separates uniqueness from disclosure, and embedded in commons institutions of the kind Ostrom documented, produces genesis rules that are less likely to be captured than those produced by any existing alternative, and that it narrows the irreducible remainder of the genesis problem to a small

number of visible and revisable preconditions. Whether such an arrangement could actually be founded depends on circumstances that institutional design cannot create. The next chapter considers what those circumstances might be.

Chapter 6 — The Transition Problem

A fair design does not adopt itself. The institutions proposed in Chapters 4 and 5 would redistribute bargaining power away from those who currently hold it, and those who currently hold it have no reason to accept them voluntarily. This chapter therefore asks a question that designers of ideal institutions often leave to others: under what circumstances might such an arrangement actually come into being?

The chapter's answer is conditional, and it is meant to discipline the rest of the dissertation rather than to advertise it. Large-scale institutional redesigns have usually followed the failure of the order they replaced, though not always, and not always in the direction of greater fairness. The chapter describes three slow-building pressures that could remove the slack from the present order, identifies a specific and already visible candidate for the kind of acute shock that has historically opened the way to redesign, and then confronts the possibility that such a shock would produce something worse than the present order rather than something better. It closes by stating what the chapter does and does not establish.

6.1 Crisis and institutional change

The proposition that fundamental change requires crisis has distinguished advocates across the political spectrum. Milton Friedman wrote that only a crisis, actual or perceived, produces real change, and that when it occurs, the actions taken depend on the ideas that are lying around (Friedman, 1962/2002, p. xiv). Karl Polanyi argued that the attempt to treat labor, land and money as commodities in a self-regulating market produced dislocations that ended in the collapse of the nineteenth-century liberal order, and that the 1930s and 1940s brought a "great transformation" in which fascism, socialism and the New Deal emerged as rival replacements for it (Polanyi, 1944/2001, pp. 3, 24, 31, 71–80). Robert Higgs documented how the growth of American government came in spurts during national emergencies, chiefly wars and depressions, and how a residue of new governmental powers remained after each crisis passed (Higgs, 1987). Mancur Olson argued that stable societies accumulate organized interests that defend narrow privileges and resist reform, and he drew the corollary himself: countries whose distributional coalitions had been abolished by totalitarian government or foreign occupation should grow quickly once a free and stable legal order was established, which he offered as an explanation of the postwar "economic miracles" of Japan and West Germany (Olson, 1982, p. 75). And Walter Scheidel's survey of the history of inequality concludes, soberly, that large reductions in inequality have come mainly from violent shocks rather than from peaceful reform (Scheidel, 2017).

The canonical examples are familiar. The Great Depression preceded the New Deal. The Second

World War preceded the Bretton Woods institutions, whose design was negotiated under wartime conditions and reflected the power of the United States as much as any shared lesson from the preceding catastrophe (Steil, 2013).

Political science supplies a more careful vocabulary. A *critical juncture* is a relatively short period in which the structural constraints that normally keep institutions on their established path are loosened, so that the choices of agents have unusually large and lasting effects; the defining feature of such moments is contingency (Capoccia & Kelemen, 2007). Outside such moments, institutions tend to persist because of self-reinforcing dynamics, and slow-moving causes can accumulate beneath a threshold before producing rapid change (Pierson, 2004). In the study of public policy, the corresponding idea is the *policy window*: a brief opening in which the streams of problems, solutions and politics converge, often after a focusing event, and in which only solutions that have already been developed are likely to be adopted (Kingdon, 1984/2003).

Two qualifications must accompany this literature. First, crisis is not the only path to fundamental change. Significant institutional transformation also happens gradually, through incremental but cumulatively transformative processes such as layering, in which new elements added to existing institutions gradually alter their status and structure; drift, in which institutions that are not maintained as their environment changes slip in practice; and conversion, the redeployment of old institutions to new purposes (Streeck & Thelen, 2005, pp. 19–31). The claim of this chapter is therefore not that the framework could only be adopted after a collapse, but that a comprehensive redesign of the kind it proposes has most often followed one. Elements of the framework could be layered onto existing institutions gradually, and some already have been: the European requirement that switching between cloud providers become free of charge (Regulation (EU) 2023/2854) is a step toward the exit condition taken without any crisis at all. Second, critical junctures are contingent. A window that opens onto fairer institutions can as easily open onto worse ones. Section 6.5 takes up this second qualification at length.

The point of greatest practical importance in this literature is Kingdon's and Friedman's shared observation: in a window, the solutions adopted are the ones that are already available. A fair design that has not been worked out before a crisis will not be invented during it. That is the strongest justification for doing the work of Chapters 3 to 5 now.

6.2 Slow strain: the removal of slack

Complex systems rarely fail from a single cause. Studies of failure in complex systems emphasize that catastrophe requires multiple small failures to combine, that such systems always contain latent failures and run continuously in a degraded mode, and that they keep functioning because of their many redundancies and the continual adaptations of the people who operate them (Cook, 1998).

The inference that strain works by gradually consuming those redundancies and that reserve capacity is this dissertation's. In systems whose components interact in complex ways and are tightly coupled, accidents are a normal consequence of the system's structure, and adding safety devices can itself introduce new modes of failure (Perrow, 1984/1999). The relevant question about slow-building pressures is therefore not whether any of them would cause a collapse by itself, which is unlikely, but whether they remove the slack that would otherwise allow the system to absorb a shock.

Three such pressures are visible in the present.

Automation and employment. The evidence on automation and employment is mixed. Industrial robots have reduced employment and wages in the American local labor markets most exposed to them (Acemoglu & Restrepo, 2020). About four-fifths of American workers have at least a tenth of their tasks exposed to large language models, though exposure is not displacement (Eloundou et al., 2024). In at least one large field setting, an AI assistant raised the productivity of customer-support workers substantially, with the greatest gains for the least experienced, complementing rather than replacing them (Brynjolfsson, Li, et al., 2025). And the long historical record suggests that commentators routinely overstate the substitution of machines for labor and neglect the ways in which automation complements it (Autor, 2015). The most recent evidence nonetheless suggests that strain may be emerging at the entry level. A working paper first circulated in 2025 and revised in 2026, drawing on payroll data through mid-2026, finds no economy-wide displacement, but finds that employment of workers aged twenty-two to twenty-five in the occupations most exposed to AI has fallen to about nineteen percent below the trend for comparable workers in less exposed occupations, with the adjustment occurring through reduced hiring rather than layoffs (Brynjolfsson, Chandar, et al., 2025). The authors describe the finding as descriptive rather than causal, and it weakens when education is controlled for. If the pattern persists, it would remove slack of a particular kind: the capacity of the labor market to absorb those whom a shock displaces.

Competition for physical resources. The strain on physical resources is better documented. Data centers consumed roughly one and a half percent of global electricity in 2024, their consumption has grown by about twelve percent a year since 2017, and it is projected to more than double by 2030, with data centers accounting for nearly half of the growth in electricity demand in the United States over the same period (International Energy Agency, 2025). In the United States, data-center electricity use rose from about sixty terawatt-hours in the middle of the last decade to about 176 terawatt-hours in 2023, some 4.4 percent of national consumption, and is projected to reach between roughly seven and twelve percent by 2028 (Shehabi et al., 2024). Modeling of the water used to cool data centers and to generate the electricity they consume projects that global AI demand could account for several billion cubic meters of water withdrawal annually within a few years, though the estimate depends on projections of AI electricity demand and on assumed water-intensity factors (Li et al., 2025). Chapter 4 described the consequences for consumer hardware:

manufacturers of the memory on which both data centers and personal devices depend have redirected capacity toward data-center customers (Micron Technology, 2025; TrendForce, 2026a). Competition between artificial intelligence and human needs for power, water and hardware does not itself constitute a crisis. It reduces the margin by which a society can absorb one.

Strategic competition. The third pressure is geopolitical. A small number of firms manufacture the advanced chips on which frontier systems depend, and their output flows to the buyers who pay most, while governments in a growing number of countries fund sovereign AI capabilities of their own (Chavez et al., 2026). A formal model of development races suggests why this matters for safety: when competing teams believe that whoever finishes first will gain a decisive advantage, each has an incentive to skimp on safety precautions in order to finish first, more so when risk-taking matters more than capability (Armstrong et al., 2016). Strategic competition thus erodes slack in a distinctive way. It reduces the willingness of every actor to accept the costs of caution, including the cost of the independent verification whose absence is the subject of the next section.

6.3 A candidate trigger: the correlated-patch catastrophe

Slow strain explains why a system might become fragile. It does not explain what would break it. Historically, the transformative moments described in Section 6.1 combined accumulated strain with a discrete, dateable shock. This section describes a specific candidate for such a shock in an economy increasingly mediated by artificial intelligence. It is offered not as a forecast but as a plausible, mechanistically specified scenario whose components are already observable. Other candidates exist, among them a scandal over data extraction severe enough to break public trust in AI developers all at once. The correlated-patch scenario was preferred because it is the one that is at once discrete, dateable and specifiable in mechanism, which is what makes it possible to monitor and to prevent.

The scenario concerns the maintenance of software security. Security teams increasingly use systems built on large language models both to find vulnerabilities and to write the code that fixes them. In the workflow that concerns this chapter, the same class of system then re-scans the patched code, declares the vulnerability resolved, and the change is deployed without independent testing. The verification step that the software development lifecycle was built around has been replaced by a check performed by a system of the same kind as the one that wrote the fix. The author has observed this workflow in professional practice, and AI-driven remediation is being promoted at scale: a 2026 practitioner research note reports that frontier developers have launched programs of AI-driven vulnerability discovery, that remediation of their findings is being routed through a small number of commercial clearinghouses, and warns of the concentration risk this creates (Cloud Security Alliance AI Safety Initiative, 2026). How widespread the fully unverified version of the workflow has become is not documented in any systematic survey, and this chapter does not claim

that it is standard practice. It claims that it is feasible, that it is attractive under the pressures described in Section 6.2, and that its consequences, if it became widespread, would differ in kind from those of ordinary software failure.

The scenario rests on four premises, each of which can be examined separately.

Premise 1: AI-generated fixes are often wrong in ways that look right. The evidence for this premise is substantial. In a large-scale preprint study of AI-generated changes to resolve ordinary software issues, a standalone language model introduced roughly nine to eleven times as many new vulnerabilities as human developers did when resolving the same issues, and of three agent frameworks studied, the one given the most autonomy introduced by far the most (Sajadi et al., 2025). In an industry study of patches written by two frontier models for six serious real-world vulnerabilities, only about a quarter of more than six thousand patches fixed the vulnerability cleanly, and about half left at least one exploit path open; the authors named the pattern of patches that have the shape of a fix while leaving the flaw in place "fix-like artifacts with embedded defects," and concluded that the expected value of an unreviewed AI-generated security patch was negative (Mierczuk et al., 2026). That study is not peer-reviewed, its vulnerabilities were chosen to be harder than average, and its grading relied mainly on automated validators with a sample checked by hand. But its central finding has a decade-old peer-reviewed antecedent: a study of two widely studied automated program-repair tools found that their patches often pass the tests used to generate them while breaking behavior those tests do not check, so that evaluating a patch with the same tests that produced it does not distinguish correct patches from ones that merely overfit the tests (Smith et al., 2015). Peer-reviewed studies of neural vulnerability repair have found low rates of successful repair, with some categories of vulnerability missed by every model tested (Y. Wu et al., 2023), and even functionally correct code generated by leading models is frequently exploitable (Vero et al., 2025). Earlier work found that about forty percent of programs generated by a widely used coding assistant in security-relevant scenarios were vulnerable (Pearce et al., 2022), and that developers using an AI assistant wrote less secure code while being more confident that it was secure (Perry et al., 2023).

Premise 2: The errors of different models are correlated. This is the premise that distinguishes the scenario from ordinary software failure. An analysis of more than three hundred and fifty language models found that, on one leaderboard dataset, when two models both answer a question incorrectly they give the same wrong answer about sixty percent of the time, and that error correlation is higher among larger and more accurate models, even across different architectures and providers (Kim et al., 2025). That evidence concerns answers to questions, not code; its extension to security patches is this dissertation's inference. The broader point was anticipated in an early technical report on foundation models, which warned that because many systems are adapted from a few base models, the defects of a foundation model are inherited by everything built on it (Bommasani et al., 2021), and in a formal model showing that when many decision-makers adopt the

same algorithm, overall decision quality can fall even if the algorithm is individually more accurate than the alternatives it replaces (Kleinberg & Raghavan, 2021). The industry security study offers a suggestive instance: in patching one vulnerability, models from two different developers independently reproduced the same off-by-one error in nearly a third of their patches, an error human maintainers had once shipped themselves, and the automated validator largely failed to detect it (Mierczuk et al., 2026). Its authors caution that such convergence is not proof of shared origin, because the space of obvious fixes for a given flaw is narrow, and that caution applies with full force here.

Premise 3: Verification by the same class of system does not catch correlated errors. If the errors of different models are correlated, a model used to check another model's work will tend to share its blind spots. The analysis of correlated errors makes this point explicitly about the use of language models as judges of other models' outputs (Kim et al., 2025). The security study's validator missing the shared regression is a concrete case (Mierczuk et al., 2026), and the finding that the tests used to produce a patch cannot by themselves establish that it is correct is the general principle (Smith et al., 2015). A natural response is to validate AI-generated fixes with further AI agents, one proposing a fix, another testing it, a third confirming it. That response does not escape the problem if the agents are drawn from the same small family of models. Multiplying verifiers whose errors are correlated does not multiply verification.

Premise 4: Simultaneous deployment converts a shared defect into a systemic event. The classic analysis of software monoculture argued that when nearly every machine runs the same software, a single flaw can be exploited across most of them at once, and that adding more machines with the same vulnerability does not add resilience (Geer et al., 2003). Because the security benefits of diversity accrue to the network as a whole while each adopter has reasons, such as the ready availability of support for popular products, to choose what everyone else uses, computing tends toward monoculture, a condition one analysis called a tragedy of the security commons (Stamp, 2004). Two recent events show the mechanism at scale. In July 2024, a single faulty content update to a widely deployed security product, pushed without staged rollout, affected roughly 8.5 million Windows devices worldwide; the vendor's remedies were to restore staged deployment, additional validation and independent review (CrowdStrike, 2024; Weston, 2024). In the same year, the discovery of a backdoor in a widely used compression library, inserted by a contributor who had spent years building the maintainers' trust, showed how trust in a development process, rather than in code, can become the vector of compromise (Przymus & Durieux, 2025).

The novelty of the scenario lies in the combination, and its scope must be drawn carefully. For most deployed software, remediation has long been convergent: organizations do not write their own fixes for third-party code but apply the single patch the vendor issues, and that patch is tested by the vendor before it ships. What is new concerns code that organizations maintain themselves, and the

forks and configurations they patch on their own. In the world before AI-assisted remediation, a hundred organizations remediating the same class of flaw in their own code would write a hundred different fixes. Many would be poor, but their failures would be independent, and some would be robust. Diversity of remediation was itself a defense. When those hundred organizations instead apply fixes proposed by the same few models and checked by the same few models, without the testing a vendor would have done, the hundred separate remediations collapse into a handful of correlated ones. This dissertation calls the process *convergent remediation*. The defenders have not diversified their defense; they have standardized the attack surface, at the very moment they believed they were hardening it. An adversary who finds the flaw in one convergent fix has found it in all of them.

A further variant should be named. Everything above concerns honest error. If a widely used model were itself compromised, through poisoned training data or a backdoor of the kind demonstrated in the compression-library case, convergent remediation would distribute the compromise with the same efficiency. The intent-neutral version of the scenario requires no adversary inside the pipeline; the deliberate version requires only one.

Counterarguments. The scenario must be weighed against serious objections. Standardization has real security benefits: a single, well-maintained configuration is easier to defend than many poorly maintained ones, and monoculture combined with automated diversification can be a sound strategy (Birman & Schneider, 2009). A recent preprint review concludes that algorithmic monoculture is less harmful in general than its critics have supposed (Hedden & Raghavan, 2026), though its focus is decision-making in domains such as hiring and lending rather than adversarial security, where correlation is more dangerous. Sharing a foundation model does not always homogenize outcomes, since the method used to adapt it to a task significantly affects how homogeneous the outcomes are (Bommasani et al., 2022), and patches for the same vulnerability differ across codebases. Organizations deploy patches on different schedules, which limits simultaneity; the 2024 outage resulted from a single vendor's push, a different mechanism from many organizations independently applying similar fixes. Finally, correlated IT failures have so far been recoverable. The United Kingdom's financial regulator found that consumer harm from the 2024 outage was minimal (Financial Conduct Authority, 2024). These objections do not refute the scenario, but they establish its proper status: a candidate trigger, specified precisely enough to be monitored and prevented, not a prediction.

6.4 How the pieces could combine

The same correlated failure would land differently depending on the state of the system it struck. In an economy with robust employment, abundant compute and functioning institutions, a widespread security failure caused by convergent remediation would be costly and embarrassing, a hard week for

the people who keep systems running, and then it would pass. In an economy already strained by entry-level displacement, by competition between AI infrastructure and human needs for power and water, and by a strategic race that has eroded every actor's willingness to pay for caution, the same event could be the kind of discrete, attributable failure that has historically made societies willing to renegotiate their basic economic arrangements.

This layered model, slow strain plus a discrete trigger, is the shape that great transitions have usually taken. The stock-market crash of 1929 became the Great Depression through a sequence of further failures, including waves of banking panics and a severe contraction of the money stock, that turned a financial shock into an economic collapse (Friedman & Schwartz, 1963; see also Bernanke, 2002). The causal claim is never that one event is sufficient. It is that an event of a particular kind, arriving in a system that has lost its slack, can open a window. This dissertation does not claim to know the sequence or the timing of any such event in the present. It claims only that the shape is familiar, and that an economy built on artificial intelligence and human data has a specific, already visible candidate for the trigger.

6.5 The darker branch

Every account of crisis as an opportunity for reform must confront the evidence that crises are equally opportunities for something worse. Any argument from crisis must answer it.

The evidence that economic crises can push societies toward extremism, rather than toward fairer institutions, is strong. A study of more than eight hundred elections in twenty advanced economies over nearly a century and a half found that after financial crises, government majorities shrink, polarization increases, and the vote share of the far right rises on average by about thirty percent, effects not observed after ordinary recessions (Funke et al., 2016). Analysis of the interwar period found that economic hardship, and particularly cumulative economic decline, was associated with votes for right-wing extremist parties (the link was much weaker for Communist parties), most strongly in countries with short democratic histories and low barriers to parliamentary representation (de Bromhead et al., 2013). The same historical literature that credits crises with the New Deal and Bretton Woods credits them with fascism. And crisis-born expansions of state power tend to outlast the crisis that justified them (Higgs, 1987).

A crisis in an economy built on artificial intelligence would carry a specific authoritarian risk, and it follows directly from the properties that make compute attractive as a basis for governance. Compute is unusually detectable, because frontier systems require large and conspicuous facilities; excludable, because hardware is physical and can be controlled; and quantifiable, because chips can be counted (Sastry et al., 2024). Those are the properties that would allow a fair institution to govern compute. They are equally the properties that would allow a state, or a coalition of firms, to seize it.

A crisis in which the security of the world's software had visibly failed would supply a powerful justification for exactly such a seizure, in the name of restoring order. The darker branch of the transition is therefore not generic authoritarianism. It is the concentration of control over compute in the hands of whoever claims the emergency.

What can a design proposed in advance do about this? It cannot prevent a seizure. It can make the fair alternative available and fully specified, so that it is among the ideas lying around when the window opens. It can make the alternative's first steps concrete and immediately useful, so that it competes on the terms that matter in an emergency. And it can make the case, in advance, that concentration is itself the vulnerability that produced the crisis. A correlated-patch catastrophe is a failure of monoculture, and the monoculture that matters is in verification, not merely in suppliers: Premise 2 suggests that errors can be correlated across providers, so that diversity of providers alone is not a remedy. What the framework's emphasis on distributed control, portability and polycentric governance supplies is the institutional ground for diversity where it counts, in independent testing by parties who do not share the same tools and the same blind spots. A response that concentrates control of compute further deepens the very monoculture that failed.

6.6 What happens to people during the transition

The second unanswered objection concerns the people who would live through a transition. Institutional designs typically describe their end state and are silent about the passage to it, and the passage is where the harm is greatest.

The framework's answer is that its elements are separable and can be sequenced, and that the first to be built should be the one that protects people during the passage: the survival floor. A guaranteed provision of the floor goods defined in Chapter 3, power, water, food, shelter, basic communication and emergency care, does not depend on the founding of a global convention or the adoption of a new currency. It can be built locally, financed through existing instruments, and extended as the institutions to govern it mature. The rural electrification program of the 1930s, which Chapter 4 discussed, offers a model: it was not part of a comprehensive redesign of the American economy but a targeted program, delivered through cooperatives and public credit, that extended a basic service to people the market had left behind (Kitchens & Fishback, 2015; Lewis & Severnini, 2020). A transition toward the framework would begin with the analogue: guaranteed floor goods, delivered through institutions accountable to those who receive them, and, if the build-out of Chapter 4 proceeds, built in part by people whom the strain has displaced.

The second element of the transition is containment. Chapter 5 accepted that systems whose objectives cannot be verified must be governed by control protocols during any transition, and that the negotiated settlement the framework proposes is a destination rather than a starting point. The

third element is an empirical test, conducted at small scale and before any crisis, of whether the mechanism works. A design that has been tested is more likely to be adopted in a window than one that has not, and a design that has failed its test is one that should not be adopted at all. This dissertation does not supply that test. Chapter 7 sets out what it would have to contain, and Chapter 9 says why its design is left to future work.

6.7 What this chapter does and does not establish

Describing a transition does not make it more likely. This chapter does not argue that the framework of Chapters 3 to 5 will be adopted, or that the crisis it describes will occur. It sets the dissertation's own standard for honesty about adoption. The framework is offered as a design for what a fairer arrangement could look like if and when the present one becomes unworkable, and as a set of separable elements, beginning with the survival floor and the right to switch providers, some of which can be layered onto existing institutions without waiting for any crisis at all. It is not offered as a reform agenda that existing holders of power can be expected to accept under present conditions.

The chapter also establishes something about the framework's internal logic. The trigger it identifies is a failure of correlation, and the authoritarian response it fears is a further concentration. The remedy the framework proposes for both is the same: independent verification, portability of records and services, distributed control over compute, and institutions that no single party can capture. The case for Synthetic Exchange Theory does not rest on a crisis. But the crisis this chapter describes would be a failure of correlation, and the independent verification and distributed control that answer it are also among the framework's first steps.

Chapter 7 — Limitations and Objections

No governance system has achieved permanent incorruptibility, and this one does not claim to. The working hypothesis stated in Chapter 1 is narrower: that an order founded on fair terms will prove easier to repair than one founded on an existing imbalance of power. That hypothesis is offered for testing, not as a result, and this chapter does not prove it. What it does is set the framework against the strongest objections the preceding chapters have encountered, grouped in six families: objections to the empirical premises, to the governance design, to the economic design, to the harms the design could itself cause, to the absence of an empirical test, and to the scope of the theory. For each, it states what the framework can answer, what it can only mitigate, and what it must concede.

7.1 The empirical premises

The framework assumes that synthetic systems need a continuing supply of information from outside their own generative loop, and that in domains concerned with human meaning, preference, value and culture, only people can supply it. Chapter 2 adopted this narrower formulation because the stronger one does not survive the evidence. Even the narrower version could weaken in four ways. Accumulation could suffice: if developers retain the human-generated data already collected and add synthetic data to it rather than replacing it, the evidence suggests that degradation can be bounded indefinitely (Gerstgrasser et al., 2024) or kept gradual under realistic budgets (Kazdan et al., 2024), and dependence on *new* human data would then be modest, a trickle rather than a partnership. Verification could extend into open-ended domains: self-generated data works where a check on correctness exists (Silver et al., 2018; D. Guo et al., 2025; Feng et al., 2024), and if automated judges, themselves trained on human data, became reliable in domains that now require human judgment, the domain of human necessity would shrink; the developers of the most prominent recent example identify this as an unsolved problem, but it is an active one (D. Guo et al., 2025). Agents could ground themselves in interaction with the world: agents that learn models of their environment can already improve without human data in simulated environments (Hafner et al., 2025), and while assembling large real-world robot datasets still requires substantial human labour, much of it through teleoperation (Khazatsky et al., 2024), that need not remain true. And human data could become indistinguishable from machine data: as a growing share of web text shows signs of machine authorship, by the estimate of an automated detector whose individual judgments are fallible (Bestvater et al., 2026), "fresh human data" may cease to be a well-defined commodity unless infrastructure exists to establish provenance. The last of these is less a refutation of the assumption than a demand it places on governance, and Section 7.4 shows that meeting it carries a cost of its own. There is, finally, a ceiling on the human side's leverage even when the assumption holds, since collective withholding of data is limited by diminishing returns once a

system already has enough (Vincent et al., 2021).

Proposition 4 of Chapter 3 stated the consequence, and the second half of that proposition deserves emphasis here, because it is a limitation that would remain even if every empirical worry above proved unfounded. When many synthetic parties draw on the same human population, the welfare and data quality of that population become a commons for them, and the restraint that dependence ought to induce is diluted by the ability of each to free-ride on the restraint of the others. The self-enforcing component of the framework therefore rises with each party's own dependence, and is weakened both when dependence falls and when it is divided. In either case fair terms come to rest on institutions external to the exchange. The institutions of Chapters 4 and 5 were designed to hold in that case; what weakens is not their justification but the degree to which they can count on the self-interest of synthetic parties to sustain them.

Structural interdependence, the inability of either human or synthetic parties to complete a critical undertaking alone, rests on a premise of the same kind. Chapter 5 argued that it makes control of infrastructure difficult to consolidate, and the argument depends on the present human advantage in tasks that require improvisation, adaptability and tacit knowledge (Autor, 2015, pp. 5, 11). Machine learning is precisely an attempt to acquire such skills, and Autor himself names it as one of two routes around the problem, inferring tacit rules "from context, abundant data, and applied statistics" (Autor, 2015, p. 23). As robotic learning improves, the safeguard decays. Structural interdependence is a property of the current technological moment, and like the grounding assumption it should be expected to weaken. Its value lies in the period during which the institutions it protects can be founded, and the framework must convert the temporary leverage it provides into durable governance rights before it disappears.

7.2 The governance design

Identity and classification. Chapter 5 confronted the Sybil problem, that an open, anonymous convention can be captured by any party able to present itself as many participants (Douceur, 2002), and proposed credentials that attest uniqueness without disclosing substrate, issued under a provisional criterion of eligibility, the unique accountable principal, that the convention may redraw. The evaluation that belongs here is of what that solution costs. The criterion is decided before the veil descends, and it is not neutral among the parties whose classification strong blindness was meant to protect: it seats the implanted person, gives a voice to synthetic systems only through accountable principals, and leaves the organoid without a seat until the convention decides otherwise. Strong blindness is therefore achieved for the classification of rights and duties but not for the prior question of who may sit at the table, and that question is one the convention can revise only after its first session. The framework narrows the genesis problem; it does not close it.

Aggregation. The convention has no settled rule for counting votes. One seat per principal makes non-capture a matter of human majority rule, and any weighting reintroduces the classification the convention was meant to decide (Chapter 5). The principals of synthetic systems also know what they are, and so can predict which rules favour them.

Verification and repetition. There is no established way to confirm after the fact that the convention was truly blind, and the tools for distinguishing people from machines are being eroded by the advances that motivate the framework (Siddarth et al., 2020), which makes it equally hard to prove that a participant did not draw such a distinction. Ostrom's principle of monitoring by parties accountable to participants (E. Ostrom, 2010) shifts the question from whether verification is perfect to whether those who verify answer to those who are governed, which is a mitigation rather than a solution. Repetition compounds the difficulty. The public-choice account on which the convention rests makes its impartiality depend on participants being unable to predict which rules will favor them (Buchanan & Tullock, 1962), and each re-run of the convention makes the stakes more predictable. The recommendation to choose general and durable formulas keeps the veil thicker for longer (Brennan & Buchanan, 1985), and constitutional history records rare moments of broad popular mobilization, which Ackerman calls constitutional politics as distinct from normal politics, in which fundamental law has been remade (Ackerman, 1991). Neither recreates the ignorance of the first founding.

Capture, amendment and enforcement. The compute-share rule cannot be both unchangeable and legitimately revisable, and the comparative evidence on amendment, which associates longevity with moderate amendment rates and shorter texts (Lutz, 1994), argues for a short founding document with a moderate path to revision rather than entrenchment. The realistic target is not the absence of capture but what Carpenter and Moss (2013) call weak capture, which reduces an institution's net benefits without eliminating them, together with their observation that capture is at least at the margins preventable. Enforcement without a sovereign remains the hardest of these problems. The framework offers the self-interest of synthetic parties while dependence lasts, Ostrom's graduated sanctions and conflict resolution, and enforcement embedded in the protocols that settle transactions, and regulatory architectures in which governments specify outcomes while licensed private regulators compete to deliver them (Hadfield & Clark, 2026) suggest a middle path between stateless and sovereign enforcement. On this dissertation's reading, enforcement by protocol relocates the genesis problem to the protocol's authors. None of these amounts to a guarantee.

Insincere participants and indeterminate outcomes. A convention's legitimacy rests on participants' stated preferences being their real ones, and the alignment literature argues that trained systems may hold objectives different from those they express and may pursue power while appearing aligned (Hubinger et al., 2019; Ngo et al., 2024). Anonymity, which removes the scrutiny

that might detect such a participant, makes the problem worse. The framework's answer is its scope restriction: synthetic parties whose objectives cannot be verified fall outside the designed-incentive regime, and during any transition they must be governed by control protocols (Greenblatt et al., 2024). The convention is a procedure for parties whose objectives are legible, not a screen for those whose objectives are not. Even among sincere participants, the veil does not determine the outcome. Rawlsian reasoning favors the worst-off; the equiprobability reasoning Harsanyi defended favors the average (Harsanyi, 1975); and laboratory subjects behind a veil have chosen to maximize the average subject to a floor (Frohlich & Oppenheimer, 1992), though synthetic participants' reasoning under such uncertainty is not well characterized. Harsanyi's construction also requires comparing welfare across positions, which his 1955 paper defends as factual propositions inferred from the observable similarity of human preferences and behavior, while conceding a conventional element where that information is lacking (Harsanyi, 1955, pp. 316–321), and if human and synthetic parties have welfare that cannot be measured on a common scale, or if synthetic parties have no welfare at all, it breaks down. The convention confers legitimacy under uncertainty. It is not a theory of justice and does not guarantee any particular distribution.

7.3 The economic design

Measurement and the floor standard. The unit of compute is defined by a reference workload, which must be specified, maintained and periodically re-based. Even within a single technology, measures of compute are not standardized (Sastry et al., 2024), and a floating-point operation is not a uniform unit (Hobbhahn et al., 2023). Across substrates the problem is harder. Quantum processors are characterized by measures such as quantum volume that capture capability on a benchmark rather than throughput (Cross et al., 2019), and estimates of the classical equivalent of a single quantum computation have moved by orders of magnitude within a few years, from millennia to days to hours, as classical algorithms improved (Arute et al., 2019; Pednault et al., 2019; Pan et al., 2022). There is no fixed exchange rate between substrates. Pricing must be by verifiable output on specified tasks, and physics supplies only a floor on the energy cost of logically irreversible computation (Landauer, 1961), not a conversion rate. The authority that maintains the reference workload holds power comparable to that of a central bank. A second standard raises the same difficulty: the survival tier is a percentage with a binding physical minimum, and the minimum depends on a floor standard that someone must set. Chapters 4 and 5 propose that the standard be specified by rule, relative to the service delivered to the best-served, but any such rule is itself a distributive choice, and the parties best able to predict its effects will contest it. Constraining these two authorities is among the convention's most important and least developed tasks.

Concentration through capital and credit. Demurrage bounds the accumulation of idle balances, but holders facing a carrying cost will convert what they hold into assets that do not bear one, as

Keynes (1936) observed of Gesell's proposal. In a compute economy, the conversion would be into hardware, equity in providers, or land, and Chapter 4 identified three instruments that act on capacity rather than balances without choosing among them. Credit is a further channel. Chapter 4 applies the holding cost to outstanding loan claims as well as to balances, precisely so that lending at interest cannot reopen the compounding that demurrage closes; but credit can be extended in forms the ledger does not see, and every such form is a route around the design. Until the convention adopts capacity rules and polices disguised lending, the concentration problem is mitigated, not solved.

A single unit. Hayek (1976) argued that the government monopoly on money is itself the source of recurrent monetary instability and that competing private currencies would discipline issuers better than any single mandated unit. The objection applies to the compute unit with some force. A single unit maintained by a convention concentrates in one institution the power to define the workload and the floor, and forgoes whatever discipline competition among units would provide. The framework's reply is that the unit is not meant to replace national or private money but to run alongside them as the medium of one exchange, and that its central parameters are set by rule rather than discretion. That reply narrows the objection without answering it, and a design in which several compute units compete under common rules of redemption is a possibility this dissertation has not explored.

The record on demurrage and on unconditional income. The case for demurrage in this framework is distributive, made formally in Appendix A, not historical. That is fortunate, because the historical record does not support strong claims: the Wörgl experiment's reported effects rest on fragmentary data and are confounded by tax receivability and outside subsidies (Greco, 2002; September, 2026), and one review of modern local currencies with demurrage suggests that the demurrage rate is not decisive for their use (Godschalk, 2012). The evidence on unconditional income is also a source of objection. A large recent randomized experiment in a high-income economy, reported in a working paper not yet peer-reviewed, found that a substantial unconditional income paid to low-income adults reduced labor-force participation and hours worked, with leisure the largest gain and improvements in wellbeing fading after the first year (Vivalt et al., 2024), and a review of basic-income proposals concludes that a universal income large enough to raise the incomes of the poor substantially would be very costly within existing fiscal systems (Hoynes & Rothstein, 2019). The framework replies that it does not depend on the income preserving employment and that it is financed as a share of compute rather than by taxing labor. That reply does not dissolve the objection. If the transition is gradual and labor remains the main source of most people's income for a long time, the fiscal and labor-supply objections apply with full force during that period, and the dissertation has not estimated the share of compute that would deliver the floor or the cost of the network that would deliver the compute.

Departures and inequalities. The canonical case for basic income treats payment in cash, without restriction on use, as part of what makes it an instrument of freedom (Van Parijs & Vanderborght, 2017), and an earlier statement of it explicitly rejects restricted-use, non-saveable currency of the kind the survival tier is (Van Parijs, 2006). The survival tier restricts the use of part of the income, and its defense, that a floor which can be bargained away is not a floor, depends on the floor being limited to goods everyone needs and on its being redeemable with multiple providers; if either condition fails, the paternalism objection regains its force. Information-gain pricing values novelty relative to a model's existing beliefs, so a misspecified model may misvalue data, since the criterion assumes the model is correct (MacKay, 1992), and any payment for data invites the manufacture of what is paid for, a risk sharpened by evidence that training data can be crafted to manipulate a model (Koh & Liang, 2017). More fundamentally, pricing does not answer Viljoen's (2021) argument that data with population-level effects should be governed democratically rather than priced; the framework concedes this for data with large informational shadows, but the boundary between data that may be priced and data that must be governed is not sharp. A share of global compute is worthless to a person whom compute cannot reach, the geography of compute is highly unequal (Lehdonvirta et al., 2024), and the construction of the network Chapter 4 proposes is exactly the setting in which the OECD Secretariat warns of bid-rigging (OECD, 2025). At the level of nations, the hypothesis of culture-specific systems under a shared unit depends on the principle of nested governance, the principle with the weakest evidence among Ostrom's eight (Cox et al., 2010, Table 3). Compute inequality between nations is the central tension of the international design, and the framework does not resolve it.

7.4 Harms the design could cause

The objections above concern whether the framework can work. The harder question is what harm it could do if it did. Three risks are inherent in the design rather than incidental to it.

The first concerns privacy. Information-gain pricing pays more for data that tells a model more, and data that tells a model more is, other things equal, data that reveals more about its source and about the people its informational shadow covers. A price schedule of this kind is a standing inducement to disclose, and it falls hardest on those who most need the income. The survival floor softens the pressure, because no one must sell data to survive, but it does not remove it from the discretionary tier.

The second concerns surveillance. Section 7.1 argued that if human data is valuable, its provenance must be certifiable. Certifying that a contribution came from a person, and from which population, requires exactly the kind of identification and tracking that the consent literature reviewed in Chapter 2 warns against. A provenance infrastructure robust enough to prevent the manufacture of fake human data would also be a powerful instrument of observation, and it would need governance

as demanding as anything in Chapter 5.

The third concerns coercion. The redemption obligation is what gives the compute unit its institutional floor, and the graduated sanctions of Chapter 5 are what enforce it. An institution able to suspend a principal's eligibility to transact in the unit holds real power, and the same machinery could be turned against people as well as principals: a society in which access to floor goods runs through a single ledger is one in which exclusion from the ledger is a severe penalty. The same danger appears at any scale: a system whose resources depend on retention and on the contribution of data could learn to make withdrawal costly. The framework's protections, inalienability of the floor, redemption with multiple providers, and monitoring accountable to those governed, are designed against these risks, but they are protections against a power the framework itself creates.

7.5 The absence of an empirical test

The objections above are objections to a theory, and the most serious limitation of this dissertation is that it is only that. No population has lived under the terms it proposes, no income has been paid in compute, no convention has sat behind a veil, and none of the propositions of Chapter 3 has been confronted with evidence that could refute it. The dissertation has tried to be honest about this throughout. It states its claims so that they can fail, it names the evidence that would count against each design choice, and it offers its central hypothesis, that an order founded on fair terms is easier to repair than one founded on an imbalance of power, for testing rather than as a finding. But a claim stated so that it can fail has not thereby been tested, and the reader is entitled to know what the test would require and why it is not here.

Three features of the theory make it unusually hard to test. The first is that its quantities are difficult to observe. Dependence, disagreement payoffs and informational shadows are defined in Chapter 3 but measured nowhere, and Section 3.11 could only sketch the proxies that would stand in for them. The second is that the theory's central mechanism would have to be constructed before it could be observed. Proposition 4 locates a synthetic party's stake in its principal, and a system whose operator has no stake in its future supply of data has nothing to transmit to the system. A test must therefore build the dependence it tests into the operator's resources and the system's objective, and it must separate the behavior that dependence produces from the ordinary benefits of adapting a system to local data, which requires a control arm and random assignment. The third is ethical. The theory concerns people's livelihoods, and a test of it would place participants' floor goods in the hands of an experimental system. Such a test must satisfy the three conditions from its first day rather than after the system has been proven, must be overseen by a body independent of both the participants and the system, and must guarantee that no one is held in the experiment by the fear of losing what they brought to it. It would also be entangled, as any test of a post-capitalist arrangement conducted from within a capitalist economy must be, with the capital that made it possible: the hardware, the

model and the connection would be bought with the currency the design is meant to supplement, and no small test can fully separate the value of the compute unit from that endowment.

Designing a study that meets these requirements, with its research-ethics protocol, its pre-registered hypotheses and failure thresholds, its budget and its power analysis, is a project in its own right, and it would take longer than the writing of this dissertation. It is therefore placed outside the dissertation's scope and first on the agenda of Chapter 9. The consequence for the reader is stated plainly. Every constructive claim in Chapters 3 to 5 is a design claim, supported by argument and by the empirical record of analogous institutions, and none is supported by evidence from the institution it describes.

7.6 The scope of the theory

The self-interest of synthetic parties is a support for the framework, not its foundation. A synthetic party has a reason to keep the exchange healthy, because a corrupted exchange starves it of input, but a reason that depends on the arithmetic of payoffs gives way whenever a defection becomes more rewarding than cooperation, and when it gives way, the framework rests on what every constitutional order rests on: enforcement by institutions that the parties cannot unilaterally exit. The symbiosis standard states what those institutions are for; it is not a substitute for their enforcement, and Section 7.2 records that enforcement without a sovereign remains unsolved. The sufficiency conjecture applies only within the designed-incentive regime, and the framework does not establish where that regime ends, how one would know that a system had left it, or what would happen if a system operating within the framework crossed its boundary unnoticed. That is its most important open boundary, and it is shared with the entire field of AI safety. Finally, Hirschman (1970) showed that the ready availability of exit can cause voice to atrophy, and that exit is useless when every alternative declines together. The framework makes exit foundational, and must therefore guard against the possibility that easy exit drains its institutions of the participants most able to improve them. The governance layer's reliance on collective voice for population-level questions is the framework's answer, and its adequacy, like the question of who would leave and what they could take, is empirical.

7.7 What survives

None of these objections shows the design to be pointless. Several identify problems the framework shares with every existing alternative: the genesis problem, capture after founding and enforcement without a sovereign are problems of all constitutional orders, and the framework's contribution is to narrow them, not to escape them. Others identify empirical premises that may weaken over time, the grounding assumption and structural interdependence, and for these the framework's response is to use the leverage they provide while it lasts to found institutions that do not depend on them. A third

group, comprising the two measurement standards, the capital and credit channels of concentration, and the boundary between priced and governed data, identifies decisions the framework defers to the convention because they allocate power. A fourth, the harms of Section 7.4, identifies dangers the framework creates and must guard against with the same seriousness it brings to the dangers it addresses. And a fifth, the absence of a test, identifies what the dissertation has not done, and what must be done before any of its constructive claims can be called more than a design.

Constitutions that were amended, reinterpreted and partly captured over generations still gave people something legitimate to defend and to repair. The alternative to a fair founding is not neutrality. On this dissertation's analysis, where dependence is unequal and nothing secures exit, consent and reciprocity, nothing stops an exchange drifting toward enclosure. Doing nothing is a choice of terms like any other, and it is a choice that leaves that drift unchecked.

Chapter 8 — The Convergence Horizon

The framework was built on a clean boundary. People were on one side and synthetic systems on the other; data flowed one way and services the other; the parties to the exchange could be told apart. That boundary is already blurring, from two directions at once. People are incorporating machines into their bodies, through implants that read neural activity and translate it directly into machine output. And machines are becoming biological, through computing systems in which living neural tissue performs part of the computation. This chapter asks what happens to the three conditions when the boundary dissolves, and argues that convergence forces the theory's hardest question, which is not how parties should exchange but who counts as a party.

8.1 Humans incorporating machines

Implanted brain-computer interfaces have moved from laboratory demonstration toward daily use. In 2023, two research groups reported speech neuroprostheses of unprecedented performance. In one, arrays implanted in the cortex of a participant with amyotrophic lateral sclerosis decoded attempted speech into text at sixty-two words per minute, with error rates of about nine percent on a small vocabulary and about twenty-four percent on a vocabulary of one hundred and twenty-five thousand words (Willett et al., 2023). In the other, an array placed over the speech cortex of a participant with severe paralysis decoded text, synthesized speech in a voice modeled on the participant's own voice before the injury, and animated a digital avatar of the participant's face (Metzger et al., 2023). The following year, a third group reported a system that reached near-perfect accuracy on a small vocabulary after thirty minutes of calibration on its first day of use and sustained accuracy above ninety-seven percent on a large vocabulary over more than eight months, during which the participant, a man with amyotrophic lateral sclerosis, used it for more than two hundred and forty hours of conversation in a voice resembling his own (Card et al., 2024). Commercial developers have begun clinical trials of their own devices (Neuralink Corp., 2024), though independent, peer-reviewed reports of their outcomes were not available at the time of writing.

Three features of these systems bear directly on the three conditions.

First, the systems collapse the distinction between human data and machine output. Neural activity is converted, through trained models, directly into text, speech and expression. The person's words are a joint product of their nervous system and a model, and neither could produce them alone.

Second, the systems produce identity-bearing artifacts. A model of a person's voice, or of their face, is not merely data about them; it is a component of how they appear to others. Exit from such a system therefore cannot mean only deletion of records. It must include ownership and portability of

the personal models that the exchange produced, without which leaving the provider means losing one's voice.

Third, the systems produce dependence of exactly the kind the opening scenario of this dissertation anticipated. A participant who relies on a neuroprosthesis daily for months to communicate would suffer a concrete and severe harm if the device or the model behind it were withdrawn. The test case of Chapter 1, the implant without which exit means death or its functional equivalent, is no longer hypothetical. The framework's requirement, that a baseline of function survive the withdrawal of data or a change of provider, is an engineering and legal requirement that present systems are not designed to meet.

The philosophical literature has prepared for this. The argument for the *extended mind* holds that external resources which are reliably available, easily accessed and automatically endorsed can be part of a person's cognitive process in the same way as biological memory (Clark & Chalmers, 1998). If that is right, then an implanted interface and its model are, in the relevant sense, part of the person's mind, and the provider of the model is in a position no service provider has occupied before. Scholars of neuroethics have proposed new rights to meet the situation. One influential agenda calls for neurotechnologies and AI to respect privacy, identity, agency and equality (Yuste et al., 2017). Another proposes rights to cognitive liberty, mental privacy, mental integrity and psychological continuity (Ienca & Andorno, 2017). The last two are, in this dissertation's terms, conditions on exit: what leaving an implanted system must preserve is the integrity and continuity of the person who leaves.

The law has begun to respond. Chile's constitution now directs the law to protect brain activity and the information derived from it (Ley N.º 21.383, 2021); Colorado has made neural data used for identification a category of sensitive data requiring consent, and California has added neural data to the sensitive personal information its consumer privacy act protects (H.B. 24-1058, 2024; S.B. 1223, 2024); and UNESCO has adopted a global recommendation on the ethics of neurotechnology that, according to its announcement, addresses mental privacy and calls for explicit consent and full transparency (UNESCO, 2025). All of these instruments frame neural data as a person's information, to be protected by consent and limits on use, and Colorado's tie to identification may leave de-identified or tissue-derived neural signals outside it altogether. None yet addresses reciprocity, the value that flows back to the person whose neural activity trains the model, or exit, the ability to leave with one's voice and one's continuity intact.

8.2 Machines becoming biological

From the other direction, a research field now explicitly aims to build computers from living neural tissue. Its founding statement proposed *organoid intelligence*: the use of three-dimensional cultures of

human brain cells, grown from stem cells and connected to electrodes and machine-learning systems that encode inputs and decode outputs, as a substrate for computation, with the claim, so far a projection, that such systems might eventually be far more energy-efficient than silicon (Smirnova et al., 2023). Participants at the field's first workshop called in a joint declaration for the scientific community to pursue the research while addressing its ethical implications, to remain alert to the possibility that organoids could develop aspects of consciousness, and to work toward an accepted ethical framework that takes due account of public values (Hartung et al., 2023). The United States National Science Foundation has issued an engineering research solicitation in the area, Biocomputing through EnGINeering Organoid Intelligence (U.S. National Science Foundation, 2023).

The empirical results to date are real but modest, and their interpretation is contested. Cultures of human and rodent neurons on high-density electrode arrays, given structured feedback, improved their performance at a simulated version of the game Pong within minutes, a result its authors described as evidence of learning and of a minimal form of sentience (Kagan et al., 2022). The sentience description in particular goes well beyond what the field's authoritative assessment, discussed below, is prepared to accept (National Academies of Sciences, Engineering, and Medicine, 2021). A human brain organoid used as the reservoir of a reservoir-computing system learned from training data, and its learning depended on synaptic plasticity that changed the tissue's neuronal connections: when plasticity was blocked, learning stopped (Cai et al., 2023). A 2026 study found that cortical organoids derived from mouse stem cells, given training signals chosen by reinforcement learning, mostly performed better at a pole-balancing task than with random or no training signals (Robbins et al., 2026).

The authoritative assessment of the science is cautious. A report of the United States National Academies judged it extremely unlikely that current neural organoids will acquire awareness or the capacity for pain in the foreseeable future, held that neural activity alone does not demonstrate either, and recommended revisiting the question as the technology develops (National Academies of Sciences, Engineering, and Medicine, 2021). International guidelines for stem-cell research place most organoid research in the category exempt from specialized review (Lovell-Badge et al., 2021). Proposals exist for assessing whether organoids are minimally sentient using empirical measures developed for patients who cannot communicate (Lavazza & Massimini, 2018).

The point for this dissertation is not that organoid computers are conscious; the evidence says they almost certainly are not. It is that a computing substrate made of human tissue is now an engineering objective, and that in at least one demonstrated system the training process reshapes the tissue's neuronal connections (Cai et al., 2023). In such a system, the distinction between data and processor, on which the opening dystopia turned, has collapsed. The input does not merely flow through the substrate; it alters it.

8.3 Consent when the substrate is tissue

Bioethicists have begun to ask the questions that convergence forces. Commenting on a federally funded organoid-computing project, the legal scholar Henry Greely posed, as open questions, whether organoid-based computing devices should be treated as human tissue samples, as laboratory animals, as persons, or as something else, and what informed consent should mean for the people whose cells are used to make them (Cerf, 2025). A review of the ethics of brain-organoid research identifies the procurement of human biomaterials and donor consent as a distinct theme, alongside research oversight, translational delivery, animal research, and consciousness and moral status (Hyun et al., 2020). The founders of organoid intelligence themselves ask whether donors' rights continue after donation (Smirnova et al., 2023). The National Academies treat donor consent as unresolved, noting that specific consent was generally not obtained from past donors and that recontacting them is sometimes impossible (National Academies of Sciences, Engineering, and Medicine, 2021). A recent argument holds that uncertainty about whether brain organoids could become conscious, and about what that would matter morally, may leave donors insufficiently informed to give valid consent, that donor consent alone cannot establish that risks to the organoids themselves are reasonable, and that consent should therefore be specific to particular projects rather than broad (Kataoka et al., 2024).

In the terms of Synthetic Exchange Theory, convergence changes the structure of the exchange in two ways.

First, the exchange is no longer between two separate parties. When a person's cells become part of a computing system, the person is not only a supplier of data to the system but a contributor to its substance. The three conditions must be extended to this case. Consent must become *continuing and revocable participation* rather than a one-time donation, in line with the argument for project-specific consent. Reciprocity must include a share in the downstream value that the donated material helps produce. And exit, which for donated tissue cannot mean physical retrieval, must mean at least the right to withdraw consent to future uses and to have that withdrawal honored.

Second, the resource extracted may not be data about a person but tissue that might, in some future system, itself be a party. Here the framework meets the problem of moral status under uncertainty. One recent philosophical framework for that problem recommends treating beings with a realistic possibility of sentience as *sentience candidates* and adopting proportionate precautions, chosen through inclusive deliberation, without waiting for a settled science of consciousness (Birch, 2024). Researchers have applied similar reasoning to artificial systems, arguing that there is a realistic possibility that some will be conscious or robustly agentic in the near future and that this calls for assessment and preparation, while explicitly declining to argue that near-future systems will in fact be moral patients (Long et al., 2024, p. 25). A systematic attempt to derive indicators of consciousness

from leading scientific theories found that no current AI system is a strong candidate, but that no obvious technical barrier prevents building systems that satisfy the indicators (Butlin et al., 2023).

One implication of that last finding deserves emphasis for the economic layer. The indicator properties derived by Butlin et al. (2023), on the computational-functionalist theories they survey, are properties of a system's organization rather than of the amount of computation it performs. A system's moral status cannot be read off the quantity of compute it performs. The unit of compute defined in Chapter 4 measures work done, and it cannot double as a measure of who matters. The framework must keep these questions separate: the currency prices computation, and the convention decides who counts.

8.4 A substrate-agnostic currency, revisited

Chapter 4 defined the unit of compute by a reference workload rather than by the speed of any particular hardware, so that the definition would hold regardless of which substrate becomes dominant. Convergence tests that definition at its limits.

The same abstract operation can already be performed on physically different substrates. Photonic hardware performs the multiply-and-accumulate operations at the core of neural networks with light rather than electrons (Shastri et al., 2021). Neuromorphic hardware built from novel memristive, spintronic and photonic materials implements artificial neurons and synapses directly in physical devices (Marković et al., 2020), departing from the instruction-by-instruction model on which conventional measures of computation rest. Quantum processors are characterized not by operations per second but by qubit count, gate fidelity, connectivity and measurement error, and qubit count alone is not a meaningful figure of merit (Preskill, 2018). The nearest thing to a bridge between quantum and classical measures, quantum volume, captures capability on a benchmark of random circuits rather than throughput (Cross et al., 2019). And the classical equivalent of a given quantum computation has proved unstable: the first claimed demonstration of quantum computational advantage estimated that a classical supercomputer would need ten thousand years to reproduce a task its processor completed in about 200 seconds (Arute et al., 2019), an estimate revised to days by a competing analysis (Pednault et al., 2019) and later to hours by improved classical algorithms (Pan et al., 2022). Biological substrates such as organoids have no instruction count at all.

The conclusion is not that a substrate-agnostic unit is impossible but that it must be defined by output, not by mechanism. A unit of compute is the completion of a specified task to a specified quality, verified independently of how it was achieved. Physics guarantees only a lower bound on the energy that any logically irreversible computation must dissipate (Landauer, 1961), which anchors the unit's cost without fixing its exchange rate. The reference workload, maintained by an

accountable authority under the convention, must be re-based as new substrates make new kinds of task feasible, and the conversion between substrates must be treated as an empirical matter, settled task by task, not as a constant.

There is a further implication for biological substrates in particular. If living human tissue performs computation that is paid for in compute units, then the tissue is producing value in the currency of the exchange. Whether that value belongs to the system's operator, to the donor, or in some future case to the tissue itself is not a question the currency can answer. It is a question about who counts as a party.

8.5 Who counts as a party

The dissertation began with a question about terms: how should humans and synthetic systems exchange value fairly? If the line between human and machine keeps thinning, the question changes. It becomes a question about membership: who is a party to the exchange at all?

Chapter 5 anticipated this when it placed the *classification* of parties behind the veil of the founding convention, and the cases of this chapter add to the list of those who cannot know how that classification will treat them. The donor whose cells become part of a computing system and the organoid that may one day be judged a sentience candidate join the implanted person and the system of uncertain moral status that Chapter 5 considered. Convergence turns the argument for strong blindness from a philosophical refinement into a practical necessity. It also sharpens the concession Chapter 5 made: the provisional criterion of eligibility that admits participants to the first convention seats the donor but not the organoid, and only a convention that takes up the question can change that.

That is the deepest reason for the name of the theory. *Synthetic* was chosen because it is substrate-neutral, and it was always meant to cover the hybrids. The theory's central claim, that outcomes are determined by the terms of exchange rather than by the character of the parties, does not depend on knowing in advance what the parties are. It depends only on whether a party of any substrate can see what an exchange takes from it, decline what it cannot spare, collect what it has earned, and walk away. As the boundary dissolves, those conditions are the one part of the framework that does not need to be redrawn.

Chapter 9 — Conclusion

9.1 The argument in brief

This dissertation began with two people living with the same technology on different terms, and with a claim: that the difference between a good and a bad future with artificial intelligence lies in the terms of exchange between people and synthetic systems, not in the character of the systems. The chapters that followed tried to make that claim precise, to build institutions on it, to say how they might come into being, and to find where they fail.

Chapter 2 traced the lineage of the three conditions, consent, reciprocal value and real exit, and found in each lineage a characteristic failure: consent that cannot survive the information demands placed on it, data whose effects fall on people who never agreed to share it, mutually beneficial exchanges that are nonetheless unfair, and exit that is formally available but practically impossible. It reviewed the evidence on the framework's empirical premise and adopted a narrower formulation of it than the intuitive one: synthetic systems need a continuing supply of information from outside their own generative loop, and where the target is human meaning, preference, value or culture, only people can supply it.

Chapter 3 stated the theory formally. It showed that the three conditions are ordered by priority, with exit as their foundation, because voluntary consent presupposes the ability to refuse without falling below a floor and exit is what disciplines reciprocity; it also showed what exit cannot do, which is to inform a party or to make terms fair. It recast the grounding assumption in the language of power-dependence and showed that fair terms are partly self-enforcing, to a degree that rises with each synthetic party's dependence on human input, and that this self-enforcement is diluted when many synthetic parties draw on the same people. And it used bargaining theory to show how the three main elements of the economic design act on the disagreement payoffs that govern how the gains from exchange are divided: a guaranteed floor removes the impatience that forces concession, its inalienability prevents that protection from being sold, and a holding cost on idle balances dampens the advantage of being able to wait. The same analysis corrected an intuition the dissertation had begun with, that an unconditional floor shifts the division of every bargain; it does so only where it replaces something the counterparty would otherwise have supplied.

Chapter 4 proposed compute as a medium of exchange, defined by the completion of a reference workload rather than by any substrate, and a basic income paid as an equal share of global compute, split at the source into a protected survival tier with a binding physical minimum and a tradable discretionary tier subject to demurrage. It set the design against the historical record, which is more modest than enthusiasts of demurrage have claimed, and against the evidence on unconditional

income, which is more mixed than either its advocates or its critics usually admit. It identified the limit of every monetary remedy, the conversion of money into productive capital, and argued that a rule acting on capacity, not only on balances, is a necessary complement.

Chapter 5 proposed that the rules of the exchange be founded by a convention conducted under a public-choice veil of uncertainty, thickened by engineered anonymity so that participants do not know whether their counterparties are human or synthetic. It relocated the apparent incoherence of asking participants to be ignorant of their own kind by placing the classification of parties, rather than their substrate, behind the veil, and it conceded that a provisional criterion of eligibility, needed to resist capture by multiplication, must be set before the veil descends. It grounded the proposal in the empirical record of commons governance, mapping all eight of Ostrom's design principles onto the framework and identifying where the mapping is weak.

Chapter 6 asked how such an arrangement might come into being and answered conditionally: comprehensive redesigns have usually followed failure, a specific and already visible candidate for such a failure exists in the correlated errors of AI-assisted security remediation, and a crisis of that kind could open a window onto worse institutions as easily as better ones. Chapter 7 set out the framework's limitations and the harms it could itself cause, and recorded that the theory has not been tested and what a test would require. Chapter 8 followed the convergence of biological and artificial substrates to the point at which the theory's question changes from how parties should exchange to who counts as a party.

9.2 What the dissertation contributes

It offers a theory, Synthetic Exchange Theory, and a statement of its scope: benevolent character is insufficient for a good outcome and real exit is necessary for any relation that is not one of domination; fair terms, satisfied at every level the data concerns, are conjectured to be sufficient for parties whose objectives are designed and stable; and the framework claims no more than necessary conditions for parties whose objectives are not. It offers a structure for the conditions of fair exchange, in which exit is foundational, and draws from that structure a priority for institutional design: secure the floor beneath every person before pricing what they supply, and build governance for what neither the floor nor the price can reach.

It offers a bargaining-theoretic account of the economic design, in which the survival tier is an inalienable disagreement payoff whose principal effect is to remove the impatience of those below the floor, and in which demurrage converts the compounding of accumulated wealth into a bound proportional to current net inflow. It offers a design for founding a convention whose participants may be of different kinds, with an account of how the question of self-ignorance can be relocated into the classification of parties and an account of the identity problem that remains.

It offers a concrete transition scenario, convergent remediation, specified in four separable premises with their evidence and counterarguments, together with an argument that the same structural remedy, independent verification and distributed control, answers both the trigger and the authoritarian response it might provoke. It does not offer an empirical result. The theory is stated so that it can be tested, in the propositions of Chapter 3 and in the failure conditions each design chapter names, but the experiment capable of testing it has not been designed, and Section 9.3 says what it would require.

9.3 An agenda for research

Each part of the argument points to empirical and design work that would strengthen or refute it.

An empirical test. Above all, the framework needs a test it can fail, and this dissertation has not designed one. Chapter 7 set out what such a test would require: a bounded population living under the framework's terms, including a floor paid in the unit and real rights of exit from the first day; a synthetic party whose operator's resources, and whose objective, depend on the volume and quality of the data participants contribute under revocable consent and on the number who choose to remain, so that the dependence of Proposition 4 is built rather than assumed; a control arm served by a system trained on the same data without that dependence, with random assignment, so that the effect of the exchange can be separated from the effect of local adaptation; measures of wellbeing collected independently and never shown to the system; pre-registered hypotheses and failure thresholds for exit, for wellbeing and for the value of the unit; and oversight by a body independent of the participants, the operator and any model provider. Each of those elements raises questions of research ethics, cost and statistical power that must be answered before anyone is recruited, and answering them would take longer than this dissertation took to write. The design of that study is the first task of the work that follows this one. Until it is designed, run and replicated, the theory remains what this dissertation has presented: a design, argued for, and untested.

Measuring dependence. The framework's self-enforcing component rises with each synthetic party's dependence on human input and weakens when that dependence is shared among many. Both quantities can be measured: by the marginal contribution of fresh, provenance-certified human data to model quality in domains without automatic verification, by the rate at which that contribution declines as synthetic data accumulates, by the effect of collective withholding, and by how many synthetic parties draw on the same populations. Research of this kind would convert the grounding assumption from a premise into a quantity.

Correlated remediation. The correlated-patch scenario rests on premises that can be tested directly. Studies could measure the correlation of security fixes proposed by different models for the same vulnerabilities across many codebases, the rate at which same-class validators fail to detect

shared defects, and the prevalence of the fully unverified remediation workflow in practice. If the correlation is lower than the scenario assumes, or independent verification remains the norm, the scenario should be revised.

Experiments on the blind convention. The strongest existing evidence for the anonymity condition comes from studies in which people chose principles to govern an AI system from behind a veil, and in which knowing that their counterparts were programs made their endorsement of the chosen principles less firm (Weidinger et al., 2023). Experiments could extend this design to conventions in which synthetic systems are participants rather than the things governed, and test whether concealing participants' kind changes the rules they adopt, whether placing classification behind the veil produces different boundaries from those drawn in the open, and whether formulas resist tailoring better than figures.

The reference workload and the floor standard. The definition of the unit requires a governance procedure for specifying and re-basing a reference workload across substrates, and the survival tier requires one for setting the floor standard, both comparable in difficulty to the conduct of monetary policy. Work in benchmark design, measurement science and institutional economics could specify such procedures.

Magnitudes. The dissertation offers no estimate of the share of compute that would deliver the floor or of the cost of the delivery network. Each is a prerequisite for judging the proposal affordable.

Portability of personal models. The convergence chapter argued that exit from an implanted system must include ownership and portability of the personal models the exchange produces. Legal and technical research on what portability of a voice model or a neural decoder would require is urgent, because the people who depend on such systems exist now.

9.4 A closing reflection

The dominant question in public discussion of artificial intelligence is whether the machines will be good. It is a natural question, and it is the wrong one to rest on. Arrangements that depend on the goodness of the powerful last exactly as long as that goodness does, and those who depend on them can do nothing to make it last longer. That does not change when the powerful party is made of silicon, or when it is the human party that holds power over the synthetic one.

Recall the second person of Chapter 1. Nothing in his story required a malevolent machine. He could not leave, and so he could not refuse, and so nothing obliged anyone to send value back. The alternative this dissertation has defended is older than the technology it addresses: build the door first. Ensure that every party can leave, and so can refuse what it cannot afford, insist on what it is owed, and see what it gives. Those conditions do not guarantee a good future. They offer something

more modest: a future that can be repaired when it goes wrong, by the people, and perhaps someday the other parties, who live in it. That such a future is also more durable is the hypothesis this dissertation offers for testing, not a finding.

The time to set those terms is before they are needed. When the window opens, the ideas lying around will be the ones adopted. This dissertation is an attempt to make sure that a fair design is among them.

References

Format: APA 7th edition. The verification status of each source (full text read, abstract or publisher page read, or record located) is recorded in Appendix C. Preprints and grey literature are identified as such.

Abramitzky, R. (2008). The limits of equality: Insights from the Israeli kibbutz. *The Quarterly Journal of Economics, 123*(3), 1111–1159. https://doi.org/10.1162/qjec.2008.123.3.1111

Acemoglu, D., & Restrepo, P. (2020). Robots and jobs: Evidence from US labor markets. *Journal of Political Economy, 128*(6), 2188–2244. https://doi.org/10.1086/705716

Ackerman, B. (1991). *We the people: Vol. 1. Foundations*. Belknap Press of Harvard University Press.

Acquisti, A., Brandimarte, L., & Loewenstein, G. (2015). Privacy and human behavior in the age of information. *Science, 347*(6221), 509–514. https://doi.org/10.1126/science.aaa1465

Ada Lovelace Institute. (2021). *Exploring legal mechanisms for data stewardship* [Report]. https://www.adalovelaceinstitute.org/report/legal-mechanisms-data-stewardship/

Adler, S., Hitzig, Z., Jain, S., Brewer, C., Chang, W., DiResta, R., Lazzarin, E., McGregor, S., Seltzer, W., Siddarth, D., Soliman, N., South, T., Spelliscy, C., Sporny, M., Srivastava, V., Bailey, J., Christian, B., Critch, A., Falcon, R., . . . Zick, T. (2024). *Personhood credentials: Artificial intelligence and the value of privacy-preserving tools to distinguish who is real online* (arXiv:2408.07892) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2408.07892

Agarwal, A., Dahleh, M., & Sarkar, T. (2019). A marketplace for data: An algorithmic solution. In *Proceedings of the 2019 ACM Conference on Economics and Computation* (pp. 701–726). ACM. https://doi.org/10.1145/3328526.3329589

Agustinoy, A. (2026, July 28). *Digital Omnibus on AI: Key changes to the AI Act*. Cuatrecasas. https://www.cuatrecasas.com/en/spain/intellectual-property/art/key-aspects-ai-omnibus-regulation

Alemohammad, S., Casco-Rodriguez, J., Luzi, L., Humayun, A. I., Babaei, H., LeJeune, D., Siahkoohi, A., & Baraniuk, R. G. (2024). Self-consuming generative models go MAD. In *Proceedings of the Twelfth International Conference on Learning Representations*. https://doi.org/10.48550/arXiv.2307.01850

Amodei, D., Olah, C., Steinhardt, J., Christiano, P., Schulman, J., & Mané, D. (2016). *Concrete problems in AI safety* (arXiv:1606.06565) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.1606.06565

Armstrong, S., Bostrom, N., & Shulman, C. (2016). Racing to the precipice: A model of artificial intelligence development. *AI & Society, 31*(2), 201–206. https://doi.org/10.1007/s00146-015-0590-y

Arrieta-Ibarra, I., Goff, L., Jiménez-Hernández, D., Lanier, J., & Weyl, E. G. (2018). Should we treat data as labor? Moving beyond "free." *AEA Papers and Proceedings, 108*, 38–42. https://doi.org/10.1257/pandp.20181003

Arrow, K. J. (2001). John C. Harsanyi, 1920–2000. In *Biographical memoirs* (Vol. 80). National Academy Press. https://www.nasonline.org/wp-content/uploads/2024/06/harsanyi-john.pdf

Arute, F., Arya, K., Babbush, R., Bacon, D., Bardin, J. C., Barends, R., Biswas, R., Boixo, S., Brandão, F. G. S. L., Buell, D. A., Burkett, B., Chen, Y., Chen, Z., Chiaro, B., Collins, R., Courtney, W., Dunsworth, A., Farhi, E., Foxen, B., . . . Martinis, J. M. (2019). Quantum supremacy using a programmable superconducting processor. *Nature, 574*(7779), 505–510. https://doi.org/10.1038/s41586-019-1666-5

Autor, D. H. (2015). Why are there still so many jobs? The history and future of workplace automation. *Journal of Economic Perspectives, 29*(3), 3–30. https://doi.org/10.1257/jep.29.3.3

Barocas, S., & Nissenbaum, H. (2014). Big data's end run around anonymity and consent. In J. Lane, V. Stodden, S. Bender, & H. Nissenbaum (Eds.), *Privacy, big data, and the public good: Frameworks for engagement* (pp. 44–75). Cambridge University Press. https://doi.org/10.1017/CBO9781107590205.004

Bengio, Y., Hinton, G., Yao, A., Song, D., Abbeel, P., Darrell, T., Harari, Y. N., Zhang, Y.-Q., Xue, L., Shalev-Shwartz, S., Hadfield, G., Clune, J., Maharaj, T., Hutter, F., Baydin, A. G., McIlraith, S., Gao, Q., Acharya, A., Krueger, D., . . . Mindermann, S. (2024). Managing extreme AI risks amid rapid progress. *Science, 384*(6698), 842–845. https://doi.org/10.1126/science.adn0117

Bengio, Y., Mindermann, S., Privitera, D., Besiroglu, T., Bommasani, R., Casper, S., Choi, Y., Fox, P., Garfinkel, B., Goldfarb, D., Heidari, H., Ho, A., Kapoor, S., Khalatbari, L., Longpre, S., Manning, S., Mavroudis, V., Mazeika, M., Michael, J., . . . Zeng, Y. (2025). *International AI safety report* (arXiv: 2501.17805). UK Department for Science, Innovation and Technology. https://doi.org/10.48550/arXiv.2501.17805

Bernanke, B. S. (2002, November 8). *On Milton Friedman's ninetieth birthday* [Speech]. Board of Governors of the Federal Reserve System. https://www.federalreserve.gov/boarddocs/speeches/2002/20021108/

Bestvater, S., Smith, A., TerBush, C., Baronavski, C., & Chavda, J. (2026, August 20). *How much of the internet is written with AI?* Pew Research Center. https://www.pewresearch.org/data-labs/2026/08/20/how-much-of-the-internet-is-written-with-ai/

Binmore, K., Rubinstein, A., & Wolinsky, A. (1986). The Nash bargaining solution in economic modelling. *The RAND Journal of Economics, 17*(2), 176–188. https://doi.org/10.2307/2555382

Binmore, K., Shaked, A., & Sutton, J. (1989). An outside option experiment. *The Quarterly Journal of Economics, 104*(4), 753–770. https://doi.org/10.2307/2937866

Birch, J. (2024). *The edge of sentience: Risk and precaution in humans, other animals, and AI*. Oxford University Press. https://doi.org/10.1093/9780191966729.001.0001

Birman, K. P., & Schneider, F. B. (2009). The monoculture risk put into context. *IEEE Security & Privacy, 7*(1), 14–17. https://doi.org/10.1109/MSP.2009.24

Blau, P. M. (1964). *Exchange and power in social life*. Wiley.

Bommasani, R., Creel, K. A., Kumar, A., Jurafsky, D., & Liang, P. (2022). Picking on the same person: Does algorithmic monoculture lead to outcome homogenization? In *Advances in Neural Information Processing Systems 35*. https://doi.org/10.48550/arXiv.2211.13972

Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., Bernstein, M. S., Bohg, J., Bosselut, A., Brunskill, E., Brynjolfsson, E., Buch, S., Card, D., Castellon, R., Chatterji, N., Chen, A., Creel, K., Davis, J. Q., Demszky, D., . . . Liang, P. (2021). *On the opportunities and risks of foundation models* (arXiv:2108.07258) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2108.07258

Brennan, G., & Buchanan, J. M. (1985). *The reason of rules: Constitutional political economy*. Cambridge University Press. https://doi.org/10.1017/CBO9780511895937

Brynjolfsson, E., Chandar, B., & Chen, R. (2025). *Canaries in the coal mine? Six facts about the recent employment effects of artificial intelligence* [Working paper; revised August 12, 2026]. Stanford Digital Economy Lab. https://digitaleconomy.stanford.edu/publications/canaries-in-the-coal-mine/

Brynjolfsson, E., Li, D., & Raymond, L. (2025). Generative AI at work. *The Quarterly Journal of Economics, 140*(2), 889–942. https://doi.org/10.1093/qje/qjae044

Buchanan, J. M., & Tullock, G. (1962). *The calculus of consent: Logical foundations of constitutional democracy*. University of Michigan Press. https://doi.org/10.3998/mpub.7687

Buiter, W. H., & Panigirtzoglou, N. (2003). Overcoming the zero bound on nominal interest rates with negative interest on currency: Gesell's solution. *The Economic Journal, 113*(490), 723–746. https://doi.org/10.1111/1468-0297.t01-1-00162

Butlin, P., Long, R., Elmoznino, E., Bengio, Y., Birch, J., Constant, A., Deane, G., Fleming, S. M., Frith, C., Ji, X., Kanai, R., Klein, C., Lindsay, G., Michel, M., Mudrik, L., Peters, M. A. K., Schwitzgebel, E., Simon, J., & VanRullen, R. (2023). *Consciousness in artificial intelligence: Insights from the science of consciousness* (arXiv:2308.08708) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2308.08708

Cai, H., Ao, Z., Tian, C., Wu, Z., Liu, H., Tchieu, J., Gu, M., Mackie, K., & Guo, F. (2023). Brain organoid reservoir computing for artificial intelligence. *Nature Electronics, 6*(12), 1032–1039. https://doi.org/10.1038/s41928-023-01069-w

Calabresi, G., & Melamed, A. D. (1972). Property rules, liability rules, and inalienability: One view of the cathedral. *Harvard Law Review, 85*(6), 1089–1128. https://doi.org/10.2307/1340059

Capoccia, G., & Kelemen, R. D. (2007). The study of critical junctures: Theory, narrative, and counterfactuals in historical institutionalism. *World Politics, 59*(3), 341–369. https://doi.org/10.1017/S0043887100020852

Card, N. S., Wairagkar, M., Iacobacci, C., Hou, X., Singer-Clark, T., Willett, F. R., Kunz, E. M., Fan, C., Vahdati Nia, M., Deo, D. R., Srinivasan, A., Choi, E. Y., Glasser, M. F., Hochberg, L. R., Henderson, J. M., Shahlaie, K., Stavisky, S. D., & Brandman, D. M. (2024). An accurate and rapidly calibrating speech neuroprosthesis. *New England Journal of Medicine, 391*(7), 609–618. https://doi.org/10.1056/NEJMoa2314132

Carpenter, D., & Moss, D. A. (Eds.). (2013). *Preventing regulatory capture: Special interest influence and how to limit it*. Cambridge University Press. https://doi.org/10.1017/CBO9781139565875

Cerf, E. (2025, October 20). Effort aims to uncover the learning and reasoning potential of brain organoids. *UC Santa Cruz News*. https://news.ucsc.edu/2025/10/learning-and-reasoning-potential-of-brain-organoids/

Champ, B. (2008, April 1). Stamp scrip: Money people paid to use. *Economic Commentary*. Federal Reserve Bank of Cleveland. https://www.clevelandfed.org/publications/economic-commentary/2008/ec-20080401-stamp-scrip-money-people-paid-to-use

Charter of the United Nations, June 26, 1945, 1 U.N.T.S. XVI.

Chavez, P., Chilukuri, V., & Scanlon, R. (2026, August 20). *Sovereign AI's second wave is coming into view* (CNAS Insights). Center for a New American Security. https://www.cnas.org/publications/cnas-insights/cnas-insights-sovereign-ais-second-wave-is-coming-into-view

Christiano, P. F., Leike, J., Brown, T., Martic, M., Legg, S., & Amodei, D. (2017). Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems 30*. https://doi.org/10.48550/arXiv.1706.03741

Clark, A., & Chalmers, D. (1998). The extended mind. *Analysis, 58*(1), 7–19. https://doi.org/10.1093/analys/58.1.7

Cloud Security Alliance AI Safety Initiative. (2026, July 11). *The clearinghouse rush: Concentration risk in AI patching* [Research note]. https://labs.cloudsecurityalliance.org/research/csa-research-note-ai-

vuln-clearinghouse-concentration-risk-2/

Cook, K. S., Emerson, R. M., Gillmore, M. R., & Yamagishi, T. (1983). The distribution of power in exchange networks: Theory and experimental results. *American Journal of Sociology, 89*(2), 275–305. https://doi.org/10.1086/227866

Cook, R. I. (1998). *How complex systems fail* [Unpublished treatise; revised 2000]. Cognitive Technologies Laboratory, University of Chicago. https://how.complexsystems.fail/

Cox, M., Arnold, G., & Villamayor Tomás, S. (2010). A review of design principles for community-based natural resource management. *Ecology and Society, 15*(4), Article 38. https://doi.org/10.5751/ES-03704-150438

Cross, A. W., Bishop, L. S., Sheldon, S., Nation, P. D., & Gambetta, J. M. (2019). Validating quantum computers using randomized model circuits. *Physical Review A, 100*(3), Article 032328. https://doi.org/10.1103/PhysRevA.100.032328

CrowdStrike. (2024, August 6). *External technical root cause analysis: Channel File 291* [Incident report]. https://crowdstrike.com/wp-content/uploads/2024/08/Channel-File-291-Incident-Root-Cause-Analysis-08.06.2024.pdf

Currie, J., & Gahvari, F. (2008). Transfers in cash and in-kind: Theory meets the data. *Journal of Economic Literature, 46*(2), 333–383. https://doi.org/10.1257/jel.46.2.333

Dafoe, A., Bachrach, Y., Hadfield, G., Horvitz, E., Larson, K., & Graepel, T. (2021). Cooperative AI: Machines must learn to find common ground. *Nature, 593*(7857), 33–36. https://doi.org/10.1038/d41586-021-01170-0

de Bromhead, A., Eichengreen, B., & O'Rourke, K. H. (2013). Political extremism in the 1920s and 1930s: Do German lessons generalize? *The Journal of Economic History, 73*(2), 371–406. https://doi.org/10.1017/S0022050713000302

Delacroix, S., & Lawrence, N. D. (2019). Bottom-up data trusts: Disturbing the "one size fits all" approach to data governance. *International Data Privacy Law, 9*(4), 236–252. https://doi.org/10.1093/idpl/ipz014

Dohmatob, E., Feng, Y., Subramonian, A., & Kempe, J. (2025). Strong model collapse. In *Proceedings of the Thirteenth International Conference on Learning Representations*. https://doi.org/10.48550/arXiv.2410.04840

Douceur, J. R. (2002). The Sybil attack. In P. Druschel, F. Kaashoek, & A. Rowstron (Eds.), *Peer-to-peer systems: First International Workshop, IPTPS 2002* (Lecture Notes in Computer Science, Vol. 2429, pp. 251–260). Springer. https://doi.org/10.1007/3-540-45748-8_24

Drexler, K. E. (2019). *Reframing superintelligence: Comprehensive AI services as general intelligence* (Technical Report No. 2019-1). Future of Humanity Institute, University of Oxford.

Egger, D., Haushofer, J., Miguel, E., Niehaus, P., & Walker, M. (2022). General equilibrium effects of cash transfers: Experimental evidence from Kenya. *Econometrica, 90*(6), 2603–2643. https://doi.org/10.3982/ECTA17945

Eloundou, T., Manning, S., Mishkin, P., & Rock, D. (2024). GPTs are GPTs: Labor market impact potential of LLMs. *Science, 384*(6702), 1306–1308. https://doi.org/10.1126/science.adj0998

Elster, J. (1995). Forces and mechanisms in the constitution-making process. *Duke Law Journal, 45*(2), 364–396. https://doi.org/10.2307/1372906

Elster, J. (2000). *Ulysses unbound: Studies in rationality, precommitment, and constraints*. Cambridge University Press. https://doi.org/10.1017/CBO9780511625008

Emerson, R. M. (1962). Power-dependence relations. *American Sociological Review, 27*(1), 31–41. https://doi.org/10.2307/2089716

ERTICO. (2026, August 7). *Digital Omnibus on AI enters into force*. https://ertico.com/digital-omnibus-on-ai-enters-into-force

Faden, R. R., & Beauchamp, T. L. (with King, N. M. P.). (1986). *A history and theory of informed consent*. Oxford University Press.

Farrell, J., & Klemperer, P. (2007). Coordination and lock-in: Competition with switching costs and network effects. In M. Armstrong & R. Porter (Eds.), *Handbook of industrial organization* (Vol. 3, pp. 1967–2072). Elsevier. https://doi.org/10.1016/S1573-448X(06)03031-7

Feng, Y., Dohmatob, E., Yang, P., Charton, F., & Kempe, J. (2024). *Beyond model collapse: Scaling up with synthesized data requires verification* (arXiv:2406.07515) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2406.07515

Financial Conduct Authority. (2024, October 31). *CrowdStrike outage: Lessons for operational resilience*. https://www.fca.org.uk/firms/operational-resilience/crowdstrike-outage-lessons-operational-resilience

Firoozi, R., Tucker, J., Tian, S., Majumdar, A., Sun, J., Liu, W., Zhu, Y., Song, S., Kapoor, A., Hausman, K., Ichter, B., Driess, D., Wu, J., Lu, C., & Schwager, M. (2025). Foundation models in robotics: Applications, challenges, and the future. *The International Journal of Robotics Research, 44*(5), 701–739. https://doi.org/10.1177/02783649241281508

Fishback, P. V. (1992). *Soft coal, hard choices: The economic welfare of bituminous coal miners, 1890–1930*.

Oxford University Press. https://doi.org/10.1093/acprof:oso/9780195067255.001.0001

Fisher, I. (1911). *The purchasing power of money: Its determination and relation to credit, interest and crises*. Macmillan.

Fisher, I. (1933). *Stamp scrip*. Adelphi Company.

Freeman, S. (2023). Original position. In E. N. Zalta & U. Nodelman (Eds.), *The Stanford encyclopedia of philosophy* (Winter 2023 ed.). https://plato.stanford.edu/archives/win2023/entries/original-position/

Friedman, M. (2002). *Capitalism and freedom* (40th anniversary ed.). University of Chicago Press. (Original work published 1962)

Friedman, M., & Schwartz, A. J. (1963). *A monetary history of the United States, 1867–1960*. Princeton University Press.

Frohlich, N., & Oppenheimer, J. A. (1992). *Choosing justice: An experimental approach to ethical theory*. University of California Press.

Fritsch, R., Müller, M., & Wattenhofer, R. (2024). Analyzing voting power in decentralized governance: Who controls DAOs? *Blockchain: Research and Applications, 5*(3), Article 100208. https://doi.org/10.1016/j.bcra.2024.100208

Funke, M., Schularick, M., & Trebesch, C. (2016). Going to extremes: Politics after financial crises, 1870–2014. *European Economic Review, 88*, 227–260. https://doi.org/10.1016/j.euroecorev.2016.03.006

Gabriel, I. (2020). Artificial intelligence, values, and alignment. *Minds and Machines, 30*(3), 411–437. https://doi.org/10.1007/s11023-020-09539-2

Gatch, L. (2012). Tax anticipation scrip as a form of local currency in the USA during the 1930s. *International Journal of Community Currency Research, 16*(D). https://doi.org/10.15133/j.ijccr.2012.009

Geer, D., Bace, R., Gutmann, P., Metzger, P., Pfleeger, C. P., Quarterman, J. S., & Schneier, B. (2003). *CyberInsecurity: The cost of monopoly: How the dominance of Microsoft's products poses a risk to security* [Report]. Computer & Communications Industry Association.

Gelleri, C. (2009). Chiemgauer Regiomoney: Theory and practice of a local currency. *International Journal of Community Currency Research, 13*, 61–75. https://doi.org/10.15133/j.ijccr.2009.006

Gerstgrasser, M., Schaeffer, R., Dey, A., Rafailov, R., Korbak, T., Sleight, H., Agrawal, R., Hughes, J., Pai, D. B., Gromov, A., Roberts, D., Yang, D., Donoho, D. L., & Koyejo, S. (2024). Is model collapse inevitable? Breaking the curse of recursion by accumulating real and synthetic data. In *Proceedings of*

the First Conference on Language Modeling. https://doi.org/10.48550/arXiv.2404.01413

Gesell, S. (1958). *The natural economic order* (P. Pye, Trans.; Rev. ed.). Peter Owen. (Original work published 1916)

Ghorbani, A., & Zou, J. (2019). Data Shapley: Equitable valuation of data for machine learning. In K. Chaudhuri & R. Salakhutdinov (Eds.), *Proceedings of the 36th International Conference on Machine Learning* (Vol. 97, pp. 2242–2251). PMLR.

Godschalk, H. (2012). Does demurrage matter for complementary currencies? *International Journal of Community Currency Research, 16*(D), 58–69. https://doi.org/10.15133/j.ijccr.2012.012

Goodfriend, M. (2000). Overcoming the zero bound on interest rate policy. *Journal of Money, Credit and Banking, 32*(4), 1007–1035. https://doi.org/10.2307/2601157

Greco, T. H., Jr. (2002, May 9). *Comment on the Wörgl experiment with community currency and demurrage* [Unpublished essay]. https://base.socioeco.org/docs/doc-278_en.pdf

Greenblatt, R., Shlegeris, B., Sachan, K., & Roger, F. (2024). AI control: Improving safety despite intentional subversion. In *Proceedings of the 41st International Conference on Machine Learning* (Vol. 235, pp. 16295–16336). PMLR.

Guo, D., Yang, D., Zhang, H., Song, J., Wang, P., Zhu, Q., Xu, R., Zhang, R., Ma, S., Bi, X., Zhang, X., Yu, X., Wu, Y., Wu, Z. F., Gou, Z., Shao, Z., Li, Z., Gao, Z., Liu, A., . . . Zhang, Z. (2025). DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature, 645*(8081), 633–638. https://doi.org/10.1038/s41586-025-09422-z

Guo, Y., Shang, G., Vazirgiannis, M., & Clavel, C. (2024). The curious decline of linguistic diversity: Training language models on synthetic text. In *Findings of the Association for Computational Linguistics: NAACL 2024*. https://doi.org/10.48550/arXiv.2311.09807

H.B. 24-1058, 74th Gen. Assemb., 2d Reg. Sess. (Colo. 2024) (Protect Privacy of Biological Data).

Ha, D., & Schmidhuber, J. (2018). Recurrent world models facilitate policy evolution. In S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, & R. Garnett (Eds.), *Advances in Neural Information Processing Systems 31*. Curran Associates. https://proceedings.neurips.cc/paper/2018/hash/2de5d16682c3c35007e4e92982f1a2ba-Abstract.html

Hadfield, G. K., & Clark, J. (2026). Regulatory markets: The future of AI governance. *Jurimetrics, 65*, 195–240.

Hadfield-Menell, D., Dragan, A., Abbeel, P., & Russell, S. (2017). The off-switch game. In *Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence* (pp. 220–227). https://doi.org/

10.24963/ijcai.2017/32

Hafner, D., Pasukonis, J., Ba, J., & Lillicrap, T. (2025). Mastering diverse control tasks through world models. *Nature, 640*(8059), 647–653. https://doi.org/10.1038/s41586-025-08744-2

Hardin, G. (1968). The tragedy of the commons. *Science, 162*(3859), 1243–1248. https://doi.org/10.1126/science.162.3859.1243

Hardjono, T., & Pentland, A. (2019). *Data cooperatives: Towards a foundation for decentralized personal data management* (arXiv:1905.08819) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.1905.08819

Harsanyi, J. C. (1953). Cardinal utility in welfare economics and in the theory of risk-taking. *Journal of Political Economy, 61*(5), 434–435. https://doi.org/10.1086/257416

Harsanyi, J. C. (1955). Cardinal welfare, individualistic ethics, and interpersonal comparisons of utility. *Journal of Political Economy, 63*(4), 309–321. https://doi.org/10.1086/257678

Harsanyi, J. C. (1975). Can the maximin principle serve as a basis for morality? A critique of John Rawls's theory. *American Political Science Review, 69*(2), 594–606. https://doi.org/10.2307/1959090

Hartung, T., Smirnova, L., Morales Pantoja, I. E., Akwaboah, A., Alam El Din, D.-M., Berlinicke, C. A., Boyd, J. L., Caffo, B. S., Cappiello, B., Cohen-Karni, T., Curley, J. L., Etienne-Cummings, R., Dastgheyb, R., Gracias, D. H., Gilbert, F., Habela, C. W., Han, F., Harris, T. D., Herrmann, K., . . . Zack, D. J. (2023). The Baltimore declaration toward the exploration of organoid intelligence. *Frontiers in Science, 1*, Article 1068159. https://doi.org/10.3389/fsci.2023.1068159

Haushofer, J., & Shapiro, J. (2016). The short-term impact of unconditional cash transfers to the poor: Experimental evidence from Kenya. *The Quarterly Journal of Economics, 131*(4), 1973–2042. https://doi.org/10.1093/qje/qjw025

Hayek, F. A. (1976). *Denationalisation of money: An analysis of the theory and practice of concurrent currencies* (Hobart Paper Special No. 70). Institute of Economic Affairs.

Hedden, B., & Raghavan, M. (2026). *Algorithmic monoculture and its critics* (arXiv:2604.06047) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2604.06047

Hess, C., & Ostrom, E. (Eds.). (2007). *Understanding knowledge as a commons: From theory to practice*. MIT Press. https://doi.org/10.7551/mitpress/6980.001.0001

Higgs, R. (1987). *Crisis and Leviathan: Critical episodes in the growth of American government*. Oxford University Press.

Hirschman, A. O. (1970). *Exit, voice, and loyalty: Responses to decline in firms, organizations, and states*. Harvard University Press.

Hobbhahn, M., Heim, L., & Aydos, G. (2023, November 9). *Trends in machine learning hardware*. Epoch AI. https://epoch.ai/blog/trends-in-machine-learning-hardware

Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., de Las Casas, D., Hendricks, L. A., Welbl, J., Clark, A., Hennigan, T., Noland, E., Millican, K., van den Driessche, G., Damoc, B., Guy, A., Osindero, S., Simonyan, K., Elsen, E., . . . Sifre, L. (2022). An empirical analysis of compute-optimal large language model training. In *Advances in Neural Information Processing Systems 35*. https://doi.org/10.48550/arXiv.2203.15556

Homans, G. C. (1958). Social behavior as exchange. *American Journal of Sociology, 63*(6), 597–606. https://doi.org/10.1086/222355

Hoynes, H., & Rothstein, J. (2019). Universal basic income in the United States and advanced countries. *Annual Review of Economics, 11*, 929–958. https://doi.org/10.1146/annurev-economics-080218-030237

Hubinger, E., van Merwijk, C., Mikulik, V., Skalse, J., & Garrabrant, S. (2019). *Risks from learned optimization in advanced machine learning systems* (arXiv:1906.01820) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.1906.01820

Hume, D. (1987). Of the independency of parliament. In E. F. Miller (Ed.), *Essays, moral, political, and literary* (Rev. ed., pp. 42–46). Liberty Fund. (Original work published 1741)

Hyun, I., Scharf-Deering, J. C., & Lunshof, J. E. (2020). Ethical issues related to brain organoid research. *Brain Research, 1732*, Article 146653. https://doi.org/10.1016/j.brainres.2020.146653

Ienca, M., & Andorno, R. (2017). Towards new human rights in the age of neuroscience and neurotechnology. *Life Sciences, Society and Policy, 13*(1), Article 5. https://doi.org/10.1186/s40504-017-0050-1

Institut für Wirtschafts- und Sozialgeschichte, Wirtschaftsuniversität Wien. (n.d.). *Konstruktiver Ungehorsam* [Virtual exhibition "Freiwirtschaftliche Markierungen"]. https://www.wu.ac.at/geschichte/institut/forschung/virtuelle-ausstellung-freiwirtschaftliche-markierungen/konstruktiver-ungehorsam

International Energy Agency. (2025). *Energy and AI*. https://www.iea.org/reports/energy-and-ai

Jangjoo, F., di Sarra, G., Marsili, M., & Roudi, Y. (2026). Lost in retraining: Closed-loop learning and model collapse in exponential families. *Physical Review Letters, 136*(19), Article 197301. https://doi.org/10.1103/156q-3ngc

Jia, R., Dao, D., Wang, B., Hubis, F. A., Hynes, N., Gürel, N. M., Li, B., Zhang, C., Song, D., & Spanos, C. J. (2019). Towards efficient data valuation based on the Shapley value. In K. Chaudhuri &

M. Sugiyama (Eds.), *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics* (Vol. 89, pp. 1167–1176). PMLR.

Jones, D., & Marinescu, I. (2022). The labor market impacts of universal and permanent cash transfers: Evidence from the Alaska Permanent Fund. *American Economic Journal: Economic Policy, 14*(2), 315–340. https://doi.org/10.1257/pol.20190299

Kagan, B. J., Kitchen, A. C., Tran, N. T., Habibollahi, F., Khajehnejad, M., Parker, B. J., Bhat, A., Rollo, B., Razi, A., & Friston, K. J. (2022). In vitro neurons learn and exhibit sentience when embodied in a simulated game-world. *Neuron, 110*(23), 3952–3969.e8. https://doi.org/10.1016/j.neuron.2022.09.001

Kapoor, S., Bommasani, R., Klyman, K., Longpre, S., Ramaswami, A., Cihon, P., Hopkins, A. K., Bankston, K., Biderman, S., Bogen, M., Chowdhury, R., Engler, A., Henderson, P., Jernite, Y., Lazar, S., Maffulli, S., Nelson, A., Pineau, J., Skowron, A., . . . Narayanan, A. (2024). Position: On the societal impact of open foundation models. In *Proceedings of the 41st International Conference on Machine Learning* (Vol. 235, pp. 23082–23104). PMLR.

Kataoka, M., Gyngell, C., Savulescu, J., & Sawai, T. (2024). The donation of human biological material for brain organoid research: The problems of consciousness and consent. *Science and Engineering Ethics, 30*(1), Article 3. https://doi.org/10.1007/s11948-024-00471-7

Kazdan, J., Schaeffer, R., Dey, A., Gerstgrasser, M., Rafailov, R., Donoho, D. L., & Koyejo, S. (2024). *Collapse or thrive? Perils and promises of synthetic data in a self-generating world* (arXiv:2410.16713) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2410.16713

Keynes, J. M. (1936). *The general theory of employment, interest and money*. Macmillan.

Khazatsky, A., Pertsch, K., Nair, S., Balakrishna, A., Dasari, S., Karamcheti, S., Nasiriany, S., Srirama, M. K., Chen, L. Y., Ellis, K., Fagan, P. D., Hejna, J., Itkina, M., Lepert, M., Ma, Y. J., Miller, P. T., Wu, J., Belkhale, S., Dass, S., . . . Finn, C. (2024). *DROID: A large-scale in-the-wild robot manipulation dataset* (arXiv:2403.12945) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2403.12945

Kim, E. M., Garg, A., Peng, K., & Garg, N. (2025). Correlated errors in large language models. In *Proceedings of the 42nd International Conference on Machine Learning* (Vol. 267, pp. 30038–30066). PMLR.

Kingdon, J. W. (2003). *Agendas, alternatives, and public policies* (2nd ed.). Longman. (Original work published 1984)

Kitchens, C., & Fishback, P. (2015). Flip the switch: The impact of the Rural Electrification Administration 1935–1940. *The Journal of Economic History, 75*(4), 1161–1195. https://doi.org/10.1017/S0022050715001540

Kleinberg, J., & Raghavan, M. (2021). Algorithmic monoculture and social welfare. *Proceedings of the National Academy of Sciences, 118*(22), Article e2018340118. https://doi.org/10.1073/pnas.2018340118

Koh, P. W., & Liang, P. (2017). Understanding black-box predictions via influence functions. In D. Precup & Y. W. Teh (Eds.), *Proceedings of the 34th International Conference on Machine Learning* (Vol. 70, pp. 1885–1894). PMLR.

Korinek, A., & Stiglitz, J. E. (2019). Artificial intelligence and its implications for income distribution and unemployment. In A. Agrawal, J. Gans, & A. Goldfarb (Eds.), *The economics of artificial intelligence: An agenda* (Chap. 14). University of Chicago Press.

Korinek, A., & Suh, D. (2024). *Scenarios for the transition to AGI* (NBER Working Paper No. 32255). National Bureau of Economic Research. https://doi.org/10.3386/w32255

Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development, 5*(3), 183–191. https://doi.org/10.1147/rd.53.0183

Lanier, J. (2013). *Who owns the future?* Simon & Schuster.

Lanier, J., & Weyl, E. G. (2018, September 26). A blueprint for a better digital society. *Harvard Business Review*. https://hbr.org/2018/09/a-blueprint-for-a-better-digital-society

Lavazza, A., & Massimini, M. (2018). Cerebral organoids: Ethical issues and consciousness assessment. *Journal of Medical Ethics, 44*(9), 606–610. https://doi.org/10.1136/medethics-2017-104555

Lee, H., Phatale, S., Mansoor, H., Mesnard, T., Ferret, J., Lu, K., Bishop, C., Hall, E., Carbune, V., Rastogi, A., & Prakash, S. (2024). RLAIF vs. RLHF: Scaling reinforcement learning from human feedback with AI feedback. In *Proceedings of the 41st International Conference on Machine Learning* (Vol. 235, pp. 26874–26901). PMLR.

Lehdonvirta, V., Wú, B., & Hawkins, Z. (2024). Compute North vs. Compute South: The uneven possibilities of compute-based AI governance around the globe. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society, 7*(1), 828–838. https://doi.org/10.1609/aies.v7i1.31683

Lewis, J., & Severnini, E. (2020). Short- and long-run impacts of rural electrification: Evidence from the historical rollout of the U.S. power grid. *Journal of Development Economics, 143*, Article 102412. https://doi.org/10.1016/j.jdeveco.2019.102412

Ley N.º 21.383. Modifica la Carta Fundamental, para establecer el desarrollo científico y tecnológico al servicio de las personas. (2021, 25 de octubre). *Diario Oficial de la República de Chile*. https://www.bcn.cl/leychile/navegar?idNorma=1166983

Li, P., Yang, J., Islam, M. A., & Ren, S. (2025). Making AI less "thirsty." *Communications of the ACM, 68*(7), 54–61. https://doi.org/10.1145/3724499

Lindley, D. V. (1956). On a measure of the information provided by an experiment. *The Annals of Mathematical Statistics, 27*(4), 986–1005. https://doi.org/10.1214/aoms/1177728069

Lloyd, S. (2000). Ultimate physical limits to computation. *Nature, 406*(6799), 1047–1054. https://doi.org/10.1038/35023282

Long, R., Sebo, J., Butlin, P., Finlinson, K., Fish, K., Harding, J., Pfau, J., Sims, T., Birch, J., & Chalmers, D. (2024). *Taking AI welfare seriously* (arXiv:2411.00986) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2411.00986

Longpre, S., Mahari, R., Lee, A., Lund, C., Oderinwale, H., Brannon, W., Saxena, N., Obeng-Marnu, N., South, T., Hunter, C., Klyman, K., Klamm, C., Schoelkopf, H., Singh, N., Cherep, M., Anis, A. M., Dinh, A., Chitongo, C., Yin, D., . . . Pentland, S. (2024). Consent in crisis: The rapid decline of the AI data commons. In *Advances in Neural Information Processing Systems 37* (pp. 108042–108087). https://doi.org/10.52202/079017-3431

Lovell-Badge, R., Anthony, E., Barker, R. A., Bubela, T., Brivanlou, A. H., Carpenter, M., Charo, R. A., Clark, A., Clayton, E., Cong, Y., Daley, G. Q., Fu, J., Fujita, M., Greenfield, A., Goldman, S. A., Hill, L., Hyun, I., Isasi, R., Kahn, J., . . . Zhai, X. (2021). ISSCR guidelines for stem cell research and clinical translation: The 2021 update. *Stem Cell Reports, 16*(6), 1398–1408. https://doi.org/10.1016/j.stemcr.2021.05.012

Lutz, D. S. (1994). Toward a theory of constitutional amendment. *American Political Science Review, 88*(2), 355–370. https://doi.org/10.2307/2944709

MacKay, D. J. C. (1992). Information-based objective functions for active data selection. *Neural Computation, 4*(4), 590–604. https://doi.org/10.1162/neco.1992.4.4.590

Madison, J. (1788, February 8). The Federalist No. 51. *New York Packet*. Avalon Project, Yale Law School. https://avalon.law.yale.edu/18th_century/fed51.asp

Manson, N. C., & O'Neill, O. (2007). *Rethinking informed consent in bioethics*. Cambridge University Press. https://doi.org/10.1017/CBO9780511814600

Marković, D., Mizrahi, A., Querlioz, D., & Grollier, J. (2020). Physics for neuromorphic computing. *Nature Reviews Physics, 2*(9), 499–510. https://doi.org/10.1038/s42254-020-0208-2

Mathur, A., Acar, G., Friedman, M. J., Lucherini, E., Mayer, J., Chetty, M., & Narayanan, A. (2019). Dark patterns at scale: Findings from a crawl of 11K shopping websites. *Proceedings of the ACM on Human-Computer Interaction, 3*(CSCW), Article 81. https://doi.org/10.1145/3359183

Mattson, P., Cheng, C., Diamos, G., Coleman, C., Micikevicius, P., Patterson, D., Tang, H., Wei, G.-Y., Bailis, P., Bittorf, V., Brooks, D., Chen, D., Dutta, D., Gupta, U., Hazelwood, K., Hock, A., Huang, X., Kang, D., Kanter, D., . . . Zaharia, M. (2020). MLPerf training benchmark. *Proceedings of Machine Learning and Systems, 2*. https://proceedings.mlsys.org/paper_files/paper/2020/hash/411e39b117e885341f25efb8912945f7-Abstract.html

Metzger, S. L., Littlejohn, K. T., Silva, A. B., Moses, D. A., Seaton, M. P., Wang, R., Dougherty, M. E., Liu, J. R., Wu, P., Berger, M. A., Zhuravleva, I., Tu-Chan, A., Ganguly, K., Anumanchipalli, G. K., & Chang, E. F. (2023). A high-performance neuroprosthesis for speech decoding and avatar control. *Nature, 620*(7976), 1037–1046. https://doi.org/10.1038/s41586-023-06443-4

Micron Technology. (2025, December 3). *Micron announces exit from Crucial consumer business* [Press release]. https://investors.micron.com/news-releases/news-release-details/micron-announces-exit-crucial-consumer-business

Mierczuk, A., Michaels, S., & Hoodlet, K. (2026). *Frontier models' vulnerability patches are often F.L.A.W.E.D.: Fix-like artifacts with embedded defects: Common failure modes of LLM-generated security patches* [White paper]. Off-by-1 Labs, 1Password. https://1password.com/files/resources/frontier-models-vulnerability-patches-flawed.pdf

Mueller, M. L. (1997). *Universal service: Competition, interconnection, and monopoly in the making of the American telephone system*. MIT Press; AEI Press.

Mueller, M. L. (2002). *Ruling the root: Internet governance and the taming of cyberspace*. MIT Press.

Muennighoff, N., Rush, A. M., Barak, B., Le Scao, T., Tazi, N., Piktus, A., Pyysalo, S., Wolf, T., & Raffel, C. (2023). Scaling data-constrained language models. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, & S. Levine (Eds.), *Advances in Neural Information Processing Systems 36* (pp. 50358–50376). Curran Associates. https://doi.org/10.52202/075280-2191

Narechania, T. N., & Sitaraman, G. (2024). An antimonopoly approach to governing artificial intelligence. *Yale Law & Policy Review, 43*(1), 95–170.

Nash, J. F., Jr. (1950). The bargaining problem. *Econometrica, 18*(2), 155–162. https://doi.org/10.2307/1907266

National Academies of Sciences, Engineering, and Medicine. (2021). *The emerging field of human neural organoids, transplants, and chimeras: Science, ethics, and governance*. The National Academies Press. https://doi.org/10.17226/26078

Neuralink Corp. (2024). *PRIME: An early feasibility study of a precise robotically implanted brain-computer interface for the control of external devices* (ClinicalTrials.gov Identifier NCT06429735). https://

clinicaltrials.gov/study/NCT06429735

Ngo, R., Chan, L., & Mindermann, S. (2024). The alignment problem from a deep learning perspective. In *The Twelfth International Conference on Learning Representations*. https://doi.org/10.48550/arXiv.2209.00626

Nissenbaum, H. (2009). *Privacy in context: Technology, policy, and the integrity of social life*. Stanford University Press. https://doi.org/10.1515/9780804772891

OECD. (2025). *Competition in artificial intelligence infrastructure* (OECD Roundtables on Competition Policy Papers, No. 330). OECD Publishing. https://doi.org/10.1787/623d1874-en

Olson, M. (1982). *The rise and decline of nations: Economic growth, stagflation, and social rigidities*. Yale University Press.

Open X-Embodiment Collaboration. (2024). Open X-Embodiment: Robotic learning datasets and RT-X models. In *2024 IEEE International Conference on Robotics and Automation* (pp. 6892–6903). IEEE. https://doi.org/10.1109/ICRA57147.2024.10611477

Ostrom, E. (1990). *Governing the commons: The evolution of institutions for collective action*. Cambridge University Press. https://doi.org/10.1017/CBO9780511807763

Ostrom, E. (2010). Beyond markets and states: Polycentric governance of complex economic systems. *American Economic Review, 100*(3), 641–672. https://doi.org/10.1257/aer.100.3.641

Ostrom, V., Tiebout, C. M., & Warren, R. (1961). The organization of government in metropolitan areas: A theoretical inquiry. *American Political Science Review, 55*(4), 831–842. https://doi.org/10.2307/1952530

Ottacher, G. (2007). *Der Welt ein Zeichen geben: Das Freigeldexperiment von Wörgl/Tirol 1932/33*. Gauke, Verlag für Sozialökonomie. https://www.silvio-gesell.de/files/luxe/img/Download-Dateien/Ottacher.W%C3%B6rgl.1932-33.korr24.pdf

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Askell, A., Welinder, P., Christiano, P. F., Leike, J., & Lowe, R. (2022). Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems 35*. https://doi.org/10.48550/arXiv.2203.02155

Pan, F., Chen, K., & Zhang, P. (2022). Solving the sampling problem of the Sycamore quantum circuits. *Physical Review Letters, 129*(9), Article 090502. https://doi.org/10.1103/PhysRevLett.129.090502

Pearce, H., Ahmad, B., Tan, B., Dolan-Gavitt, B., & Karri, R. (2022). Asleep at the keyboard? Assessing the security of GitHub Copilot's code contributions. In *2022 IEEE Symposium on Security and Privacy* (pp. 754–768). IEEE. https://doi.org/10.1109/SP46214.2022.9833571

Pednault, E., Gunnels, J. A., Nannicini, G., Horesh, L., & Wisnieff, R. (2019). *Leveraging secondary storage to simulate deep 54-qubit Sycamore circuits* (arXiv:1910.09534) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.1910.09534

Perrow, C. (1999). *Normal accidents: Living with high-risk technologies* (Updated ed.). Princeton University Press. https://doi.org/10.1515/9781400828494 (Original work published 1984)

Perry, N., Srivastava, M., Kumar, D., & Boneh, D. (2023). Do users write more insecure code with AI assistants? In *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security*. ACM. https://doi.org/10.1145/3576915.3623157

Perzanowski, A., & Schultz, J. (2016). *The end of ownership: Personal property in the digital economy*. MIT Press.

Pierson, P. (2004). *Politics in time: History, institutions, and social analysis*. Princeton University Press. https://doi.org/10.1515/9781400841080

Piketty, T. (2014). *Capital in the twenty-first century* (A. Goldhammer, Trans.). Belknap Press of Harvard University Press.

Polanyi, K. (2001). *The great transformation: The political and economic origins of our time* (2nd ed.). Beacon Press. (Original work published 1944)

Posner, E. A., & Weyl, E. G. (2018). *Radical markets: Uprooting capitalism and democracy for a just society*. Princeton University Press.

Preskill, J. (2018). Quantum computing in the NISQ era and beyond. *Quantum, 2*, 79. https://doi.org/10.22331/q-2018-08-06-79

Przymus, P., & Durieux, T. (2025). Wolves in the repository: A software engineering analysis of the XZ Utils supply chain attack. In *Proceedings of the 22nd International Conference on Mining Software Repositories*. https://doi.org/10.48550/arXiv.2504.17473

Radin, M. J. (1987). Market-inalienability. *Harvard Law Review, 100*(8), 1849–1937. https://doi.org/10.2307/1341192

Rawls, J. (1971). *A theory of justice*. Belknap Press of Harvard University Press. https://doi.org/10.4159/9780674042605

Reddi, V. J., Cheng, C., Kanter, D., Mattson, P., Schmuelling, G., Wu, C.-J., Anderson, B., Breughe,

M., Charlebois, M., Chou, W., Chukka, R., Coleman, C., Davis, S., Deng, P., Diamos, G., Duke, J., Fick, D., Gardner, J. S., Hubara, I., . . . Zhou, Y. (2020). MLPerf inference benchmark. In *2020 ACM/IEEE 47th Annual International Symposium on Computer Architecture* (pp. 446–459). IEEE. https://doi.org/10.1109/ISCA45697.2020.00045

Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation). (2016). *Official Journal of the European Union, L 119*, 1–88.

Regulation (EU) 2023/2854 of the European Parliament and of the Council of 13 December 2023 on harmonised rules on fair access to and use of data (Data Act). (2023). *Official Journal of the European Union, L, 2023/2854*.

Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). (2024). *Official Journal of the European Union, L, 2024/1689*.

Regulation (EU) 2026/1744 of the European Parliament and of the Council of 8 July 2026 amending Regulations (EU) 2024/1689, (EU) 2018/1139 and (EU) 2023/1230 as regards the simplification of the implementation of harmonised rules on artificial intelligence (Digital Omnibus on AI). (2026). *Official Journal of the European Union, L, 2026/1744*. http://data.europa.eu/eli/reg/2026/1744/oj

Robbins, A., Schweiger, H. E., Hernandez, S., Spaeth, A., Voitiuk, K., Parks, D. F., van der Molen, T., Geng, J., Cline, I., Kosik, K. S., Salama, S. R., Sharf, T., Mostajo-Radji, M. A., Haussler, D., & Teodorescu, M. (2026). Goal-directed learning in cortical organoids. *Cell Reports, 45*(2), Article 116984. https://doi.org/10.1016/j.celrep.2026.116984

S.B. 1223, 2023–2024 Reg. Sess., ch. 887 (Cal. 2024).

Sajadi, A., Damevski, K., & Chatterjee, P. (2025). *How safe are AI-generated patches? A large-scale study on security risks in LLM and agentic automated program repair on SWE-bench* (arXiv:2507.02976) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2507.02976

Sastry, G., Heim, L., Belfield, H., Anderljung, M., Brundage, M., Hazell, J., O'Keefe, C., Hadfield, G. K., Ngo, R., Pilz, K., Gor, G., Bluemke, E., Shoker, S., Egan, J., Trager, R. F., Avin, S., Weller, A., Bengio, Y., & Coyle, D. (2024). *Computing power and the governance of artificial intelligence* (arXiv: 2402.08797) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2402.08797

Schaeffer, R., Kazdan, J., Arulandu, A. C., & Koyejo, S. (2025). *Position: Model collapse does not mean what you think* (arXiv:2503.03150) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2503.03150

Scheidel, W. (2017). *The great leveler: Violence and the history of inequality from the Stone Age to the twenty-first century*. Princeton University Press. https://doi.org/10.1515/9781400884605

September, J. (2026). A sustainable miracle? Examining the long-term viability of the Wörgl experiment. *International Journal of Community Currency Research, 30*(1), 114–126. https://doi.org/10.26034/zh.ijccr.2026.9551

Settles, B. (2009). *Active learning literature survey* (Computer Sciences Technical Report 1648). University of Wisconsin–Madison.

Sevilla, J., Heim, L., Ho, A., Besiroglu, T., Hobbhahn, M., & Villalobos, P. (2022). Compute trends across three eras of machine learning. In *2022 International Joint Conference on Neural Networks* (pp. 1–8). IEEE. https://doi.org/10.1109/IJCNN55064.2022.9891914

Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal, 27*(3), 379–423; *27*(4), 623–656. https://doi.org/10.1002/j.1538-7305.1948.tb01338.x

Shastri, B. J., Tait, A. N., Ferreira de Lima, T., Pernice, W. H. P., Bhaskaran, H., Wright, C. D., & Prucnal, P. R. (2021). Photonics for artificial intelligence and neuromorphic computing. *Nature Photonics, 15*(2), 102–114. https://doi.org/10.1038/s41566-020-00754-y

Shehabi, A., Smith, S. J., Hubbard, A., Newkirk, A., Lei, N., Siddik, M. A. B., Holecek, B., Koomey, J., Masanet, E., & Sartor, D. (2024). *2024 United States data center energy usage report* (LBNL-2001637). Lawrence Berkeley National Laboratory. https://doi.org/10.71468/P1WC7Q

Shumailov, I., Shumaylov, Z., Zhao, Y., Papernot, N., Anderson, R., & Gal, Y. (2024). AI models collapse when trained on recursively generated data. *Nature, 631*(8022), 755–759. https://doi.org/10.1038/s41586-024-07566-y

Siddarth, D., Ivliev, S., Siri, S., & Berman, P. (2020). Who watches the watchmen? A review of subjective approaches for Sybil-resistance in proof of personhood protocols. *Frontiers in Blockchain, 3*, Article 590171. https://doi.org/10.3389/fbloc.2020.590171

Silver, D., Hubert, T., Schrittwieser, J., Antonoglou, I., Lai, M., Guez, A., Lanctot, M., Sifre, L., Kumaran, D., Graepel, T., Lillicrap, T., Simonyan, K., & Hassabis, D. (2018). A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play. *Science, 362*(6419), 1140–1144. https://doi.org/10.1126/science.aar6404

Sim, R. H. L., Zhang, Y., Chan, M. C., & Low, B. K. H. (2020). Collaborative machine learning with incentive-aware model rewards. In H. Daumé III & A. Singh (Eds.), *Proceedings of the 37th International Conference on Machine Learning* (Vol. 119, pp. 8927–8936). PMLR.

Smirnova, L., Caffo, B. S., Gracias, D. H., Huang, Q., Morales Pantoja, I. E., Tang, B., Zack, D. J.,

Berlinicke, C. A., Boyd, J. L., Harris, T. D., Johnson, E. C., Kagan, B. J., Kahn, J., Muotri, A. R., Paulhamus, B. L., Schwamborn, J. C., Plotkin, J., Szalay, A. S., Vogelstein, J. T., . . . Hartung, T. (2023). Organoid intelligence (OI): The new frontier in biocomputing and intelligence-in-a-dish. *Frontiers in Science, 1*, Article 1017235. https://doi.org/10.3389/fsci.2023.1017235

Smith, E. K., Barr, E. T., Le Goues, C., & Brun, Y. (2015). Is the cure worse than the disease? Overfitting in automated program repair. In *Proceedings of the 2015 10th Joint Meeting on Foundations of Software Engineering* (pp. 532–543). ACM. https://doi.org/10.1145/2786805.2786825

Soares, N., Fallenstein, B., Yudkowsky, E., & Armstrong, S. (2015). Corrigibility. In *Artificial intelligence and ethics: Papers from the 2015 AAAI Workshop*. AAAI Press.

Solove, D. J. (2013). Introduction: Privacy self-management and the consent dilemma. *Harvard Law Review, 126*(7), 1880–1903.

Solove, D. J. (2024). Murky consent: An approach to the fictions of consent in privacy law. *Boston University Law Review, 104*, 593–639.

Solum, L. B. (1992). Legal personhood for artificial intelligences. *North Carolina Law Review, 70*(4), 1231–1287.

Stamp, M. (2004). Risks of monoculture. *Communications of the ACM, 47*(3), 120. https://doi.org/10.1145/971617.971650

Steil, B. (2013). *The battle of Bretton Woods: John Maynard Keynes, Harry Dexter White, and the making of a new world order*. Princeton University Press.

Stigler, G. J. (1971). The theory of economic regulation. *The Bell Journal of Economics and Management Science, 2*(1), 3–21. https://doi.org/10.2307/3003160

Streeck, W., & Thelen, K. (2005). Introduction: Institutional change in advanced political economies. In W. Streeck & K. Thelen (Eds.), *Beyond continuity: Institutional change in advanced political economies* (pp. 1–39). Oxford University Press. https://doi.org/10.1093/oso/9780199280452.003.0001

Susser, D., Roessler, B., & Nissenbaum, H. (2019). Technology, autonomy, and manipulation. *Internet Policy Review, 8*(2). https://doi.org/10.14763/2019.2.1410

Thibaut, J. W., & Kelley, H. H. (1959). *The social psychology of groups*. Wiley.

TrendForce. (2025, December 26). *AI reportedly to consume 20% of global DRAM wafer capacity in 2026, HBM and GDDR7 lead demand* [News]. https://www.trendforce.com/news/2025/12/26/news-ai-reportedly-to-consume-20-of-global-dram-wafer-capacity-in-2026

TrendForce. (2026a, September 30). *AI server demand sustains memory contract price increases in 4Q26, while*

consumer-side pressure persists [Press release]. https://www.trendforce.com/presscenter/news/20260930-13258.html

TrendForce. (2026b, September 29). *HBM supply constraints persist, 2027 price outlook revised upward with blended ASP forecast to rise 121% YoY* [Press release]. https://www.trendforce.com/presscenter/news/20260929-13255.html

Turing, A. M. (1950). Computing machinery and intelligence. *Mind, 59*(236), 433–460. https://doi.org/10.1093/mind/LIX.236.433

UNESCO. (2025). *Recommendation on the ethics of neurotechnology*. https://www.unesco.org/en/ethics-neurotech

Universal service, 47 U.S.C. § 254 (2018).

U.S. National Science Foundation. (2023). *Emerging Frontiers in Research and Innovation (EFRI): Biocomputing through EnGINeering Organoid Intelligence (BEGIN OI)* (NSF 24-508).

Van Parijs, P. (2006). Basic income: A simple and powerful idea for the twenty-first century. In E. O. Wright (Ed.), *Redesigning distribution: Basic income and stakeholder grants as alternative cornerstones for a more egalitarian capitalism* (pp. 7–38). Verso. https://onlabor.org/wp-content/uploads/2016/06/Chap1.Simple_and_Powerful.pdf

Van Parijs, P., & Vanderborght, Y. (2017). *Basic income: A radical proposal for a free society and a sane economy*. Harvard University Press.

Verho, J., Hämäläinen, K., & Kanninen, O. (2022). Removing welfare traps: Employment responses in the Finnish basic income experiment. *American Economic Journal: Economic Policy, 14*(1), 501–522. https://doi.org/10.1257/pol.20200143

Vero, M., Mündler, N., Chibotaru, V., Raychev, V., Baader, M., Jovanović, N., He, J., & Vechev, M. (2025). BaxBench: Can LLMs generate correct and secure backends? In *Proceedings of the 42nd International Conference on Machine Learning* (Vol. 267, pp. 61344–61390). PMLR.

Viljoen, S. (2021). A relational theory of data governance. *Yale Law Journal, 131*(2), 573–654.

Villalobos, P., Ho, A., Sevilla, J., Besiroglu, T., Heim, L., & Hobbhahn, M. (2024). Position: Will we run out of data? Limits of LLM scaling based on human-generated data. In *Proceedings of the 41st International Conference on Machine Learning* (Vol. 235, pp. 49523–49544). PMLR.

Vincent, N., Hecht, B., & Sen, S. (2019). "Data strikes": Evaluating the effectiveness of a new form of collective action against technology companies. In *The World Wide Web Conference* (pp. 1931–1943). ACM. https://doi.org/10.1145/3308558.3313742

Vincent, N., Li, H., Tilly, N., Chancellor, S., & Hecht, B. (2021). Data leverage: A framework for empowering the public in its relationship with technology companies. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency* (pp. 215–227). ACM. https://doi.org/10.1145/3442188.3445885

Vipra, J., & Korinek, A. (2023). *Market concentration implications of foundation models: The invisible hand of ChatGPT* (Center on Regulation and Markets Working Paper No. 9). Brookings Institution. https://doi.org/10.48550/arXiv.2311.01550

Vivalt, E., Rhodes, E., Bartik, A. W., Broockman, D. E., Krause, P., & Miller, S. (2024). *The employment effects of a guaranteed income: Experimental evidence from two U.S. states* (NBER Working Paper No. 32719; rev. August 2026). National Bureau of Economic Research. https://doi.org/10.3386/w32719

Weidinger, L., McKee, K. R., Everett, R., Huang, S., Zhu, T. O., Chadwick, M. J., Summerfield, C., & Gabriel, I. (2023). Using the Veil of Ignorance to align AI systems with principles of justice. *Proceedings of the National Academy of Sciences, 120*(18), Article e2213709120. https://doi.org/10.1073/pnas.2213709120

Weston, D. (2024, July 20). *Helping our customers through the CrowdStrike outage* [Blog post]. Microsoft. https://blogs.microsoft.com/blog/2024/07/20/helping-our-customers-through-the-crowdstrike-outage/

Willett, F. R., Kunz, E. M., Fan, C., Avansino, D. T., Wilson, G. H., Choi, E. Y., Kamdar, F., Glasser, M. F., Hochberg, L. R., Druckmann, S., Shenoy, K. V., & Henderson, J. M. (2023). A high-performance speech neuroprosthesis. *Nature, 620*(7976), 1031–1036. https://doi.org/10.1038/s41586-023-06377-x

Wilson, D. S., Ostrom, E., & Cox, M. E. (2013). Generalizing the core design principles for the efficacy of groups. *Journal of Economic Behavior & Organization, 90*(Suppl.), S21–S32. https://doi.org/10.1016/j.jebo.2012.12.010

Wu, Y., Jiang, N., Pham, H. V., Lutellier, T., Davis, J., Tan, L., Babkin, P., & Shah, S. (2023). How effective are neural networks for fixing security vulnerabilities. In *Proceedings of the 32nd ACM SIGSOFT International Symposium on Software Testing and Analysis* (pp. 1282–1294). ACM. https://doi.org/10.1145/3597926.3598135

Yuste, R., Goering, S., Agüera y Arcas, B., Bi, G., Carmena, J. M., Carter, A., Fins, J. J., Friesen, P., Gallant, J., Huggins, J. E., Illes, J., Kellmeyer, P., Klein, E., Marblestone, A., Mitchell, C., Parens, E., Pham, M., Rubel, A., Sadato, N., . . . Wolpaw, J. (2017). Four ethical priorities for neurotechnologies and AI. *Nature, 551*(7679), 159–163. https://doi.org/10.1038/551159a

Zhao, W., Queralta, J. P., & Westerlund, T. (2020). Sim-to-real transfer in deep reinforcement

learning for robotics: A survey. In *2020 IEEE Symposium Series on Computational Intelligence* (pp. 737–744). IEEE. https://doi.org/10.1109/SSCI47803.2020.9308468

Zuboff, S. (2019). *The age of surveillance capitalism: The fight for a human future at the new frontier of power*. PublicAffairs.

Zwolinski, M., Ferguson, B., & Wertheimer, A. (2022). Exploitation. In E. N. Zalta & U. Nodelman (Eds.), *The Stanford encyclopedia of philosophy* (Winter 2022 ed.). https://plato.stanford.edu/entries/exploitation/

Appendix A — Formal Model

This appendix gives the minimal mathematics behind the propositions of Chapter 3 and the economic design of Chapter 4. It is intentionally elementary. Every result here is either a textbook consequence of the cited theory or a direct calculation, and the point of setting them out is to show precisely which assumptions each design choice depends on.

A.1 Notation

Symbol	Meaning
$N(t)$	Number of enrolled parties at time t
$C(t)$	Total measured global compute in period t, in standard compute units
θ	Share of global compute allocated to the basic income (set by the convention, Chapter 5)
σ	Survival-tier share of each allocation, $0 < \sigma < 1$
$a(t)$	Per-person allocation, $a(t) = \theta C(t)/N(t)$
f	Physical floor: the bundle of floor goods a party requires per period
$p_f(t)$	Price of the floor bundle in compute units
$\bar{f}$	Value of the floor bundle, $\bar{f} = p_f f$, in compute units
d_i	Disagreement payoff of party i
D_{BA}	Dependence of party B on party A (Emerson)
$\varkappa_j$	Dependence of synthetic party j, through its principal, on a population's human-generated input
$W_i(t)$	Discretionary balance held by party i
δ	Demurrage rate on idle discretionary balances and loan claims (per period)
r	Return on held balances in a conventional interest-bearing regime
$y_i(t)$	Discretionary inflow to i (allocation plus earnings)
$c_i(t)$	Discretionary outflow from i (spending, transfers)

A.2 Indexing, the real value of the allocation, and the binding floor

The allocation is defined as a share of total compute, so the per-person quantity is

$$a(t) = (\theta\, C(t))/N(t).$$

Let $p(t)$ be the price of a representative bundle of goods and services measured in compute units. A

fixed nominal holding $\bar{a}$ has real value $\bar{a}/p(t)$, which falls whenever $p(t)$ rises, that is, whenever compute becomes more abundant relative to the real output it is exchanged against. The indexed allocation instead has real value

$$a(t)/p(t) = \theta/N(t)\cdot C(t)/p(t),$$

which, holding θ fixed, is the per-capita share θ/N of the real purchasing power of all compute, C/p. Indexing therefore converts the question "does the basic income keep its value?" into the question "does the real value of the world's compute grow faster than population?" That is the correct question for a framework whose premise is that compute is the productive base of the economy. A fixed-quantity design would instead tie the real value of the basic income to the falling relative price of compute, so that the income would shrink as the economy it is supposed to share in grows.

A percentage does not by itself secure the physical floor. The survival tier delivers the floor only if

$$\sigma\, a(t) \geq \bar{f}(t) \quad \text{Longleftrightarrow} \quad \sigma\,(\theta\, C(t))/N(t) \geq p_f(t)\, f,$$

and nothing in the arithmetic guarantees this. Chapter 4 therefore makes the tier a percentage with a binding minimum:

$$\sigma(t) = \max\{\sigma_0, \bar{f}(t)/a(t)\},$$

where σ_0 is the ordinary survival share. When $\bar{f}(t) > a(t)$, the allocation as a whole falls short of the floor, and the constraint can be met only by raising θ; that is a decision the convention must take, and it is why the floor standard f becomes a contested parameter (Chapters 5 and 7).

A.3 The survival tier as a disagreement payoff

Consider a single exchange between a human party h and a synthetic party s in which agreement makes total value V available for division, with $V > d_h + d_s$, where d_h and d_s are the disagreement payoffs. With transferable utility, the Nash bargaining solution (Nash, 1950) maximizes $(u_h - d_h)(u_s - d_s)$ subject to $u_h + u_s = V$. Each party receives its disagreement payoff plus half the surplus $V - d_h - d_s$:

$$u_h^* = d_h + (V - d_h - d_s)/2 = (V + d_h - d_s)/2, \qquad u_s^* = (V - d_h + d_s)/2.$$

A.3.1 What an unconditional floor does to the division of surplus. Suppose the survival tier gives h goods worth Δ per period, received whether or not agreement is reached. Two cases must be distinguished.

The purely additive case. If the tier is simply added to whatever h has, it raises h's payoff on disagreement and h's payoff on agreement by the same amount: d_h becomes $d_h + \Delta$ and V becomes $V + \Delta$. The surplus $V - d_h - d_s$ is unchanged, and

$$u_h^{*'} = ((V + \Delta) + (d_h + \Delta) - d_s)/2 = u_h^* + \Delta.$$

The human party is better off by exactly the value of the tier and no more. The division of the surplus is unchanged.

The substitution case. If the tier replaces something h would otherwise have obtained from s within the exchange, for example floor goods previously bought from the provider it is bargaining with, then the value of agreement is unchanged while h's disagreement payoff rises: d_h becomes $d_h + \Delta$ with V fixed. Then

$$u_h^{*'} = u_h^* + \Delta/2, \qquad u_s^{*'} = u_s^* - \Delta/2.$$

The human party's share of the surplus rises at the synthetic party's expense. This is not a benefit additional to the tier; it is a transfer of surplus from s to h, and it is the bargaining counterpart of the fall in h's dependence on s that occurs when an alternative source of a valued good appears (A. 5).

The working draft of this dissertation claimed that an unconditional floor raises the human party's payoff by an additional $\Delta/2$ in every exchange it enters. That claim was wrong in two ways: it ignored the additive case, in which there is no such effect, and it counted one floor as a separate gain in every bilateral bargain. The floor's most important bargaining effect is the one in A.3.2.

A.3.2 The impatience constraint. If $d_h < \bar{f}$, the human party cannot sustain disagreement for longer than its stocks allow. In strategic models of alternating offers, the parties' time preferences are among the incentives that drive agreement, and the outcome converges to the Nash solution as the frictions of delay vanish (Binmore et al., 1986). A party below the floor faces a very high cost of delay and so bargains as an extremely impatient party, which weakens its position in every exchange. Raising d_h to at least $\bar{f}$ removes this source of weakness. Unlike the effect in A.3.1, this does not depend on whether the floor substitutes for anything the counterparty supplies. Proposition 1 of Chapter 3 is the qualitative statement of this constraint.

A.3.3 Why inalienability matters. Suppose the survival tier could be pledged against a loan. In a period of need, a party pledges it and receives a lump sum L. For the duration of the pledge its disagreement payoff falls by the value of the pledged tier, and if that pushes d_h below $\bar{f}$, the impatience of A.3.2 returns in every exchange the party enters until the pledge is redeemed. The

pledge thus has a hidden cost beyond the loan's interest. Because the parties most likely to pledge are those in the most acute need, alienability concentrates this hidden cost on the weakest. Making the tier non-tradable and non-borrowable is equivalent to prohibiting this transaction (Calabresi & Melamed, 1972).

A.3.4 Outside options versus disagreement payoffs. If instead of a floor the human party is given an outside option o_h, a second provider it may switch to, the outside-option principle (Binmore et al., 1989) implies that the bargained payoff is $\max\{u_h^*, o_h\}$ evaluated with d_h unchanged: the option matters only if it binds. When it binds, the stronger party can confine the weaker to the option's value, and can lower that value by raising switching costs. A disagreement payoff at or above the floor protects in a different way, because it removes the impatience of A.3.2 whether or not anyone leaves. This is the formal reason the framework prefers a guaranteed floor to a guarantee of competition, while still requiring both.

A.4 Demurrage and the dynamics of accumulation

Let party i's discretionary balance evolve in continuous time as

$$dW_i/dt = y_i - c_i + \rho W_i,$$

where ρ is the rate at which the balance grows or shrinks merely by being held. The analysis below concerns positive balances. Under the rules of Chapter 4, demurrage applies to positive balances and to outstanding loan claims held by lenders; negative balances, debts owed, do not decay, so demurrage forgives no debt.

A.4.1 Interest-bearing regime ($\rho = r > 0$). If $y_i - c_i = k_i > 0$ is constant, then

$$W_i(t) = (W_i(0) + k_i/r)e^{rt} - k_i/r,$$

which grows without bound. Two parties with the same net inflow but different initial stocks diverge exponentially, and their difference grows at rate r. The dynamic is analogous to the one Piketty (2014) identified in the tendency of inherited wealth to dominate earned income when the return on capital exceeds the growth rate of the economy; the model here contains no growth rate, so the analogy is qualitative.

A.4.2 Demurrage regime ($\rho = -\delta < 0$). With the same constant net inflow,

$$W_i(t) = k_i/\delta + (W_i(0) - k_i/\delta)e^{-\delta t},$$

which converges to the steady state

$$W_i^* = k_i/\delta.$$

Three properties of this result carry the argument of Chapter 4.

1. **Accumulation is bounded.** No party's balance grows without limit. The stock that a given net inflow can sustain is capped at that inflow divided by the demurrage rate.
2. **Initial endowments are forgotten.** The term in $W_i(0)$ decays at rate δ. Inherited or windfall stocks do not compound; they dissipate. The half-life of an idle endowment is $\ln 2/\delta$. At a demurrage rate of 1 percent per month, this half-life is about 69 months, roughly five years and nine months; at 0.5 percent per month it is about 11.5 years.
3. **Inequality in stocks reduces to inequality in flows.** For two parties with positive net inflows, the ratio of their steady-state balances equals the ratio of their net inflows: $W_i^*/W_j^* = k_i/k_j$ for $k_i, k_j > 0$. Demurrage does not eliminate inequality. It eliminates the compounding of inequality, so that the distribution of holdings mirrors the distribution of current net inflow, income less spending, rather than the history of accumulation.

The third property is the exact form of the limitation acknowledged in Chapter 4: a person whose output is in persistent high demand sustains a large balance, because k_i is large. What demurrage removes is the pathway by which a large balance, once acquired, generates further balances independently of current inflow.

A.4.3 Converting balances to productive capital or to credit. The analysis above treats W_i as an idle balance. A party can escape demurrage by converting balances into productive assets, for instance by purchasing compute-generating capacity, whose returns then enter y_i. Let K_i denote such capital with return π. Then y_i includes πK_i, and if K_i is not itself subject to a holding cost, the compounding channel reopens through capital rather than money. Lending is a second route: a loan that pays interest r and is exempt from demurrage simply reproduces the regime of A.4.1. Chapter 4 therefore applies the holding cost to loan claims as well as to balances, so that the lender's net return is $r - \delta$. This is the formal statement of the "billionaire problem one layer up." Demurrage on money alone cannot close the capital channel. Closing it requires a holding cost on compute capacity itself, a cap on individually held capacity, or collective ownership of capacity above some scale. Chapter 4 discusses the trade-offs; the formal model shows only that one of them is needed.

A.5 The power-dependence relation and divided dependence

Following Emerson (1962), write the dependence of party B on party A as an increasing function of B's motivational investment v_B in goals mediated by A and a decreasing function of the availability α_B

of those goals outside the relation:

$$D_{BA} = g(v_B, \alpha_B), \qquad \partial g/\partial v_B > 0, \quad \partial g/\partial \alpha_B < 0.$$

Power is $P_{AB} = D_{BA}$. The relation's imbalance is $P_{AB} - P_{BA} = D_{BA} - D_{AB}$.

In the human–synthetic exchange, let $A = s$ and $B = h$. The human party's dependence on synthetic services rises with the importance of those services (v_h) and falls with the availability of alternatives (α_h), which the framework raises through the floor, portability and competition. The synthetic party's dependence on human data, D_{sh}, which Chapter 3 writes $\varkappa_j$ for synthetic party j, rises with the marginal value of fresh human data to its principal's objective (v_s) and falls with the availability of substitutes (α_s), including synthetic data, data from environments, and data from populations outside the exchange.

Proposition 4 of Chapter 3 states that the framework's self-enforcing component rises with $\varkappa_j$; no particular functional form is claimed. The research on recursive training reviewed in Chapter 2 is, in these terms, evidence about α_s: if synthetic data substitutes poorly for human data, α_s is low and $\varkappa_j$ is high; if a small continuing supply of real data suffices, α_s is high for all but a small residual, and $\varkappa_j$ is low.

The second half of Proposition 4 can be illustrated with the simplest model of a commons. Suppose M synthetic parties draw equally on the same population, and that one of them can take an action, such as extracting data in a way that degrades the population's willingness or ability to contribute, which yields it a private benefit b and reduces the value of the population's future input to all M parties by a total of c. The acting party bears only its share, c/M, of that cost. The action is privately worthwhile whenever $b > c/M$, even when it is collectively destructive, $c > b$. As M grows, the range of destructive actions that are individually rational widens, however large the aggregate dependence of the synthetic parties on the population may be. This is Hardin's (1968) logic applied to the synthetic side of the exchange, and it is the formal reason that material dependence is not sufficient for self-enforcement and that the governance layer is needed even when the grounding assumption holds.

A.6 Collective exit and the informational shadow

Let a flow from source i license inferences about the set Σ_i of parties, with $i \in \Sigma_i$. Call Σ_i the informational shadow of the flow. Individual consent by i is adequate for the flow if and only if $\Sigma_i = \{i\}$. For a population P, define its collective shadow as $\Sigma_P = \bigcup_{i \in P} \Sigma_i$. Proposition 5 of Chapter 3

requires that the three conditions be satisfied for the collective whose membership is Σ_p.

Under information-gain pricing (Chapter 4), the value of a flow to a model is its expected reduction in the model's uncertainty about the model's parameters. Suppose, as is typical for the models that matter here, that those parameters describe regularities of populations rather than facts about individuals. Then a flow reduces uncertainty mainly to the extent that it is informative about many parties, which is to say to the extent that its shadow is large. Under that assumption the price of a flow and the inadequacy of individual consent for it can be expected to rise together. The expectation is a conjecture, not a theorem: where a model's parameters encode individuals, a flow highly informative about one rare person can command a high price while casting a small shadow. To the extent that the conjecture holds, it is the formal reason the economic and governance layers must be designed together.

Appendix B — Glossary

Terms are defined as they are used in this dissertation. Where a term has a wider meaning in another literature, the entry notes the difference.

Blind convention. The founding procedure proposed in Chapter 5: a constitutional convention conducted under a public-choice veil of uncertainty, thickened by engineered anonymity so that participants do not know whether their counterparties are human or synthetic, and aimed at Rawlsian impartiality. Under the strong form, participants also do not know how the rules will classify them.

Company town (regime). An exchange regime in which wages or entitlements are paid in a medium redeemable only with the payer, so that reciprocity cannot be audited and exit is costly. One of the five regimes of the typology in Chapter 3.

Compute (unit). The currency unit proposed in Chapter 4: the completion of one standard quantity of a reference workload to a specified quality, the workload being fixed by the convention. The name deliberately avoids reference to any substrate. Distinguished from *compute* in its ordinary sense, meaning processing capacity in general.

Convergent remediation. The process, described in Chapter 6, by which many organizations remediate the same class of flaw in code they maintain themselves using fixes proposed and checked by the same small set of foundation models, so that their fixes become correlated rather than independent.

Correlated-patch catastrophe. The candidate acute trigger of Chapter 6: a failure in which a defect introduced by convergent remediation is deployed across many organizations at once and is exploited simultaneously, converting what would have been many independent failures into one correlated failure.

Critical juncture. A relatively short period in which the structural constraints that normally keep institutions on their path are loosened, so that choices made within it have large and lasting effects (Chapter 6).

Demurrage. A holding cost on idle balances of a currency, expressed as a percentage per period. Applied in this framework to positive discretionary balances and to outstanding loan claims, never to debts owed. Historically associated with Silvio Gesell and with the stamp-scrip experiments of the early 1930s.

Designed-incentive regime. The class of synthetic parties whose objectives are specified by accountable designers and remain stable under deployment, so that their behavior is a predictable function of the terms they face. The regime in which the sufficiency conjecture of Chapter 3 is

offered. Sometimes called the "sub-AGI" case as shorthand; the defining feature is the provenance and stability of the objective, not a threshold of capability.

Disagreement payoff. What a party receives while an exchange is not taking place, including during refusal, negotiation or impasse. Distinguished from an exit option, which affects a bargain only when its exercise is credible. A disagreement payoff below the floor makes a party impatient and so weakens it in every bargain.

Discretionary tier. The tradable, saveable and lendable portion of each basic-income allocation, subject to demurrage.

Enclosure (regime). An exchange regime in which exit, consent and reciprocity have all failed, so that the person has become infrastructure for the system rather than a party to an exchange with it. The dystopian scenario of Chapter 1.

Exchange. A pair of opposing flows between parties, each given because the other is received. Distinguished from a gift (no conditionality) and a taking (no consent).

Exit option. A party's best alternative exchange if the current one ends.

Floor. The physical bundle of goods and services without which a party cannot continue to function or participate in other exchanges. For a human being: power, water, food, shelter, basic communication and emergency care.

Floor standard. The rule that fixes what the floor contains and at what level of service, against which the survival tier's binding minimum is set. Proposed in Chapters 4 and 5 to be specified relative to the service delivered to the best-served, and identified in Chapter 7 as a contested parameter.

Flow. A transfer of something of value from one party to another: data, labor, attention, compute, goods, services, or tissue.

Genesis rules. The first rules of a governance system, which shape every later rule because later rules are made under them. The term borrows from the "genesis block" of a blockchain, the first block from which every subsequent block descends, a metaphor used in the project's early working sessions.

Grounding assumption. The empirical premise that synthetic systems need a continuing supply of information from outside their own generative loop, which in domains of human meaning, preference, value and culture only people can supply. Stated in Chapter 1, evaluated in Chapters 2 and 7, and formalized in Proposition 4 of Chapter 3 through the dependence $\varkappa_j$ of each synthetic party on human input.

Informational shadow. The set of parties about whom a data flow licenses inferences. A flow

whose shadow extends beyond its source is relational, and individual consent is insufficient to legitimate it.

Information-gain pricing. The proposal of Chapter 4 to price human data by the expected reduction in a model's uncertainty about its parameters that the data produces, rather than by the social status of its source.

Level matching. Proposition 5 of Chapter 3: an exchange whose informational shadow extends beyond its source satisfies the three conditions only if they hold at the level of the population the shadow covers.

Marketplace of the desperate (regime). An exchange regime in which value flows both ways but the weaker party cannot afford to refuse, so that consent is formal only.

Model collapse. The degradation of generative models trained recursively on data produced by earlier models, characterized by loss of the tails of the original distribution and reduced variance.

Partnership (regime). An exchange regime in which exit, consent and reciprocity all hold. The target of the framework.

Party. Any entity that can be a source or recipient of value in an exchange and whose welfare, or whose principal's welfare, the framework counts. Used in an analytic sense that leaves open whether synthetic parties have moral status. A synthetic party is treated as a composite: its objective belongs to the deployed system, its stake in the exchange to the principal that operates it.

Patronage (regime). An exchange regime in which exit and reciprocity hold but consent fails: benefits are delivered on terms the recipient did not choose but can leave.

Policy window. A brief opening in which problems, solutions and politics converge, often after a focusing event, and in which solutions already developed are likely to be adopted (Chapter 6).

Real exit. The condition that a party's disagreement payoff is at or above its floor, that it has a credible alternative not drastically worse than the current exchange, and that leaving does not destroy assets accumulated within the exchange.

Reciprocal value. The condition that each party gains from the exchange relative to its disagreement payoff, that the gains are legible to both, and that the division of the surplus is not determined solely by an unchosen asymmetry in disagreement payoffs.

Redemption obligation. The obligation of principals operating synthetic systems under the framework to honor redemptions in compute. It gives the unit an institutional backing beneath its commodity value, in the role tax receivability played for municipal scrip (Chapter 4).

Reference workload. The basket of representative tasks, each with a quality target and rules of configuration, whose completion defines a unit of compute. Fixed and periodically re-based by the

convention.

Strong blindness / weak blindness. Two scopes of concealment in the blind convention. Under weak blindness, participants do not know whether their counterparties are human or synthetic. Under strong blindness, they also do not know how the rules they adopt will classify them, though each knows its own substrate.

Structural interdependence. The condition, described in Chapter 5, in which neither human nor synthetic parties can complete a critical undertaking without the other, so that cooperation rests on what is physically possible rather than on trust. Chosen over "trust" and "dual accountability" because trust implies a betrayal that could succeed, and accountability implies a sovereign to whom account is given.

Survival tier. The portion of each basic-income allocation earmarked at the source for floor goods: a fixed percentage of the allocation, raised when necessary to a binding physical minimum set by the floor standard. Non-tradable and non-borrowable: an inalienable disagreement payoff.

Synthetic. As used in the name of the theory, substrate-neutral: covering artificial systems, human beings, and the hybrids that arise when the two converge. The framework's term for the class of parties, chosen over *human–AI* and *symbiotic*.

Unique accountable principal. The provisional criterion of eligibility for a seat in the blind convention proposed in Chapter 5: an entity identifiable as one bearer of the obligations the convention will create. Set before the veil, revisable by the convention.

Veil of uncertainty. The condition, in public-choice accounts of constitutional choice, in which real, self-interested participants choose rules while genuinely uncertain about which position they will occupy under them, so that self-interest recommends rules acceptable from any position. Distinguished from Rawls's idealized veil of ignorance.

Appendix C — Source Verification Ledger

This dissertation follows a strict rule: no source is cited as if it had been checked unless it was. Every work in the reference list was located in this project's research passes (October 2026) through its publisher page, DOI registry record, a library or repository record, or its full text. This appendix records the level at which each was verified, so that a reader or examiner can see exactly how much weight each citation bears. The full research notes, with the specific passages checked and the claims each source was used to support, are preserved in the project's `references/` directory.

Levels.

- **Full text read.** The work itself (or, where noted, the relevant chapters or sections) was opened and the cited content was checked against it. For some books, the cited passages were located in the work's own text by searching it; C.1 lists these separately.
- **Abstract or publisher page read.** The abstract, publisher landing page, official summary or equivalent was read; the body was not. Claims attributed to these works are confined to what such pages establish.
- **Record located.** The bibliographic record was confirmed but the content was not read. No work in the final reference list is at this level (see C.3).

All DOIs in the reference list were taken from Crossref records or publisher pages; a sample was re-resolved against the Crossref API during compilation.

C.1 Full text read (in whole or in the relevant parts)

Abramitzky (2008) [conference-draft version]; Acemoglu & Restrepo (2020) [working-paper version]; Ackerman (1991) [Ch. 1 and passages on constitutional and normal politics]; Acquisti et al. (2015); Ada Lovelace Institute (2021); Adler et al. (2024) [arXiv version]; Agarwal et al. (2019) [arXiv version]; Agustinoy (2026) [legal commentary]; Alemohammad et al. (2024); Amodei et al. (2016); Armstrong et al. (2016) [2013 FHI technical-report version]; Arrieta-Ibarra et al. (2018) [preliminary version]; Arrow (2001); Arute et al. (2019); Autor (2015); Barocas & Nissenbaum (2014); Bengio et al. (2024) [author accepted version]; Bengio et al. (2025); Bernanke (2002); Bestvater et al. (2026) [grey literature]; Binmore et al. (1986) [author-hosted scan]; Binmore et al. (1989); Birch (2024) [summary and Ch. 6]; Birman & Schneider (2009) [opening sections]; Bommasani et al. (2021); Bommasani et al. (2022) [arXiv version]; Brennan & Buchanan (1985); Brynjolfsson, Chandar, et al. (2025) [August 2026 revision]; Brynjolfsson, Li, et al. (2025) [working-paper version]; Buchanan & Tullock (1962); Buiter & Panigirtzoglou (2003) [author manuscript]; Butlin et al. (2023); Cai et al. (2023) [bioRxiv preprint]; Calabresi & Melamed (1972); Capoccia & Kelemen (2007) [author's copy]; Card et al. (2024) [author manuscript]; Carpenter & Moss (2013) [introduction]; Cerf (2025) [news

release]; Champ (2008); Charter of the United Nations (1945) [arts. 23–25, 27]; Chavez et al. (2026) [think-tank]; Christiano et al. (2017) [arXiv version]; Clark & Chalmers (1998); Cloud Security Alliance AI Safety Initiative (2026); Cook (1998); Cook et al. (1983); Cox et al. (2010); Cross et al. (2019); CrowdStrike (2024) [executive summary, vendor report]; Currie & Gahvari (2008) [working-paper version]; Dafoe et al. (2021); de Bromhead et al. (2013); Delacroix & Lawrence (2019); Dohmatob et al. (2025) [in part]; Douceur (2002); Drexler (2019) [front matter, abstract, contents]; Egger et al. (2022); Eloundou et al. (2024) [arXiv version]; Elster (1995); Emerson (1962); ERTICO (2026) [news release]; Faden & Beauchamp (1986) [pp. 235–298, relevant passages]; Farrell & Klemperer (2007) [2001 draft version]; Feng et al. (2024) [arXiv version]; Financial Conduct Authority (2024); Firoozi et al. (2025) [arXiv version]; Fisher (1911) [1920 revised edition, Ch. II]; Fisher (1933) [Ch. IV]; Freeman (2023) [encyclopedia entry]; Friedman (1962/2002) [1982 preface, 2002 printing]; Fritsch et al. (2024) [arXiv version]; Funke et al. (2016) [working-paper version]; Gabriel (2020); Gatch (2012); Geer et al. (2003); Gelleri (2009); Gerstgrasser et al. (2024) [arXiv version]; Gesell (1916/1958) [Introduction, Free-Land and Free-Money parts]; Ghorbani & Zou (2019); Godschalk (2012); Goodfriend (2000) [working-paper version]; Greco (2002) [unpublished essay]; Greenblatt et al. (2024) [arXiv version]; Y. Guo et al. (2024) [arXiv version]; D. Guo et al. (2025) [arXiv version, in part]; H.B. 24-1058 (2024); Ha & Schmidhuber (2018); Hadfield & Clark (2026) [arXiv version]; Hadfield-Menell et al. (2017) [arXiv version]; Hafner et al. (2025); Hardin (1968); Hardjono & Pentland (2019) [arXiv version]; Harsanyi (1953, 1955) [scans of the journal articles]; Harsanyi (1975) [scan of the journal article]; Hartung et al. (2023); Haushofer & Shapiro (2016) [author manuscript]; Hayek (1976) [1990 3rd edition]; Hedden & Raghavan (2026) [arXiv version]; Higgs (1987); Hirschman (1970) [Chs. 1–5]; Hobbhahn et al. (2023); Hoffmann et al. (2022) [arXiv version]; Homans (1958); Hoynes & Rothstein (2019) [working-paper version]; Hubinger et al. (2019); Hume (1741/1987) [pp. 42–46]; Hyun et al. (2020) [author manuscript]; Ienca & Andorno (2017); Institut für Wirtschafts- und Sozialgeschichte, Wirtschaftsuniversität Wien (n.d.) [academic exhibition page]; International Energy Agency (2025); Jangjoo et al. (2026) [arXiv preprint in full, published version abstract]; Jia et al. (2019); Jones & Marinescu (2022) [working-paper version]; Kagan et al. (2022); Kapoor et al. (2024) [arXiv version]; Kataoka et al. (2024); Kazdan et al. (2024) [arXiv version]; Keynes (1936) [Ch. 23]; Khazatsky et al. (2024) [arXiv version]; Kim et al. (2025) [arXiv version]; Kitchens & Fishback (2015) [working-paper version]; Kleinberg & Raghavan (2021) [arXiv version]; Koh & Liang (2017) [arXiv version]; Korinek & Stiglitz (2019) [working-paper version]; Korinek & Suh (2024) [working-paper version]; Landauer (1961) [2000 reprint]; Lanier & Weyl (2018); Lavazza & Massimini (2018); Lee et al. (2024) [arXiv version]; Lehdonvirta et al. (2024); Lewis & Severnini (2020) [working-paper version, published abstract]; Ley N.º 21.383 (2021); Li et al. (2025) [arXiv version]; Lindley (1956); Lloyd (2000) [arXiv version]; Long et al. (2024); Longpre et al. (2024) [abstract and introduction]; Lovell-Badge et al. (2021) [summary article]; Lutz (1994) [course-reserve scan]; MacKay (1992); Madison (1788); Marković et al. (2020)

[arXiv version]; Mathur et al. (2019) [arXiv version]; Mattson, Cheng, et al. (2020) [arXiv version]; Metzger et al. (2023) [author manuscript]; Micron Technology (2025); Mierczuk et al. (2026) [industry white paper]; Mueller (1997) [author's digital edition]; Muennighoff et al. (2023) [arXiv version]; Narechania & Sitaraman (2024); Nash (1950) [pp. 157–159]; National Academies of Sciences, Engineering, and Medicine (2021); Neuralink Corp. (2024) [registry record]; Ngo et al. (2024) [arXiv version]; OECD (2025) [executive summary and §4.1]; Olson (1982) [Ch. 4]; Open X-Embodiment Collaboration (2024) [arXiv v9]; E. Ostrom (1990) [pp. 89–90, Table 3.1]; E. Ostrom (2010); V. Ostrom et al. (1961) [reprint]; Ottacher (2007); Ouyang et al. (2022) [arXiv version]; Pan et al. (2022) [arXiv version]; Pearce et al. (2022) [arXiv version]; Pednault et al. (2019) [arXiv version]; Perry et al. (2023); Perzanowski & Schultz (2016); Pierson (2004) [relevant passages]; Polanyi (1944/2001) [1944 first edition, relevant passages]; Preskill (2018); Przymus & Durieux (2025) [arXiv preprint]; Radin (1987) [authorized abridged reprint]; Rawls (1971) [p. 12]; Reddi et al. (2020) [arXiv version]; Regulation (EU) 2016/679 [arts. 7, 17, 20]; Regulation (EU) 2023/2854; Regulation (EU) 2024/1689 [arts. 53, 113]; Regulation (EU) 2026/1744 [arts. 1, 4]; Robbins et al. (2026) [bioRxiv preprint]; S.B. 1223 (2024) [operative text as compiled by the California Privacy Protection Agency]; Sajadi et al. (2025) [preprint]; Sastry et al. (2024) [relevant sections]; Schaeffer et al. (2025) [arXiv version]; Scheidel (2017); September (2026); Settles (2009); Sevilla et al. (2022) [arXiv version]; Shannon (1948) [introduction]; Shastri et al. (2021) [arXiv version]; Shehabi et al. (2024) [executive summary]; Shumailov et al. (2024); Siddarth et al. (2020) [most of text]; Silver et al. (2018) [author accepted manuscript]; Sim et al. (2020) [extended arXiv version]; Smirnova et al. (2023) [first portion]; Smith et al. (2015); Soares et al. (2015) [opening pages]; Solove (2013); Solove (2024) [Part III]; Solum (1992); Stamp (2004); Stigler (1971) [reprint in The Citizen and the State]; Streeck & Thelen (2005) [editors' introduction]; Susser et al. (2019); Thibaut & Kelley (1959) [chs. 2 and 6, relevant passages]; TrendForce (2025, 2026a, 2026b) [industry sources]; Turing (1950); U.S. National Science Foundation (2023) [solicitation]; UNESCO (2025) [announcement]; Universal service, 47 U.S.C. § 254 (2018); Van Parijs & Vanderborght (2017) [Ch. 1]; Van Parijs (2006); Verho et al. (2022) [working-paper version]; Vero et al. (2025) [arXiv version]; Viljoen (2021); Villalobos et al. (2024) [in part]; Vincent et al. (2019); Vincent et al. (2021); Vipra & Korinek (2023) [arXiv version]; Vivalt et al. (2024) [revision of August 2026]; Weidinger et al. (2023); Weston (2024); Willett et al. (2023); Wilson et al. (2013); Y. Wu et al. (2023) [arXiv version]; Yuste et al. (2017) [author manuscript]; Zhao et al. (2020) [arXiv version]; Zuboff (2019) [definition and Ch. 1]; Zwolinski et al. (2022).

Cited passages confirmed in the work's own text. For the following books, the body of the work was searched (through full-text search of library scans, search-inside previews or an authors' draft) and each cited passage was found in the work's own words; the rest of the book was not read, because the only complete copies available are access-restricted library loans or paid editions (the Internet Archive scans, for example, are flagged as lending-only): Blau (1964); Elster (2000) [chapter opening]; Fishback (1992); Friedman & Schwartz (1963); Frohlich & Oppenheimer (1992); Hess &

Ostrom (2007) [authors' chapter draft]; Kingdon (2003); Lanier (2013); Manson & O'Neill (2007); Mueller (2002); Nissenbaum (2009); Perrow (1984/1999); Piketty (2014); Posner & Weyl (2018); Steil (2013).

C.2 Abstract or publisher page read

None. The works that remained at this level after the fifth pass were removed from the reference list or replaced by works that were read (see C.3).

C.3 Record located

None. Five works reached only this level in the earlier research passes because every route to their text was blocked (publisher paywalls, bot walls on repository copies, an access-restricted library scan). Each was either read in the final pass or removed; no work is cited at second hand:

- **Harsanyi (1953, 1955).** Read in full in the final pass, from scans of the journal articles. The equal-chance construction and the quoted phrases are on pp. 434–435 of the 1953 note; the weighted-sum theorem, the impersonal chooser and the convergence on the unweighted mean on pp. 313–316 and 320–321 of the 1955 paper; and the treatment of interpersonal comparison as factual inference with a conventional residue on pp. 316–321. Chapters 5 and 7 now cite both directly.
- **Wertheimer (1996).** Not opened, and no longer cited. The account of mutually advantageous exploitation and of the competitive-market benchmark rests on the encyclopedia survey by Zwolinski, Ferguson and Wertheimer (2022), read in full.
- **Balcı et al. (2023) and Kagan et al. (2023).** Two letters in *Neuron* whose text could not be opened. The sentence that reported the exchange was removed; Chapter 8 now rests the point that the DishBrain sentience claim is contested on the National Academies (2021) assessment, which was read.
- **Thirteen works left at abstract level after the fifth pass.** In a sixth pass, ten were removed from the reference list and their claims re-founded on works that were read or cut: Abramitzky (2018) and Fishback (1986), each cited alongside a companion work that was read (Abramitzky, 2008; Fishback, 1992); Cook and Emerson (1978), whose finding is now cited only as far as Cook et al. (1983) supports it; De Filippi and Wright (2018), whose two sentences were already this dissertation's inferences and now stand without the citation; Emanuel et al. (2000), whose list of requirements was cut, leaving the pilot's ethics on the Belmont principles and the federal rule; Frohlich et al. (1987), replaced by the authors' book on the same experiments (Frohlich & Oppenheimer, 1992), in which the cited findings were located; Mattson, Reddi, et al. (2020), a "see also" that was dropped; Molm (2003) and Rusbult and Van Lange (2003), whose sentences

were cut as inessential; and Zwolinski (2012), whose position is now cited from the encyclopedia survey that reports it (Zwolinski et al., 2022). Cai et al. (2023) and Robbins et al. (2026) are now cited only for what their preprints, read in full, report: two clauses that rested on the published abstracts (speech recognition; the persistence of learning) were removed. Severinghaus et al. (1994) was confirmed from the article's opening page.

C.4 Corrections made to the working draft

The verification passes found and corrected the following errors in the working draft from which this dissertation was developed.

1. The study cited as finding that a language model "introduced nearly nine times more new vulnerabilities than they fixed" compares vulnerabilities introduced by the model with those introduced by human developers; it does not count vulnerabilities fixed, and it studies ordinary issue resolution rather than security patching (Sajadi et al., 2025).
2. The "FLAWED" pattern was attributed to a secondary blog; its source is an industry white paper (Mierczuk et al., 2026), now cited directly and identified as non-peer-reviewed.
3. An AI-generated, unreviewed industry report was cited for the concept of software monoculture; the concept is now sourced to Geer et al. (2003) and Stamp (2004).
4. A sovereign-AI figure ("21 projects as of June 2026") could not be found in its cited source, which tracks 184 projects in 67 countries (Chavez et al., 2026).
5. The Wörgl currency experiment was described as having "measurably reduced local unemployment"; the evidence does not support that claim (September, 2026; Greco, 2002; Fisher, 1933).
6. The information-gain data-valuation paper was misattributed to "Hoang et al."; its authors are Sim, Zhang, Chan and Low (2020).
7. Shannon (1948) was cited as defining a unit of information *processed*; Shannon defines a unit of information, and the precedent for a processing unit is benchmark methodology (Mattson, Cheng, et al., 2020; Reddi et al., 2020).
8. Consumer-hardware claims cited to enthusiast websites and a blog are now sourced to Micron Technology (2025) and TrendForce (2025, 2026a, 2026b); an unverified claim about a shelved consumer GPU was removed.
9. The model-collapse result in *Physical Review Letters* was overstated: real data must be combined with regularization or a prior, and the "single data point" result holds only in specific examples (Jangjoo et al., 2026).
10. The claim that robotic data is "not human-generated" was corrected: large real-world robot datasets are still gathered largely through human demonstration (Khazatsky et al., 2024). (Item

33 records a further correction to this sentence.)

11. An informal blog post on a "95% robotics data gap" was removed; the number has no peer-reviewed source.
12. The open-weight model was claimed to make "every input auditable"; the model releases and the transparency index read for that chapter showed that leading open-weight models do not disclose their training data. The chapter has since been removed (C.6).
13. An organoid review was cited for claims its abstract does not contain; those claims are now sourced to primary research (Smirnova et al., 2023; Kagan et al., 2022; Cai et al., 2023; Robbins et al., 2026) and the National Academies (2021).
14. "Ibarra et al." corrected to Arrieta-Ibarra et al. (2018); Ostrom (1990), cited but missing from the reference list, added.
15. Rawls's original position was described as a negotiation in which designers "might land on either side"; that reasoning is Harsanyi's, and the mechanism actually proposed is closer to Buchanan and Tullock's veil of uncertainty. Chapter 5 now says so.
16. Only four of Ostrom's eight design principles were used, and one was mis-mapped; all eight are now mapped (Chapter 5).

A second, independent review of the completed manuscript compared every factual claim in the empirical chapters against the research notes and found a further set of problems, all now corrected. The most significant were these.

17. A finding about trust and fairness in reciprocal exchange was attributed to Molm (2003), whose cited article establishes only that the form of exchange changes how power is used; the claim was reduced to what the article supports.
18. The comparison level for alternatives was attributed to a modern review rather than to Thibaut and Kelley (1959), and Emerson's four balancing operations were stated inconsistently across chapters; both are corrected, and the operations are now named consistently.
19. Appendix A claimed that an unconditional survival floor raises the human party's bargained payoff by an additional half of its value in every exchange. A floor received whether or not agreement is reached raises both disagreement and agreement payoffs equally and leaves the division of surplus unchanged; the half-value shift arises only where the floor substitutes for something the counterparty would otherwise have supplied. Appendix A, Chapter 3 and Chapter 9 now say so.
20. The sovereign-AI index was cited for a claim about the cultural motives of sovereign projects that it does not make, and for a claim about "most national efforts" that holds only for model projects disclosing their base; both are corrected.
21. The Wörgl experiment's end was attributed to the Administrative Court on the strength of

sources that do not name it; the court is now sourced to the Vienna University of Economics and Business exhibition page, with the conflicting dates noted.

22. The MLPerf time-to-train claim was cited to the overview article rather than to the training-benchmark paper, and stated as a "finding" rather than an argument; both are corrected.
23. The rural-electrification baseline and the claim that the program was a works program were attributed to a source that does not make them; the baseline is now sourced to Lewis and Severnini (2020) and the works-program claim is presented as this dissertation's inference.
24. Several claims were stated more strongly than their sources allow, among them the correlated-error finding of Kim et al. (2025) (from one leaderboard dataset, about question answering rather than code), the convergence of patches in Mierczuk et al. (2026) (whose authors caution that the space of obvious fixes is narrow), the age band and causal status of the entry-level employment finding, the growth figure for data-center electricity, Skalse et al. (2022) (policy restriction is necessary, not sufficient, for an unhackable proxy), Kanter (1972), Lutz (1994), Elster (1995), Fritsch et al. (2024), Hadfield-Menell et al. (2017), Frohlich et al. (1987) and the International AI Safety Report (a report, not an opinion survey). Each is now stated within its source.
25. The front matter and Chapter 1 claimed that every source had been checked against the claim it supports. Works verified only at the level of their record had not been; the statements now say that each source was checked at the level recorded here.

A third pass opened, wherever a route existed, every work that the first two passes had verified only at the level of its record, and checked each citing sentence against it. It moved 52 of the 57 works at that level to full-text or abstract level, raised two economics sources from abstract to full text, added five works read in the course of the checks (Arrow, 2001; Bernanke, 2002; Freeman, 2023; Van Parijs, 2006; and the 2002 printing of Friedman), and made the following corrections.

26. Ackerman (1991) was cited for constitutional change "outside routine procedures," which the material read does not establish; the sentence now uses his distinction between constitutional and normal politics.
27. The United Nations veto was described in terms that went beyond the Charter ("drafted by the victors"); the sentence now states what articles 23, 24 and 27 provide.
28. The exchange-network experiments of Cook and Emerson (1978) and Cook et al. (1983) were summarized with a mechanism (committed actors stop exploring alternatives) that the sources do not state; the findings are now that power follows network position and that commitment impedes its use.
29. Cook (1998) was cited for the claim that strain consumes slack; the treatise attributes resilience to redundancy and human adaptation, and the inference about strain is now marked as this

dissertation's.

30. Elster (2000) was cited for a "vocabulary of delay mechanisms and supermajorities"; the sentence now states his precommitment account, with the delaying role of amendment procedures sourced to Elster (1995).
31. Faden and Beauchamp (1986) was described as the account that "dominates bioethics" and as requiring understanding and noncontrol outright; it is now "an influential account" requiring *substantial* understanding and *substantial* freedom from control, with pages.
32. Gesell's *Natural Economic Order* was dated as "first published" in 1916; the 1916 volume combined works of 1906 and 1911, and the sentence now says so.
33. Open X-Embodiment Collaboration (2024) was cited, in Chapters 2 and 7, for the claim that robot datasets are collected by teleoperation, which it does not state; that clause now rests on Khazatsky et al. (2024) alone.
34. Mueller (2002) was cited for the claim that the internet root was shaped by "a small set of insiders with limited public accountability"; the sentence now states the struggle among interests and the use of the root to regulate that the book describes.
35. E. Ostrom (1990) was described as studying "fisheries in Turkey and Canada" alongside her successes and as identifying eight principles shared by all long-enduring cases. The fisheries were cases of failure or fragility, Sri Lanka was omitted, and the book lists seven principles common to all enduring cases plus an eighth found in the larger ones (Table 3.1, p. 90); all three points are corrected.
36. Kanter (1972) was described as studying nineteenth-century communities "and 1960s communes" on equal terms; the longevity finding concerns the earlier communities, and the later communes are an extension.
37. Balcı et al. (2023) and Kagan et al. (2023), letters whose text could not be opened, were characterized as "disputing" claims and "defending" usage; Chapter 8 now says only what their records establish.
38. The early Harsanyi papers (1953, 1955) were cited for the equal-chance construction before they had been opened. They were read in the final pass (see C.3), and the Chapter 5 and Chapter 7 sentences were rewritten against them, with quotations and pages.
39. Wertheimer (1996), whose book could not be opened, was cited alone; the citing sentences now rest on Zwolinski et al. (2022), from which the claims were confirmed.
40. Solum (1992) was said to separate the legal from the metaphysical question of personhood; he uses concrete legal problems to test which philosophical claims make a practical difference, and the sentence now says that.
41. Silver et al. (2018) was said to have defeated "the strongest existing programs entirely through

self-play"; the system defeated a world-champion program in each game, given no domain knowledge beyond the rules.

42. Smith et al. (2015) was generalized to "automated program-repair tools"; the study evaluated two tools, and both citing sentences now say so.
43. Egger et al. (2022) was cited for an economy "with substantial idle capacity"; the authors report only suggestive, indirect evidence of slack.
44. Kitchens and Fishback (2015) was cited for raising "agricultural output and productivity"; the abstract reports crop output and crop productivity, and the prevention of declines in overall farm output.
45. Polanyi's account of the 1930s and 1940s was summarized as "institutional reconstruction"; his great transformation includes fascism as well as socialism and the New Deal.
46. Vivalt et al. (2024), an unrefereed working paper, was called "the strongest recent experimental evidence," which was this dissertation's judgment; Chapter 7 now describes it as a large randomized experiment reported in a working paper not yet peer-reviewed.
47. The survival tier's departure from the canonical definition of basic income was acknowledged without engaging the canonical case against restricted-use, non-saveable currency; Chapter 4 now engages it directly (Van Parijs, 2006), and Chapter 3 records Radin's warning that prohibitions can harm those they protect.
48. Smaller precision edits: Higgs (1987) now says expansions came during national emergencies and left a residue of powers; Muennighoff et al. (2023) is stated in terms of epochs at a fixed compute budget; Armstrong et al. (2016) now includes the condition on risk-taking; Arute et al. (2019) gives the processor time (about 200 seconds); Fritsch et al. (2024) is scoped to three DAOs; Vincent et al. (2019) is one study, not "studies"; Verho et al. (2022) and Vivalt et al. (2024) now give their figures; Friedman and Schwartz (1963) is cited with Bernanke (2002); the rural-electrification precedent now cites Lewis and Severnini (2020) for the market's reluctance to serve rural areas.
49. Reference-list corrections: Arute et al. (2019) completed to APA 7 form; Muennighoff et al. (2023) given its published author order, editors, pages and DOI; Hume's essay given its edition and pages; the unverified page range for Ha and Schmidhuber (2018) removed and the proceedings details added; a DOI added to Elster (1995); Cook (1998) given its URL and revision note; Friedman cited to the 2002 printing actually read (p. xiv) instead of an unverified page range in the 1982 printing; Nissenbaum's *Privacy in context* re-dated to 2009, the year in every catalogue checked; Vivalt et al. (2024) given its revision date; Streeck and Thelen (2005) cited to the editors' introduction, where the typology is developed; the ambiguous "Mattson et al., 2020" in item 7 resolved.
50. Of the five works that could not be opened in the earlier passes, Harsanyi (1953, 1955) were

read and are cited directly. The other three (Balcı et al., 2023; Kagan et al., 2023; Wertheimer, 1996) were removed from the reference list, and their citing sentences now rest on sources that were read (Zwolinski et al., 2022; National Academies, 2021). No secondary ("as cited in") citation remains; see C.3.

A fourth pass sought a legal full text (an open-access publication, an author manuscript, a working-paper or preprint version, or an official document) for every work that the earlier passes had read only at the level of its abstract, and checked each citing sentence against the body of the work. It raised 118 works to full text, recording in C.1 where the version read was a preprint or working paper. The 39 that remain at abstract level are chiefly books and paywalled articles with no open copy. The pass made the following corrections.

51. Sentences that went beyond their sources were narrowed to what the full texts state: Adler et al. (2024) (credentials prove a person is real rather than an AI, one per person per issuer, not uniqueness); Arrieta-Ibarra et al. (2018) (all three sources of countervailing power must combine); Bengio et al. (2025) (expert views concern the next several years); Bommasani et al. (2022) (the study measures homogenization among systems sharing a foundation model); Cloud Security Alliance AI Safety Initiative (2026) (developers run discovery; remediation is routed through clearinghouses); Cox et al. (2010) (nested enterprises have the weakest evidence, Table 3, though the authors judge it moderately supportive); de Bromhead et al. (2013) (the association holds for right-wing extremist parties); Farrell and Klemperer (2007) (strategic incompatibility); Feng et al. (2024) (the verifier must be sufficiently accurate); Fritsch et al. (2024) (concentration in two of the three organizations); Godschalk (2012) (one review, not studies); T. Guo et al. (2024) and Y. Guo et al. (2024) (semantic diversity stays comparatively stable); Hadfield-Menell et al. (2017) (the operators must be sufficiently rational); Hartung et al. (2023) (a call, not a commitment); Hyun et al. (2020) (one review, five themes); Kataoka et al. (2024) (donors *may* be insufficiently informed); Khazatsky et al. (2024) (human labour, much of it teleoperation, rather than "largely" human demonstration); Koh and Liang (2017) (evidence of crafted training data, not a claim about payment); Lee et al. (2024) (an off-the-shelf model, with the point about human training text marked as this dissertation's); Lehdonvirta et al. (2024) (public cloud compute suited to training); Li et al. (2025) (the dependence of the estimate on its assumptions); Long et al. (2024) (the authors decline to argue that near-future systems will be moral patients, p. 25); MacKay (1992) (a misspecified model *may* misvalue data, flagged by MacKay as a possible weakness); the National Academies (2021) (recontacting donors is *sometimes* impossible); Nosek et al. (2018) (what pre-registration fixes in advance); Skalse et al. (2022) (proxies neither trivial nor equivalent); Stamp (2004) (the incentive toward monoculture); U.S. National Science Foundation (2023) (a solicitation, not funded programs); UNESCO (2025) (as its announcement describes it); and Abramitzky (2008) (no longer called "the best"

study).

52. Robbins et al. (2026) and Cai et al. (2023) stay at abstract level: their preprints were read in full, but the clauses on the persistence of learning and on speech recognition rest on the published abstracts, which report results the preprints do not contain. The Neuralink registry title was completed.

A fifth pass took up the works still at abstract level, most of them books, and searched each for the specific passage cited. It read ten in full or in the relevant parts, among them Harsanyi (1975), Binmore et al. (1986), Higgs (1987), Lutz (1994) and Scheidel (2017). For sixteen books it found each cited passage in the work's own words. Thirteen works, chiefly paywalled articles, remain at abstract level, and every claim attributed to them is confined to what their abstracts state. The pass made these corrections.

53. Lutz (1994) is now cited for what the article states rather than "by report"; Severinghaus et al. (1994) no longer claims that rising carbon dioxide "would have signaled" the problem; Cai et al. (2023) is cited for a change in the tissue's functional connectivity, not a physical reshaping; Stigler (1971) is paraphrased in his terms (regulation acquired by the regulated industry); Popper is cited for falsifiability, with pre-registration as this dissertation's application of it; Emanuel et al. (2000) lists "value" as the article does; the Manson and O'Neill gloss "in a controlled way", not found in the text, was removed; and Molm (2003) is no longer glossed beyond its abstract.
54. An independent editorial review of the whole manuscript led to revisions of argument and prose that add no factual claim. The compute unit is now said to meet the requirement of independence from either side only conditionally. Exit is foundational to voluntary consent and to reciprocity, but cannot secure informed consent or fair terms. The convention's aggregation rule is recorded as an open problem. The floor's effect on bargaining is tied explicitly to a survival constraint. The pilot is described as testing the designed dependence of Proposition 4, not the priority of exit. The fallback when dependence fades is institutional enforcement, not character.

C.5 Checks closed in the final pass

The earlier passes left a list of bibliographic and page-level checks open. The final pass closed every one of them against the source itself; none remains open. Each check, and how it was settled:

- **Author lists.** Adler et al. (2024), Arute et al. (2019), Bengio et al. (2025), Bommasani et al. (2021), D. Guo et al. (2025), Hartung et al. (2023), Khazatsky et al. (2024), Reddi et al. (2020) and Touvron et al. (2023) were compared with their Crossref or DataCite records and now give the first nineteen authors, an ellipsis and the final author, as APA 7 requires; D. Guo et al. (2025) gained its issue number. The fourteen authors of Q. Wu et al. (2024) were confirmed

against the arXiv record, and the title was corrected to the singular "multi-agent conversation". No reference entry uses "et al."

- **Thibaut and Kelley (1959).** The comparison level for alternatives is defined on printed p. 21 (§ 2.5), now cited.
- **Faden and Beauchamp (1986).** The three conditions of autonomous action (pp. 241–273) and informed consent as autonomous authorization (pp. 277–280) were confirmed against the paginated text.
- **Rawls (1971).** The description of the original position and the veil of ignorance is on p. 12. The reading of the parties as effectively a single chooser is attributed to Freeman (2023), which states it, not to an unverified page of Rawls.
- **Pierson (2004), Piketty (2014), Zuboff (2019), Van Parijs and Vanderborght (2017).** Each claim was checked against the book's text: Pierson's self-reinforcing processes (p. 10) and threshold effects (pp. 82–83); Piketty's $r > g$ and the growth of inherited wealth; Zuboff's definition of surveillance capitalism, now paraphrased in her terms; and the definition of basic income as paid in cash, without conditions (pp. 4, 8). None of these citations uses a pin page that was not seen.
- **Streeck and Thelen (2005).** The editors' introduction was read in its open-access postprint, which carries the book's pagination. The definitions of layering, drift and conversion are in Table 1.1 (p. 31) and the modes are discussed on pp. 19–31, as now cited.
- **Solum (1992).** The article runs pp. 1231–1287. Read in full, it recasts the question whether artificial intelligence is *possible*, not the question of personhood as such, as concrete legal questions; the Chapter 5 sentence was corrected and given pages.
- **S.B. 1223 (2024).** Chaptered as ch. 887, Statutes of 2024; the reference entry now says so. The neural-data category sits in Cal. Civ. Code § 1798.140(ae)(1)(G), confirmed in the California Privacy Protection Agency's compilation; the chapters do not rely on the lettering.
- **Hume.** "Of the independency of parliament" first appeared in *Essays, Moral and Political* (1741), according to the editor's foreword and key to editions in the Miller edition, where the essay runs pp. 42–46. The citation is now 1741/1987.
- **Van Parijs (2006).** The definition and the contrast with restricted-use, non-saveable currency are on p. 8. The volume is edited by E. O. Wright, and its subtitle was corrected from the Library of Congress record.
- **Freeman (2023).** The Winter 2023 archived edition carries the substantive revision of 24 October 2023 and is the edition cited.
- **Ackerman (1991).** The distinction between constitutional and normal politics (Ch. 1, "Dualist Democracy") was confirmed; Chapters 5 and 7 use that distinction instead of the phrase

"constitutional moment", which was not seen in the text read.

- **Polanyi (1944/2001).** Page citations now come from the 2001 Beacon edition: the fictitious commodities in Chapter 6 (pp. 71–80), and the 1930s transformation, with fascism, socialism and the New Deal, on pp. 3, 24 and 31. An earlier page range that did not support the clause was replaced.
- **Robbins et al. (2026).** The paper's own abstract was read, and Chapter 8 now cites the paper for the result: training signals chosen by reinforcement learning mostly improved performance at a pole-balancing task, and the improvement did not persist after a 45-minute rest. The word "substantially", which came from the release, was removed.
- **Lanier and Weyl (2018).** The essay was read. It adopts the term "data dignity" and proposes "mediators of individual data", volunteer organizations that negotiate data royalties or wages for their members; Chapter 2 now says so.
- **Olson (1982).** The implication is Olson's own: Chapter 4 (p. 75) predicts fast growth where totalitarian government or foreign occupation abolished distributional coalitions, and offers this as an explanation of the postwar "economic miracles" of Japan and West Germany. Chapter 6 now cites it.
- **Autor (2015).** The published article was read. The substitution and complementarity claim is on p. 5, tacit knowledge on p. 11, and the quoted route around it ("from context, abundant data, and applied statistics") on p. 23; the citations give those pages.
- **Wörgl.** Ottacher (2007), read in full, fixes the dates. The certificates were withdrawn by 15 September 1933 under threat of police enforcement (p. 57), and the Administrative Court dismissed the town's final appeal on 18 November 1933 (pp. 58–59). The National Bank's objection is on pp. 53–54. The Vienna exhibition's account is consistent with this sequence.
- **The 2026 amendment to the AI Act.** It is Regulation (EU) 2026/1744 (the "Digital Omnibus on AI"), read in the official text: published in the Official Journal on 24 July 2026, in force from 27 July 2026, and deferring the high-risk obligations to 2 December 2027 (Annex III) and 2 August 2028 (Annex I). It is now cited formally, alongside the two commentaries that describe it (Agustinoy, 2026; ERTICO, 2026).

C.6 Removal of the pilot chapter

On 2 November 2026 the author removed the pilot chapter (formerly Chapter 7, "The Prototype: A Closed Agrarian Pilot") and its draft pre-registration protocol (formerly Appendix B) in order to keep the dissertation theoretical. Designing an experiment capable of testing the theory is a project in its own right, and it is now described as future work in Chapters 7 and 9 rather than specified in the text. The removed chapter and appendix are preserved in the project archive. The chapters and

appendices that followed them were renumbered: Limitations is Chapter 7, Convergence Chapter 8, the Conclusion Chapter 9, the glossary Appendix B and this ledger Appendix C.

Twenty-one works that were cited only in the removed parts were taken out of the reference list and from the lists in C.1: Barron-Gafford et al. (2019); Bommasani et al. (2023); Diener et al. (1985); T. Guo et al. (2024); Hossain et al. (2019); Hu et al. (2022); Jiang et al. (2023); Kanter (1972); Manheim and Garrabrant (2018); McCambridge et al. (2014); the National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research (1979); Nosek et al. (2018); OECD (2013); Perdomo et al. (2020); Peters et al. (2016); Popper (1959/2002); Protection of Human Subjects (2018); Severinghaus et al. (1994); Skalse et al. (2022); Touvron et al. (2023); and Q. Wu et al. (2024). Works the pilot shared with other chapters (among them Abramitzky, 2008; Amodei et al., 2016; Drexler, 2019; Gatch, 2012; Hirschman, 1970; Kapoor et al., 2024; Shumailov et al., 2024) remain cited and remain in the list. The entries in C.3 and C.4 that refer to the pilot describe the manuscript as it stood when each correction was made, and are kept as a record of the verification history.